

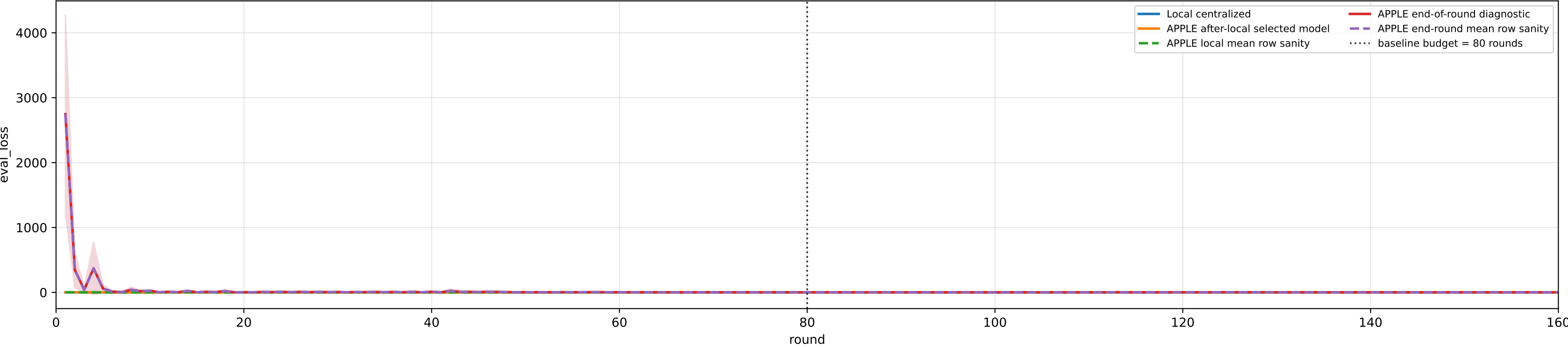
Controlled task-specialization level: Low | APPLE diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

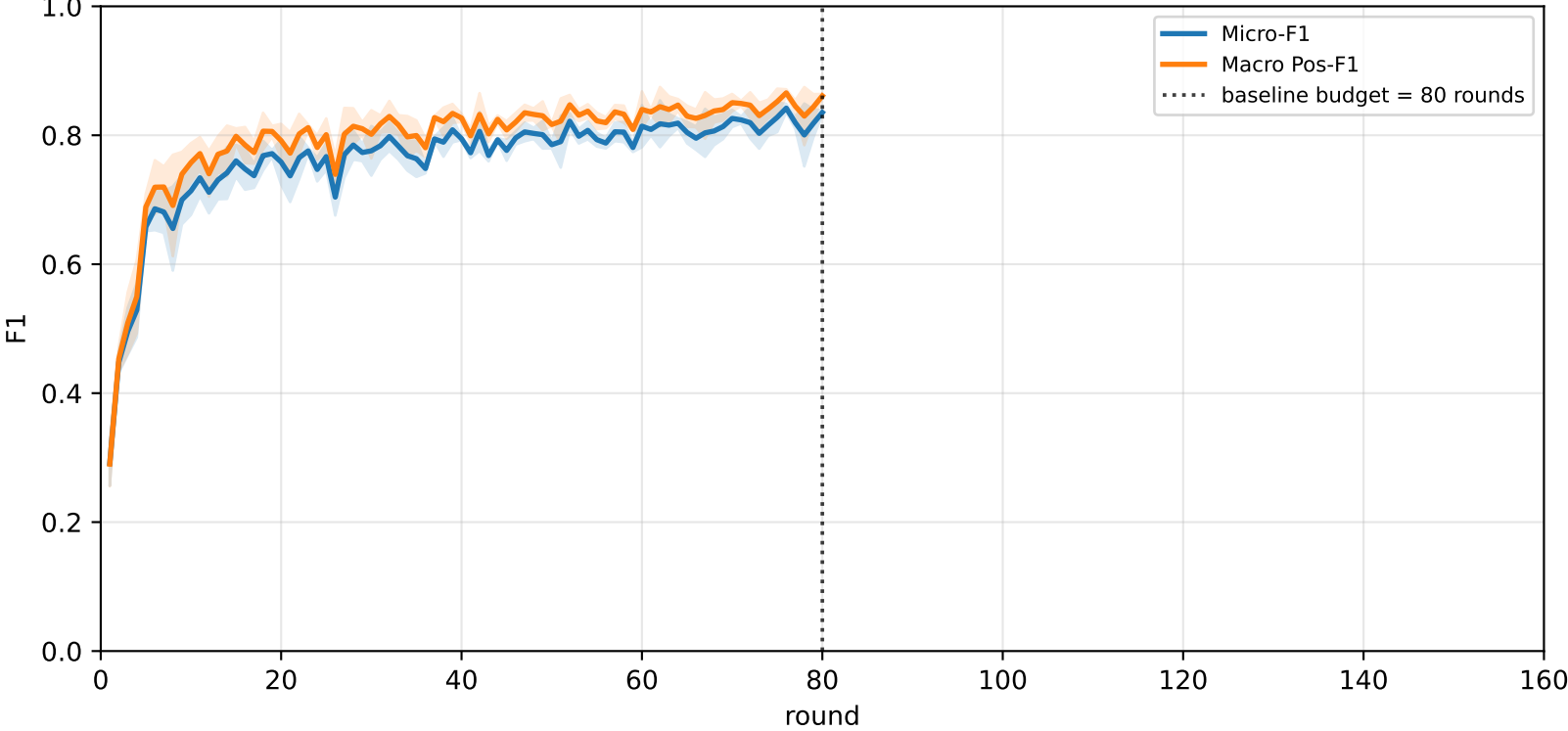
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	83.4% ± 0.3%	85.7% ± 0.3%	96.8% ± 0.6%	94.9% ± 0.4%	86.8% ± 0.2%	78.7% ± 0.7%	70.4% ± 0.6%
APPLE	80.5% ± 2.0%	82.4% ± 1.9%	95.1% ± 1.2%	90.8% ± 2.4%	80.4% ± 2.8%	75.1% ± 1.2%	70.5% ± 2.5%

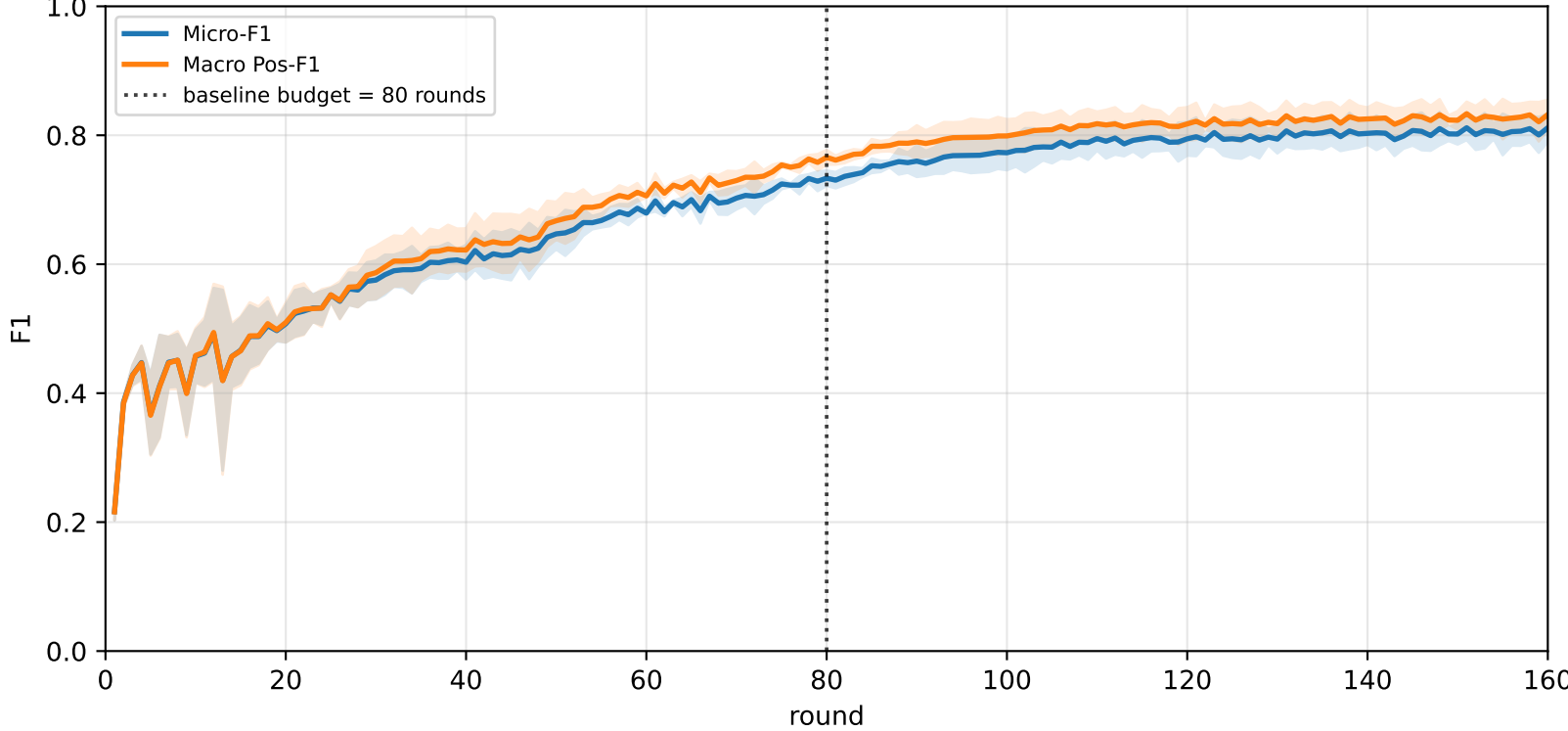
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



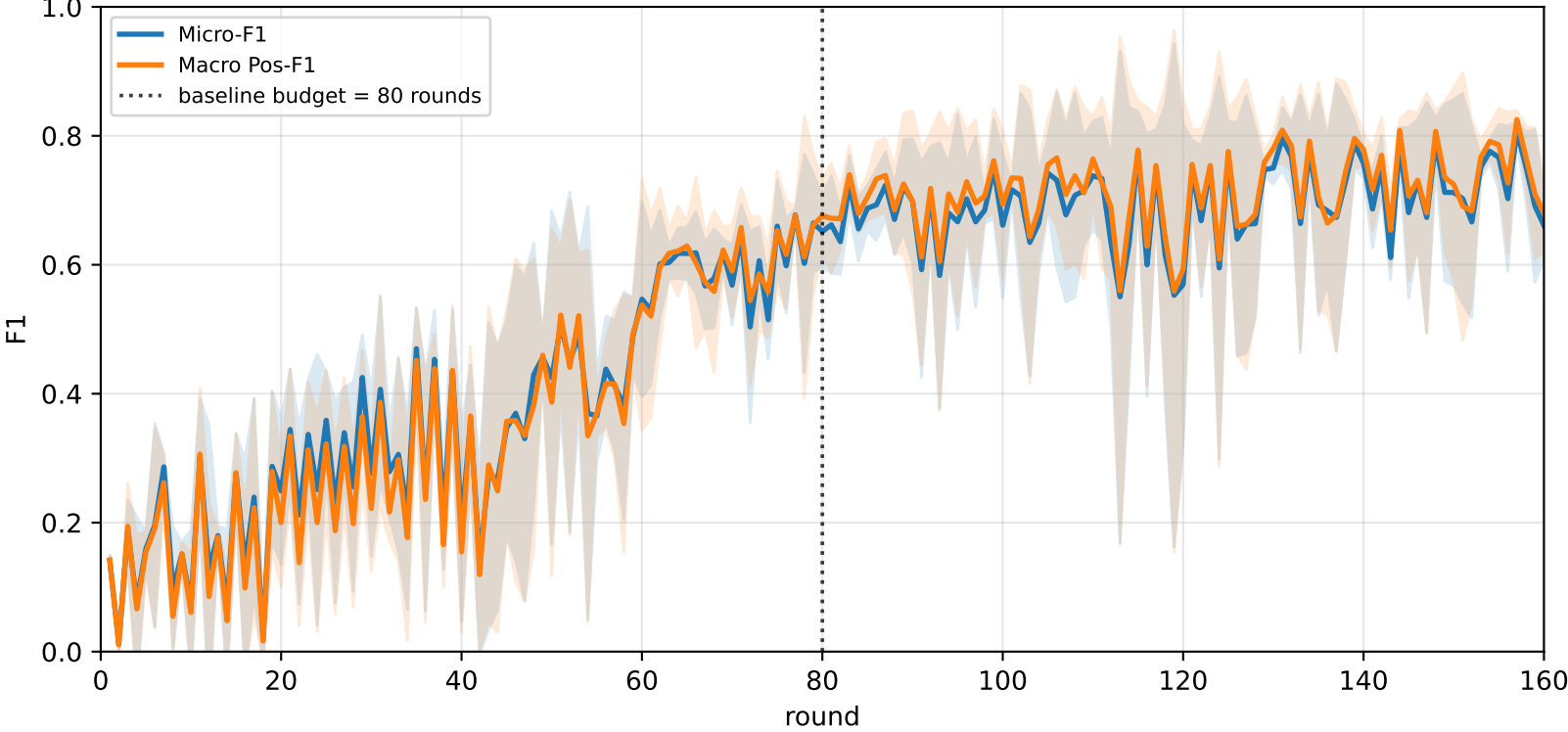
Pooled local centralized micro / macro F1



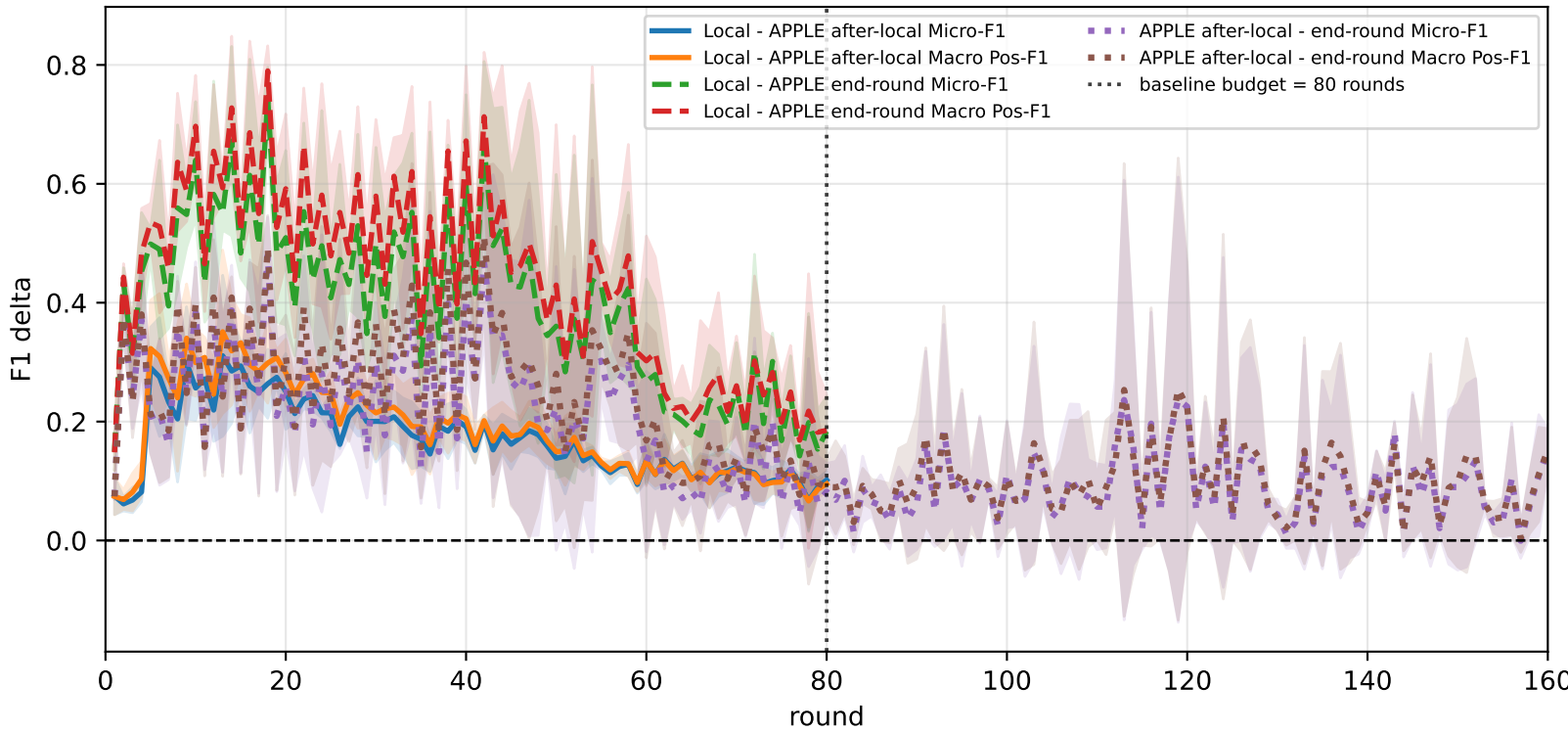
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



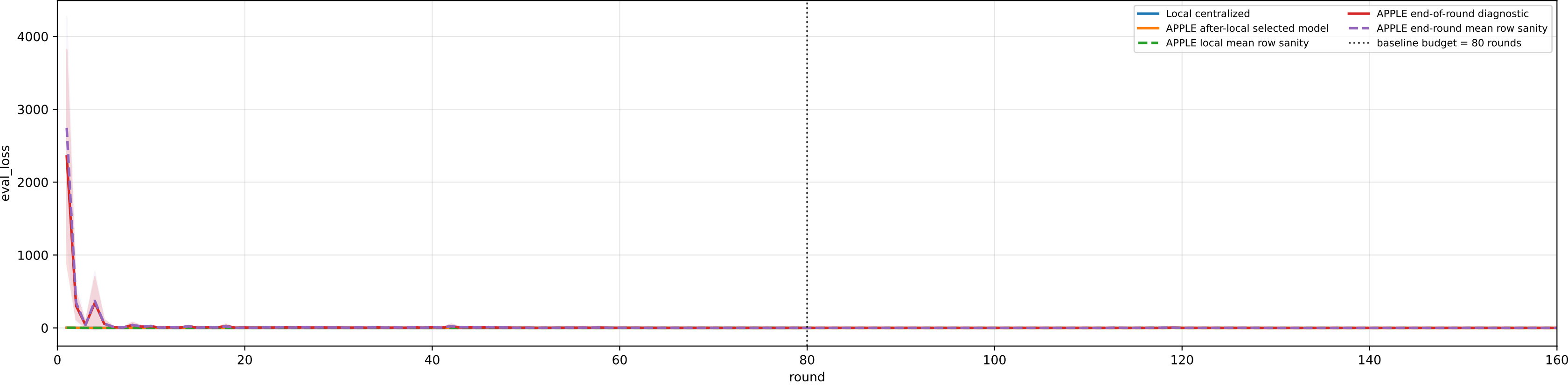
Controlled task-specialization level: Low | APPLE diagnostics | family=specialized_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

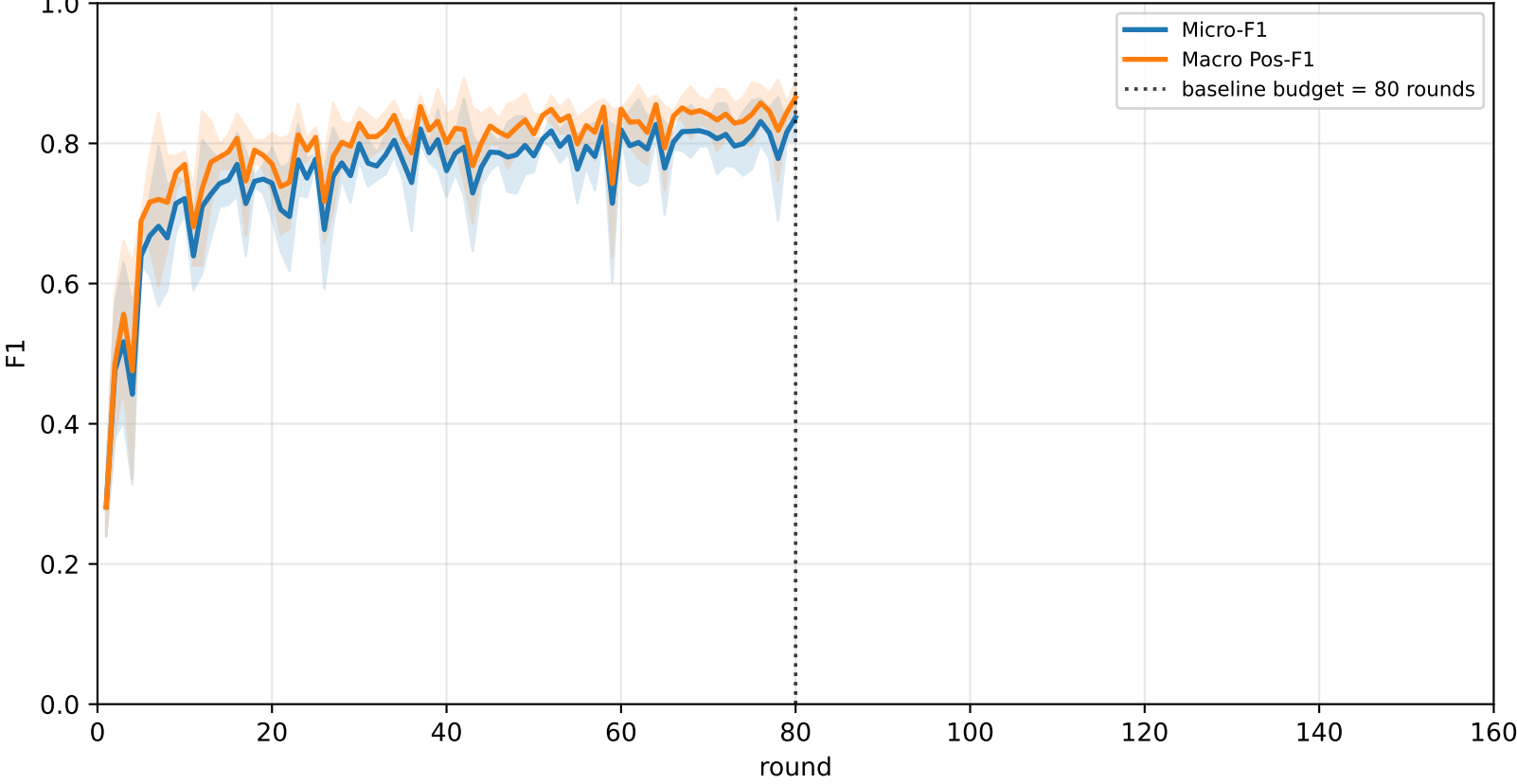
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.5% ± 1.5%	84.0% ± 1.5%	96.6% ± 1.3%	93.0% ± 1.5%	85.5% ± 2.0%	78.8% ± 2.1%	66.4% ± 2.0%
APPLE	80.8% ± 1.7%	82.6% ± 1.7%	95.5% ± 2.1%	90.1% ± 2.9%	81.0% ± 1.8%	75.1% ± 1.0%	71.3% ± 1.8%

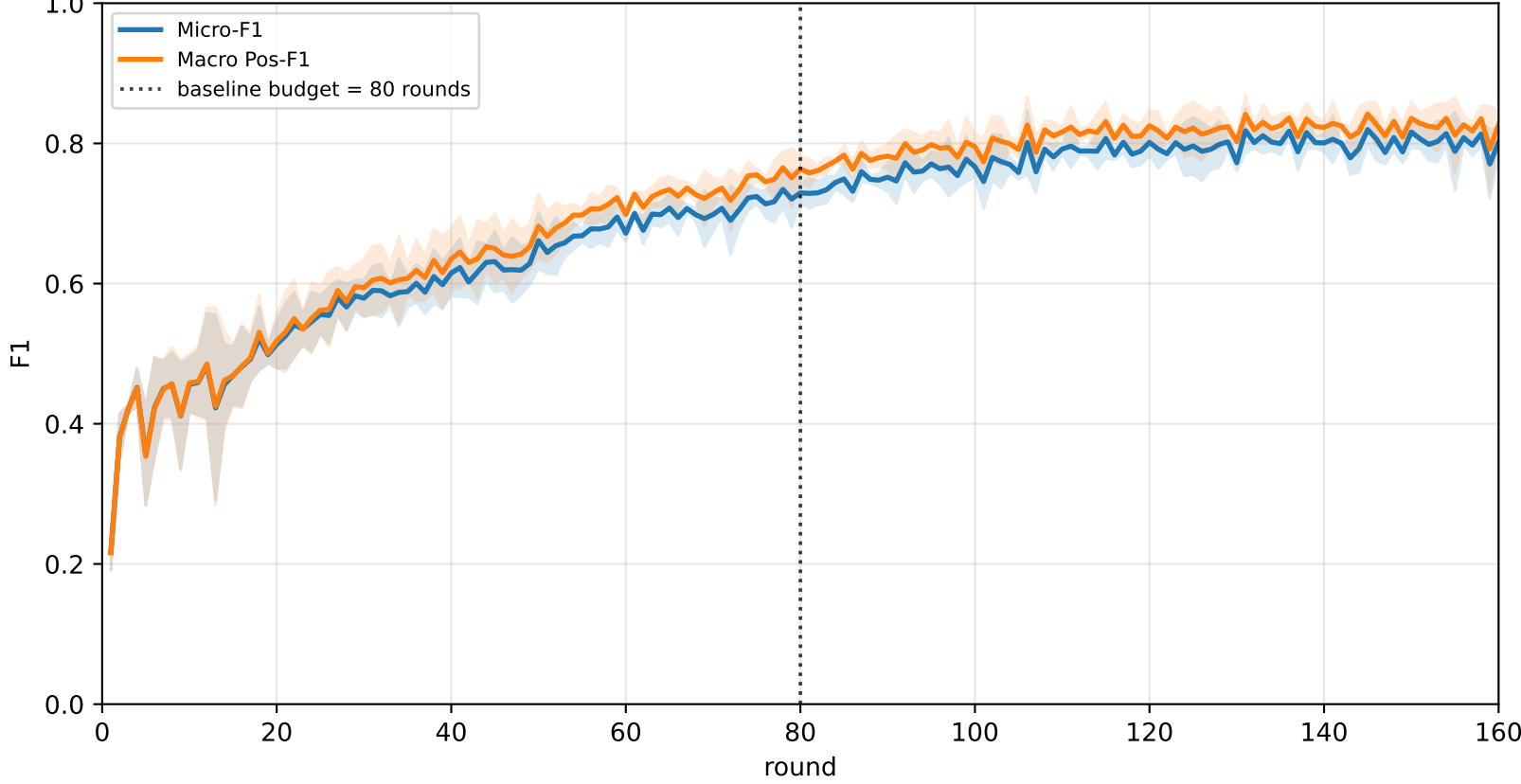
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



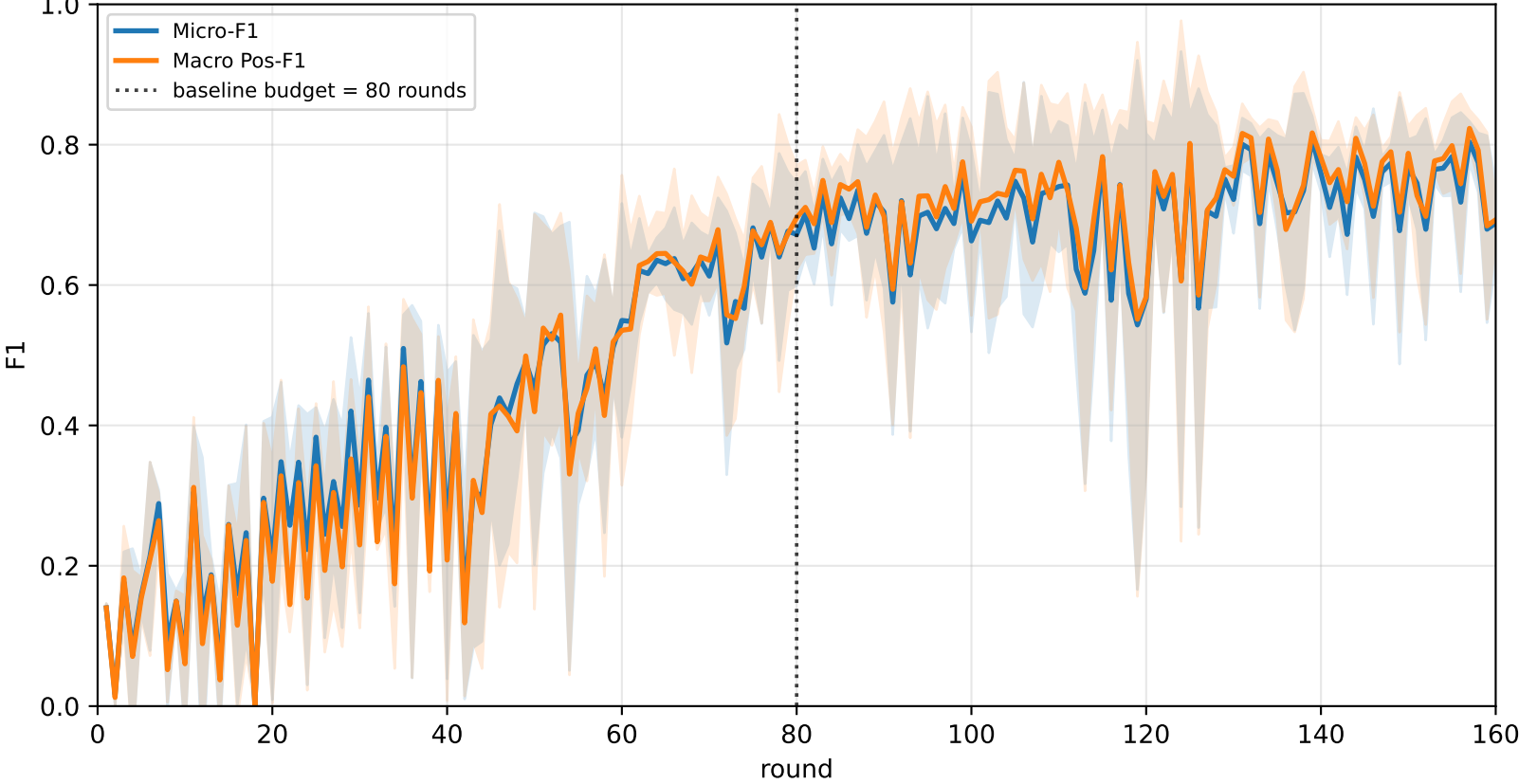
Pooled local centralized micro / macro F1



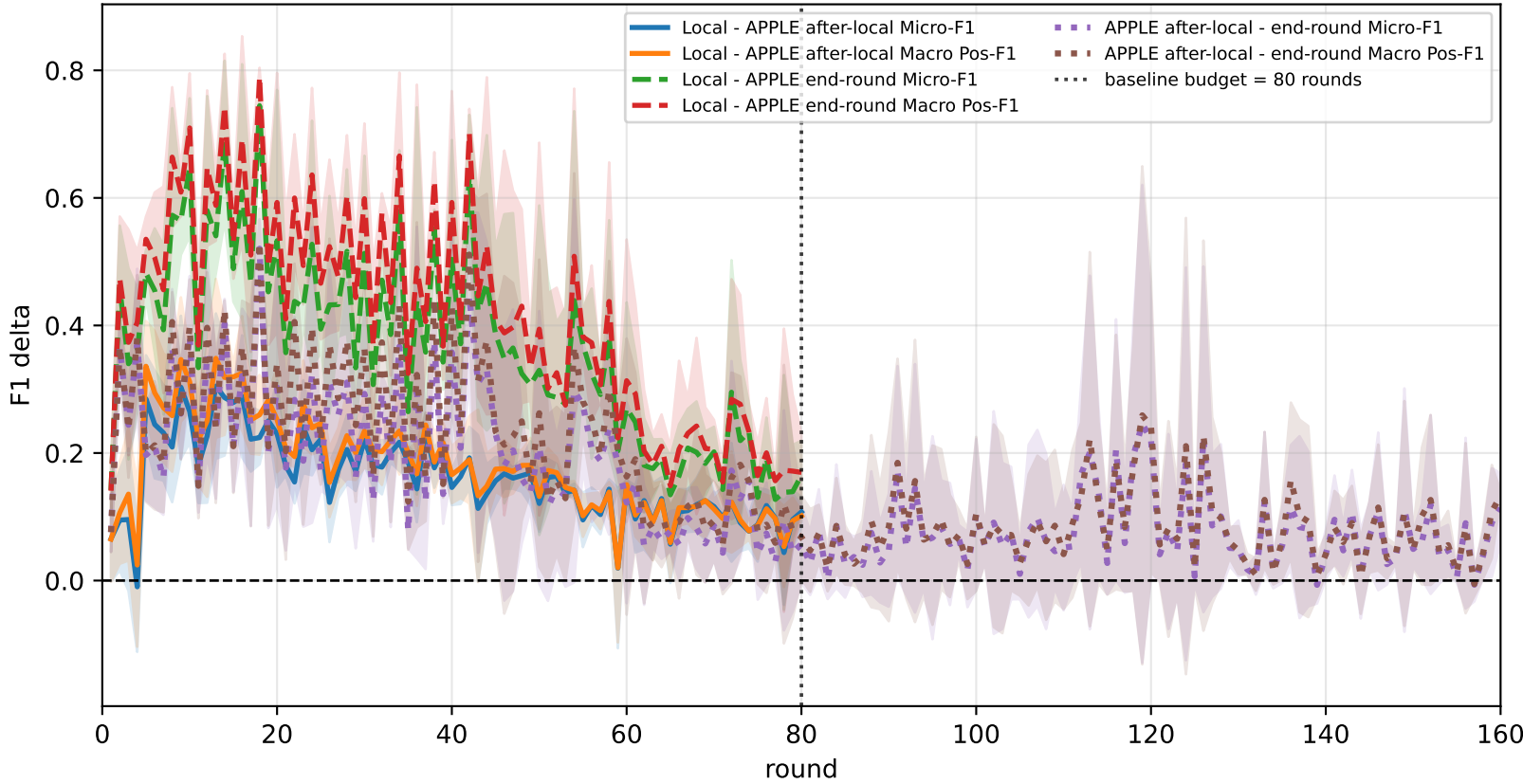
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



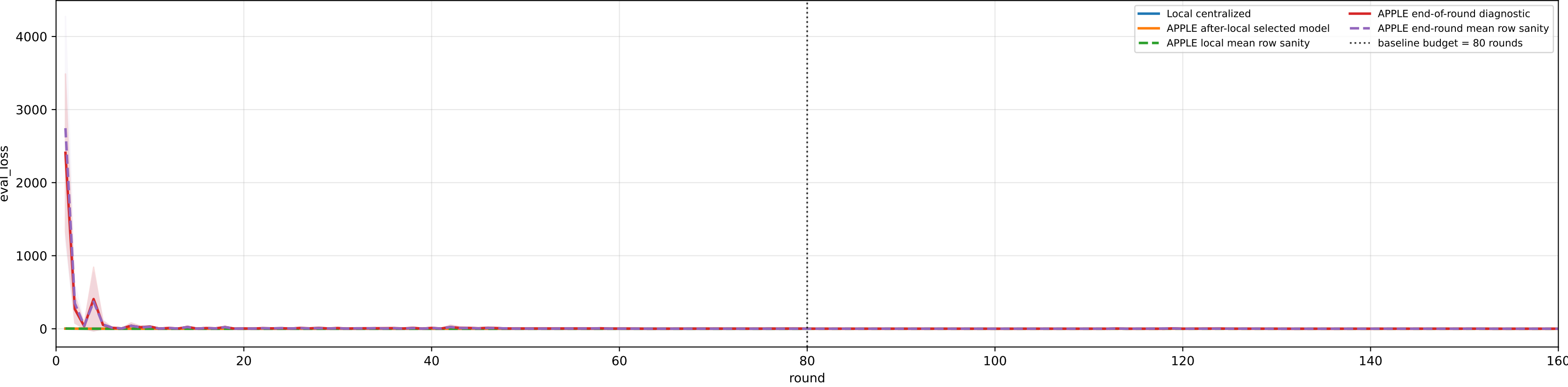
Controlled task-specialization level: Low | APPLE diagnostics | family=specialized_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

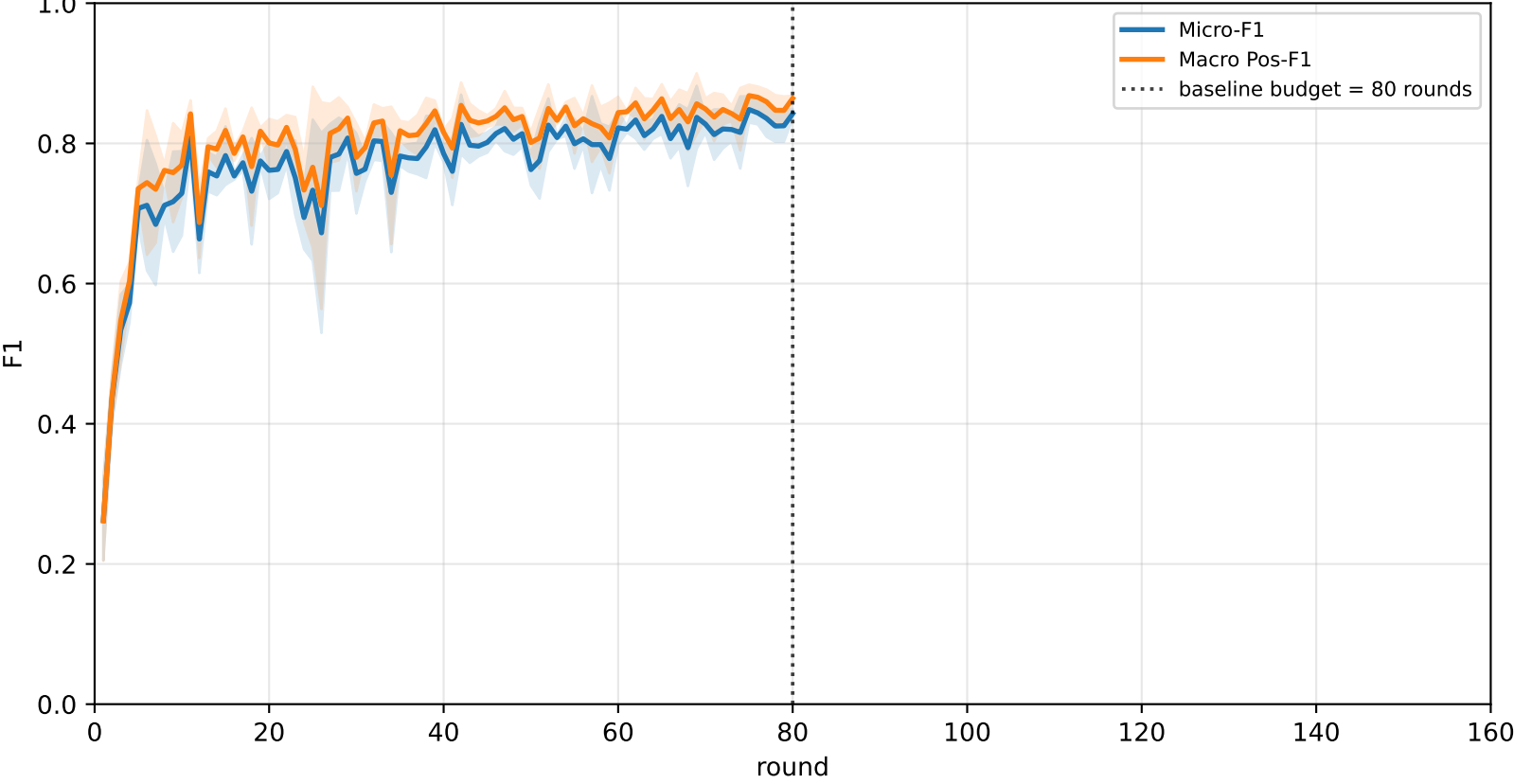
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	84.3% ± 1.4%	86.1% ± 1.1%	96.7% ± 0.5%	95.8% ± 1.1%	86.6% ± 2.6%	78.0% ± 2.0%	73.7% ± 3.9%
APPLE	80.9% ± 2.4%	82.8% ± 2.1%	94.5% ± 0.7%	92.3% ± 2.8%	80.2% ± 2.8%	76.9% ± 1.0%	70.4% ± 3.8%

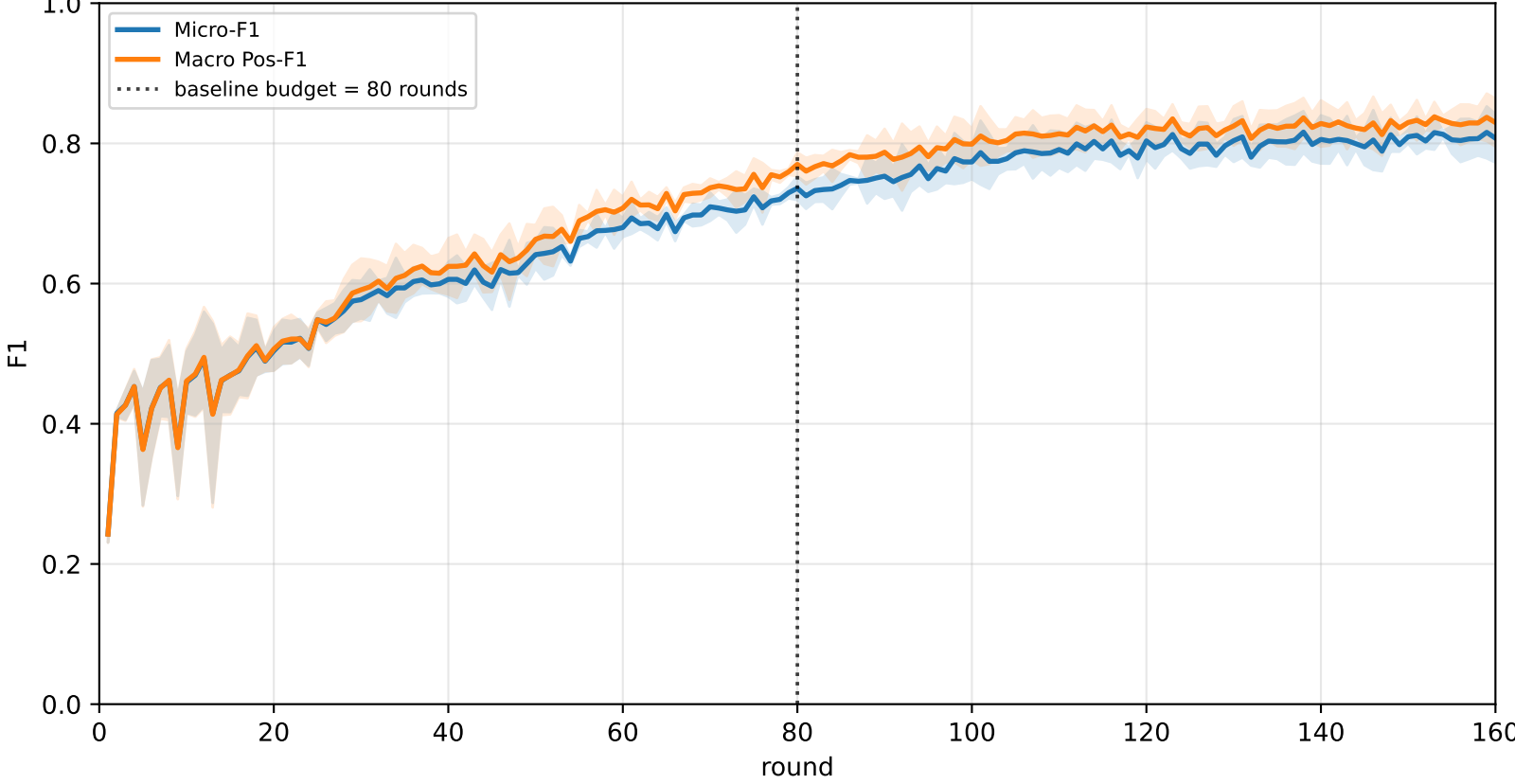
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



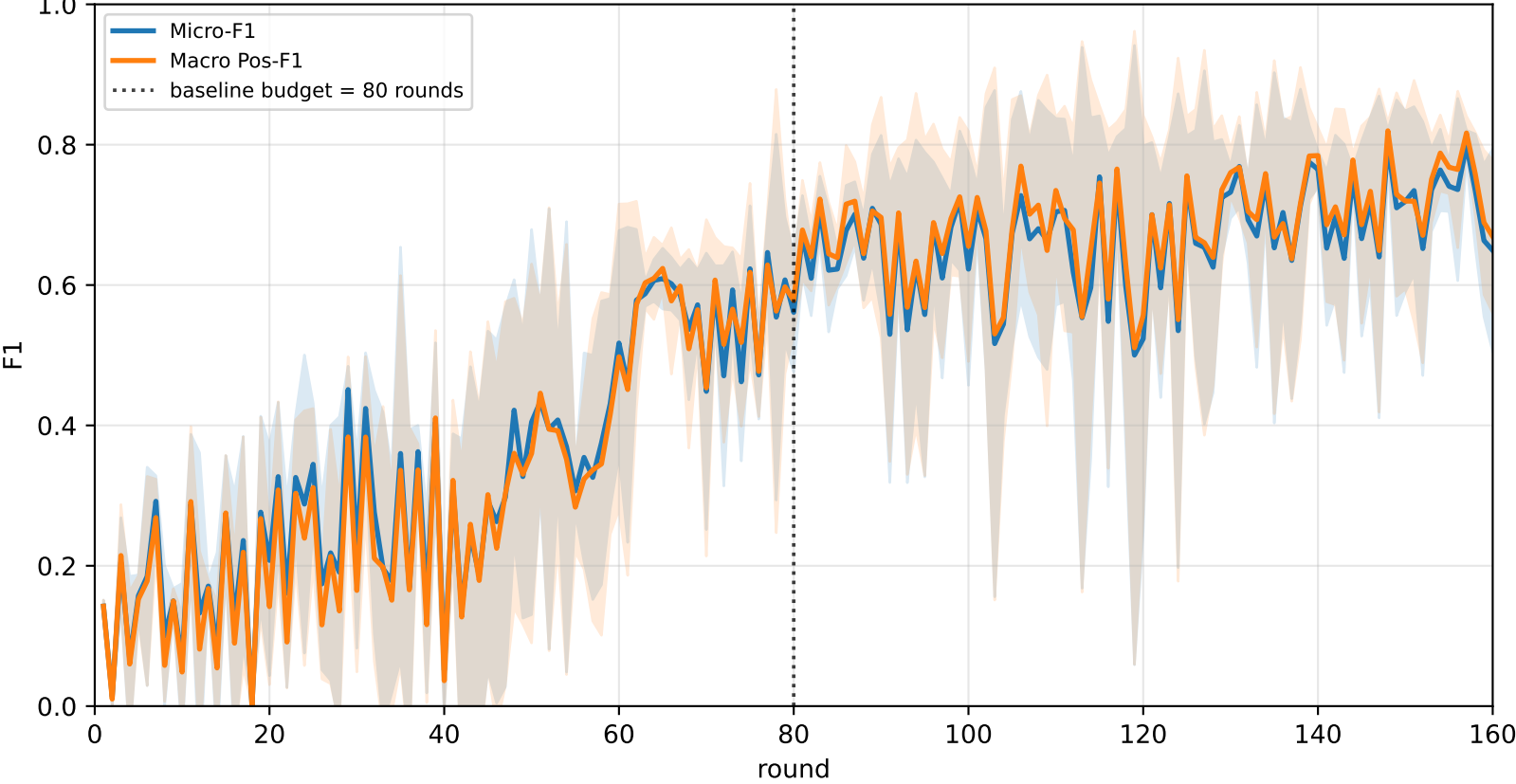
Pooled local centralized micro / macro F1



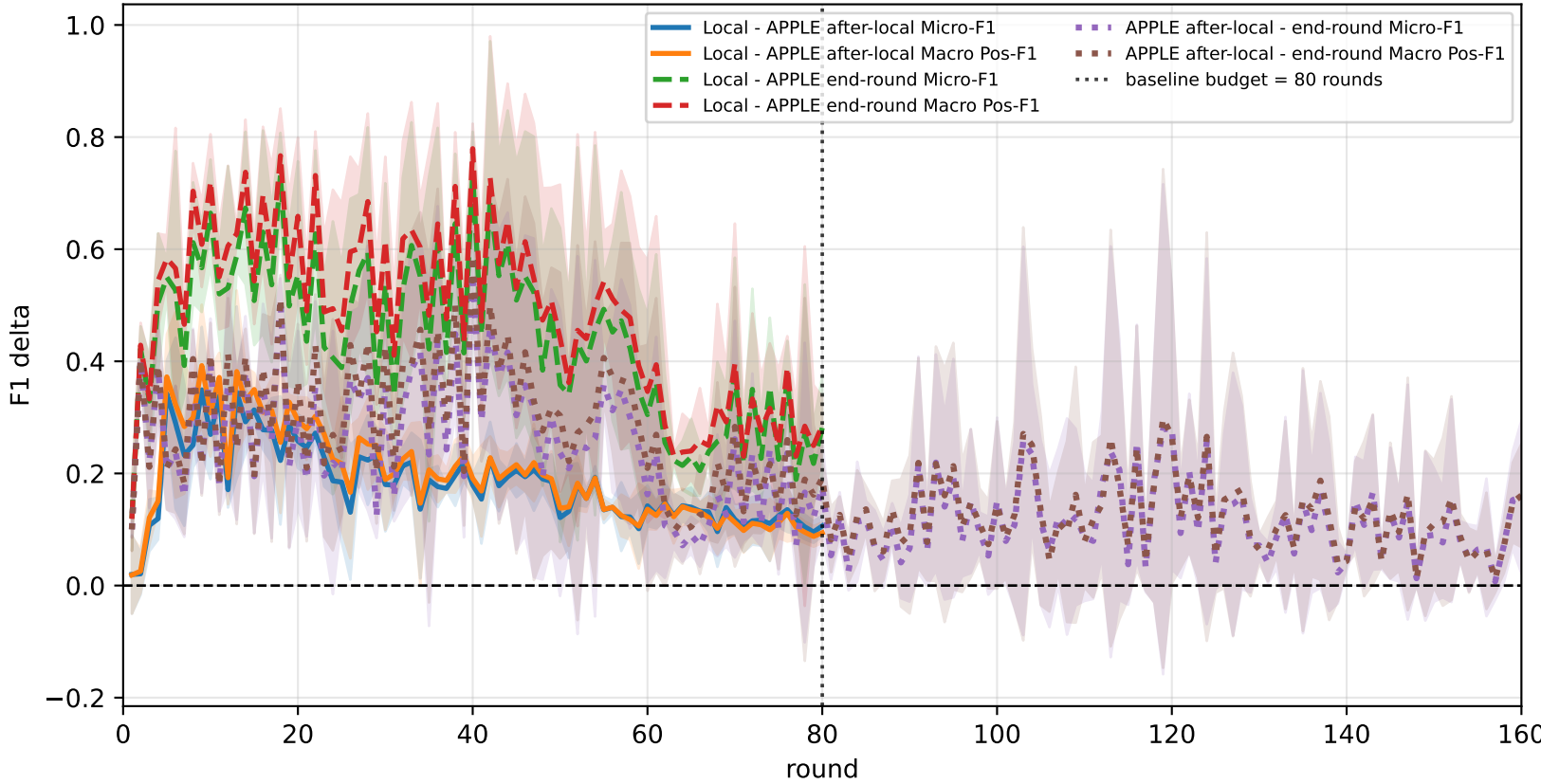
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



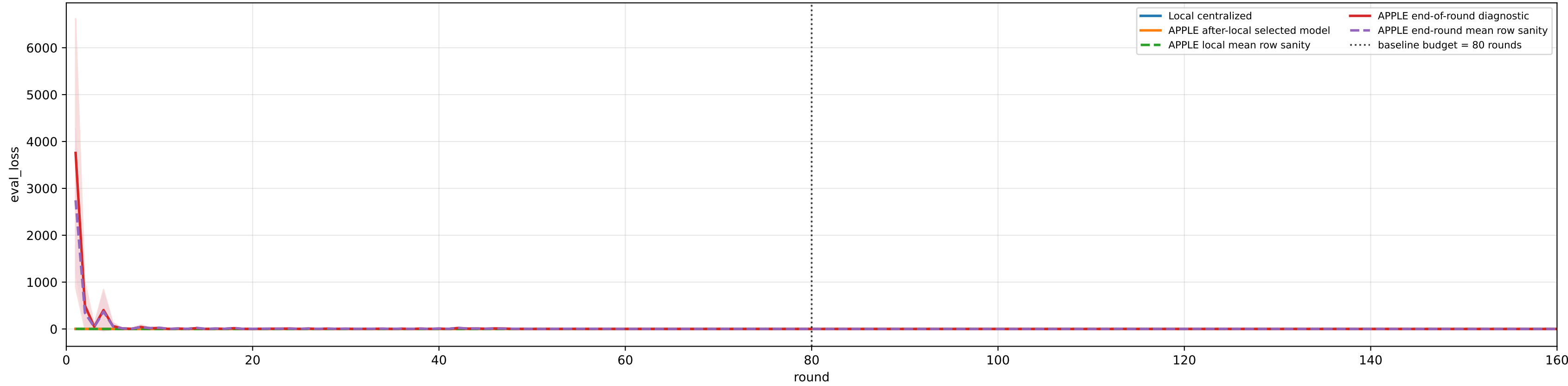
Controlled task-specialization level: Low | APPLE diagnostics | family=specialized_cycle4 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle4 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

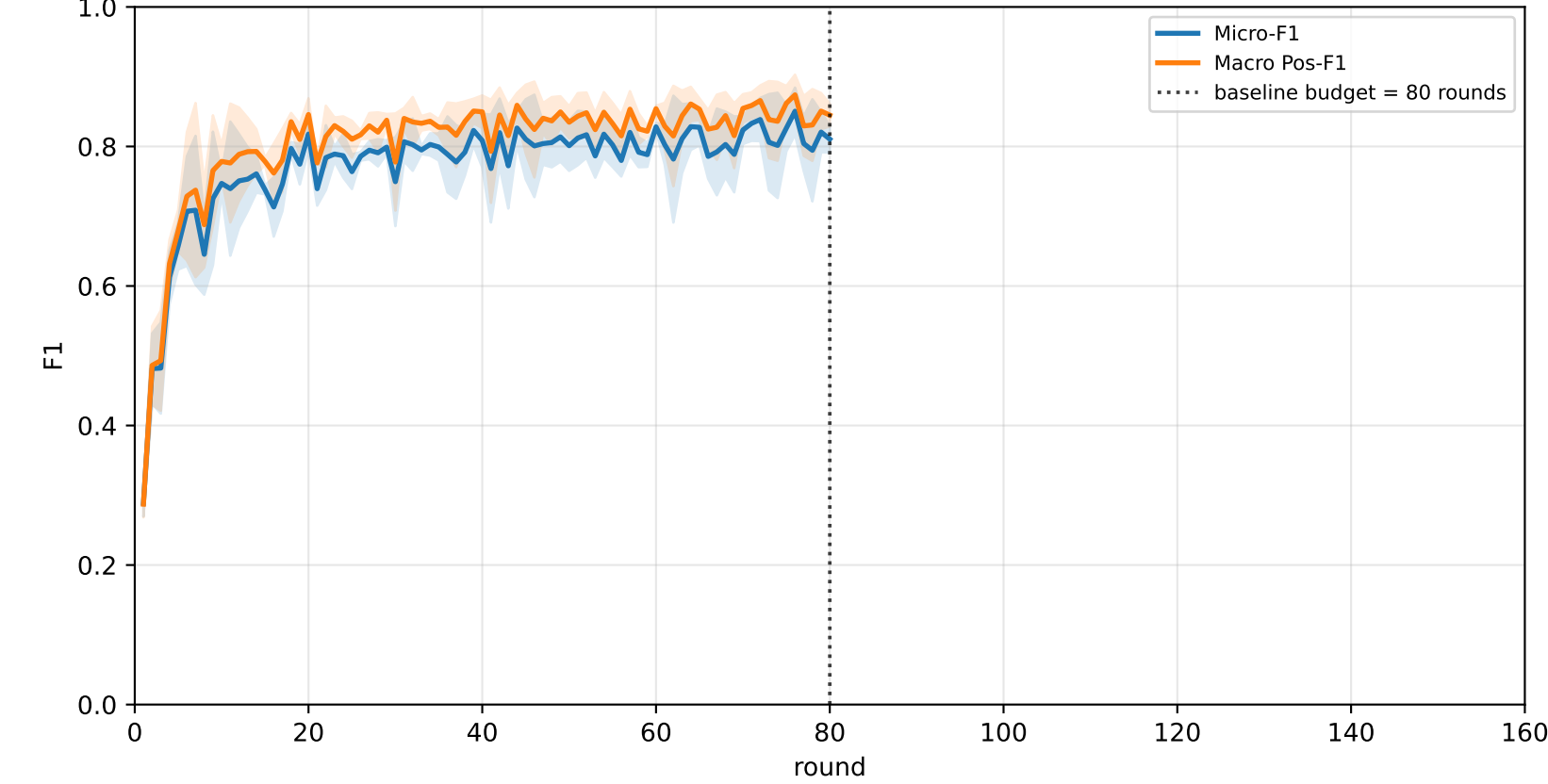
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.9% ± 2.9%	84.6% ± 2.1%	96.4% ± 1.2%	96.1% ± 0.4%	85.9% ± 3.7%	76.4% ± 0.2%	68.1% ± 7.4%
APPLE	80.4% ± 4.0%	82.5% ± 3.6%	96.0% ± 1.4%	90.4% ± 3.6%	81.5% ± 5.9%	74.6% ± 2.9%	70.0% ± 5.6%

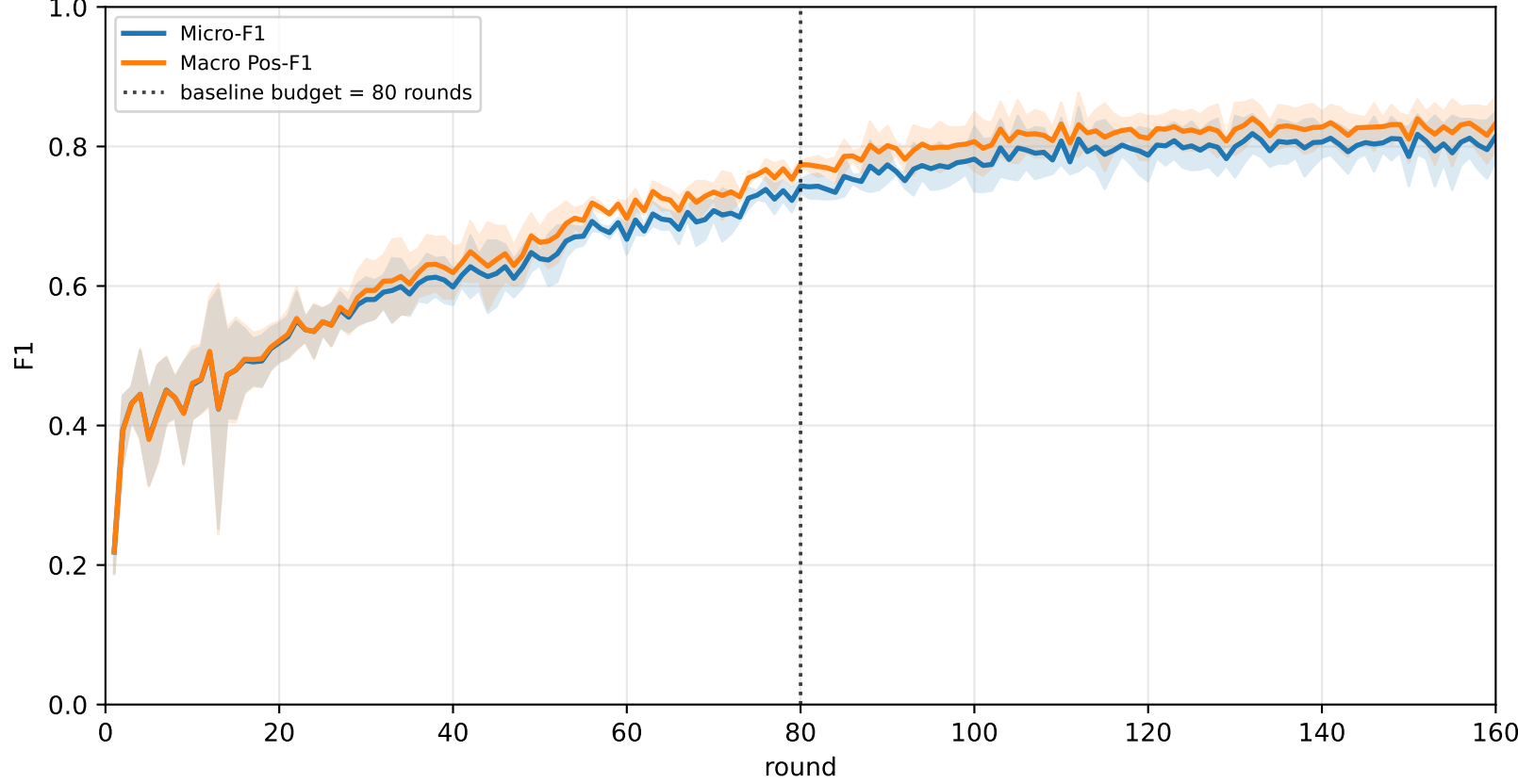
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



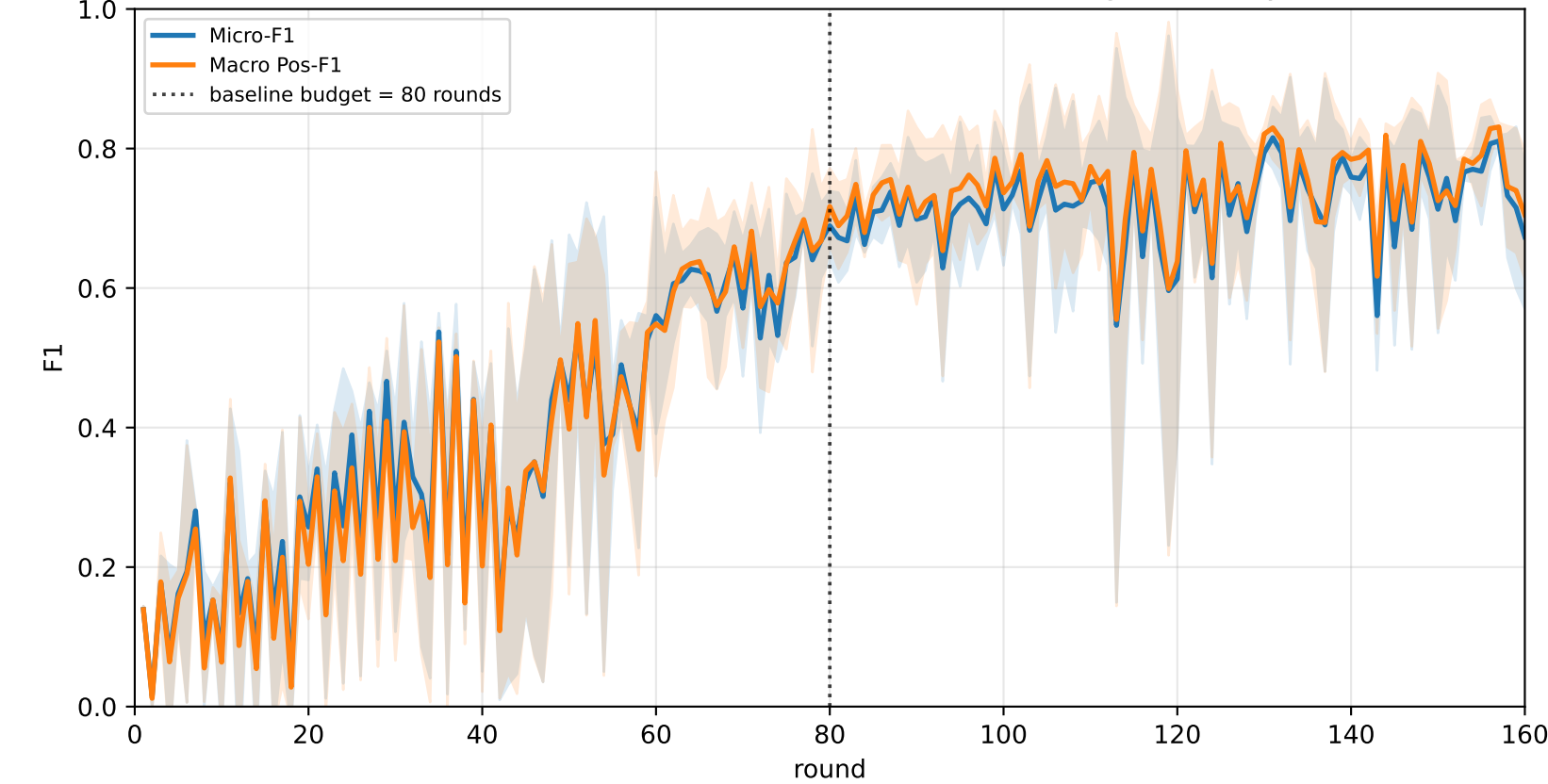
Pooled local centralized micro / macro F1



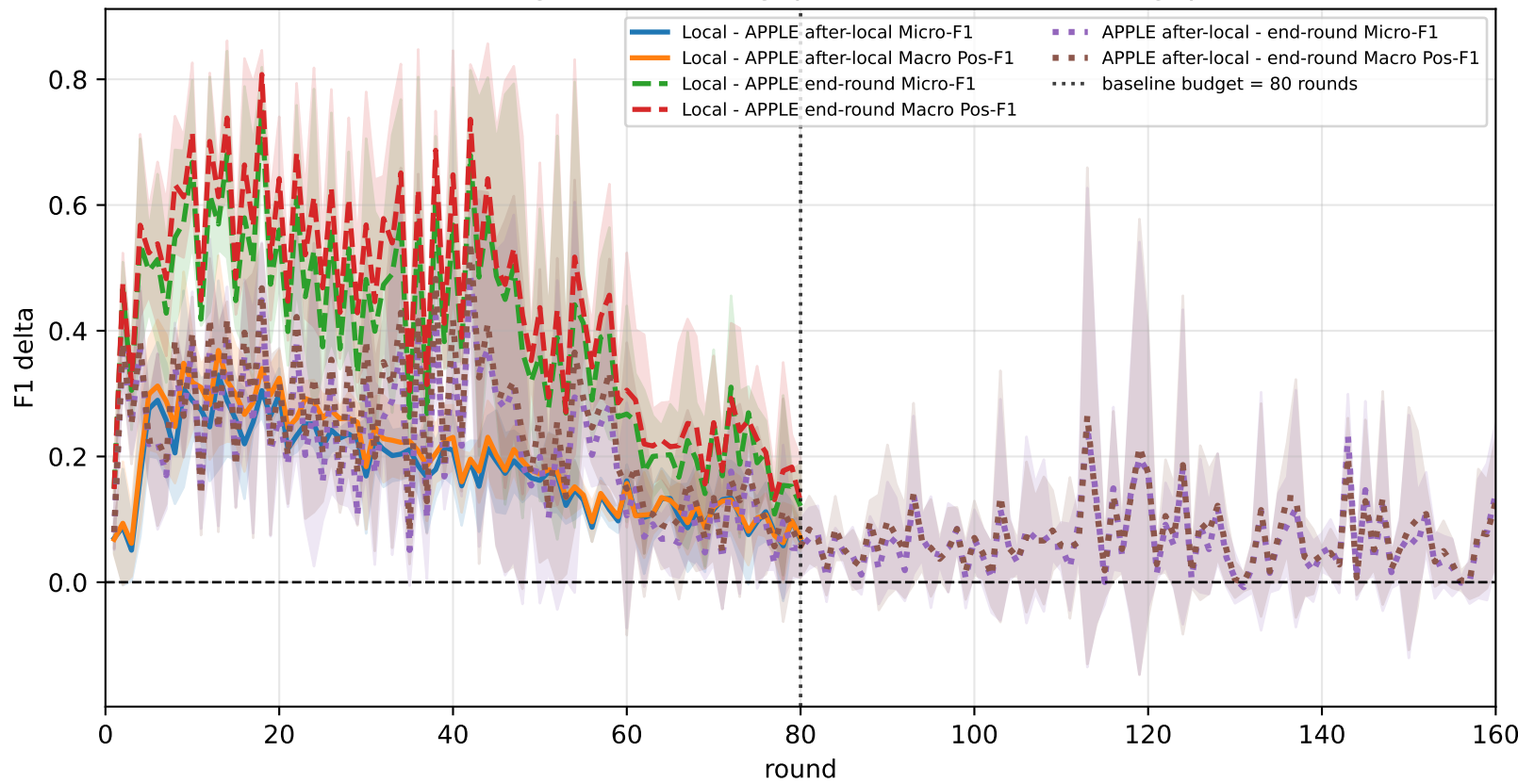
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



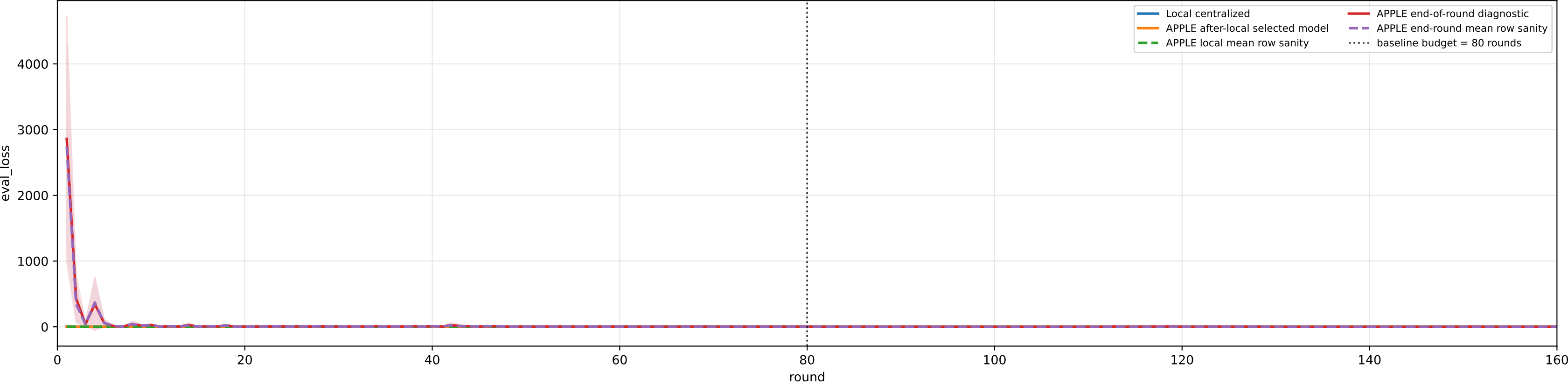
Controlled task-specialization level: Low | APPLE diagnostics | family=specialized_cycle5 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

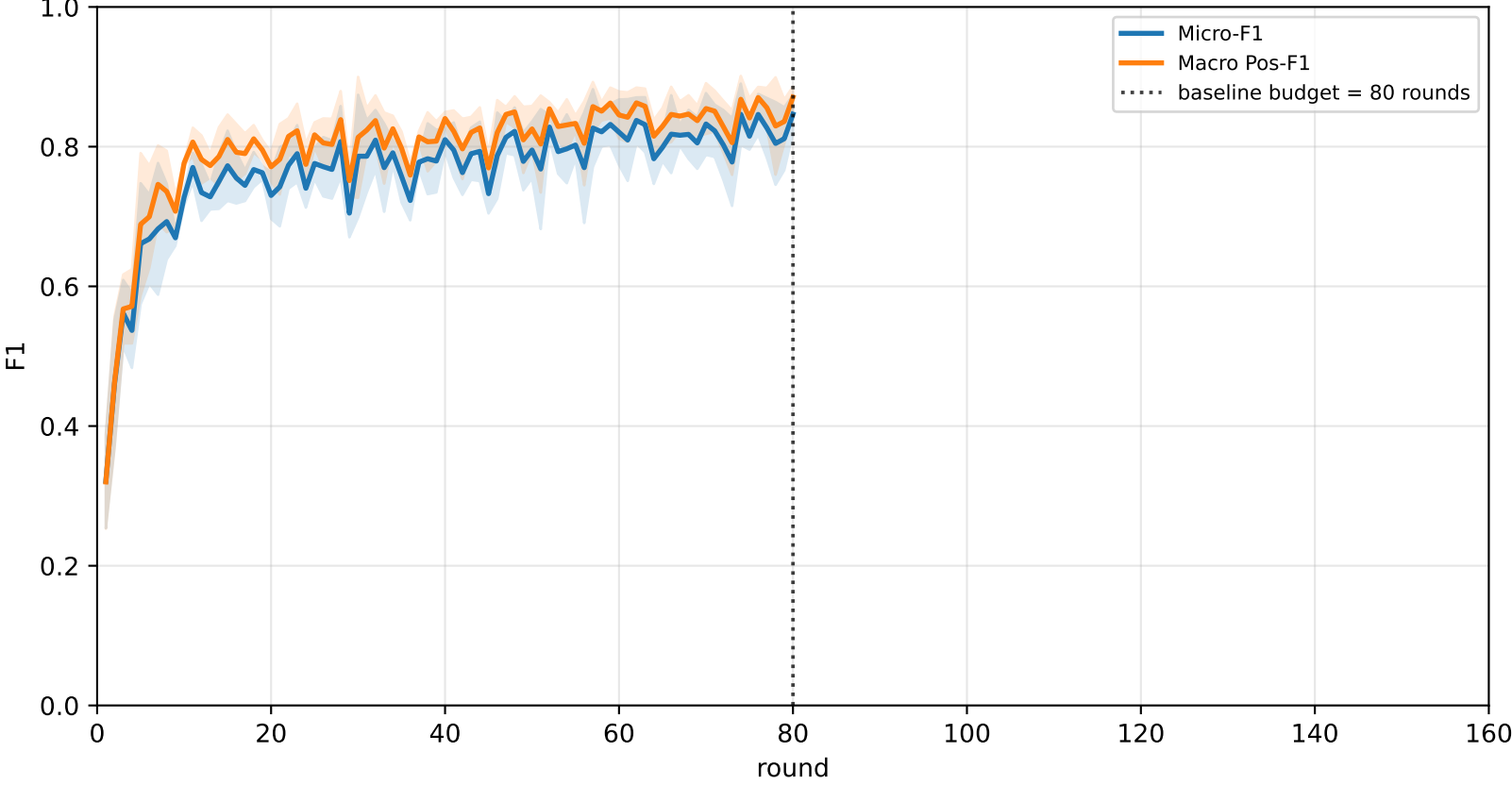
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	85.4% ± 1.5%	87.2% ± 0.5%	96.5% ± 1.7%	95.7% ± 0.7%	88.3% ± 2.6%	81.1% ± 0.7%	74.6% ± 7.4%
APPLE	80.2% ± 1.3%	82.1% ± 1.1%	95.2% ± 1.8%	91.0% ± 1.6%	79.2% ± 1.8%	74.9% ± 1.7%	70.4% ± 1.9%

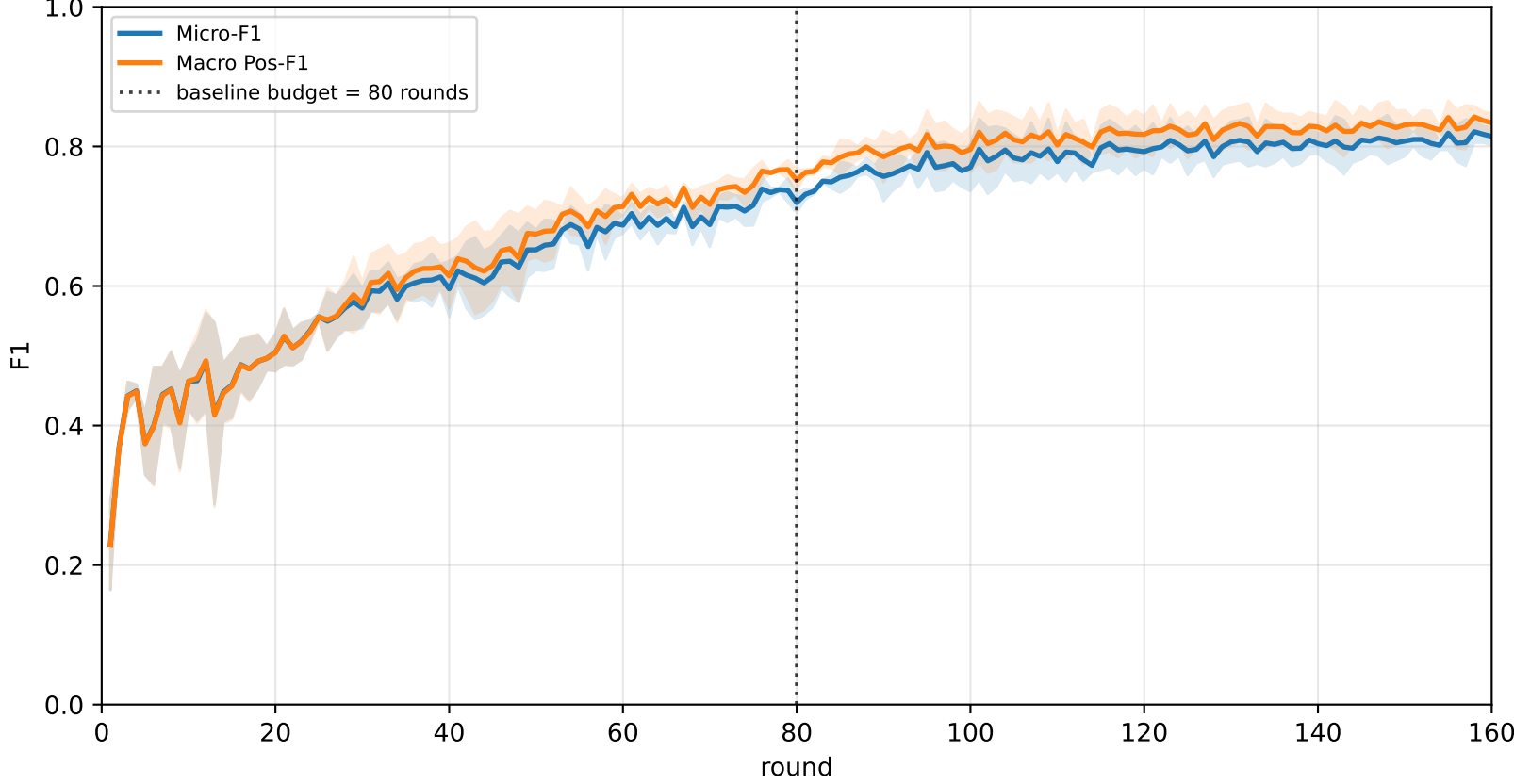
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



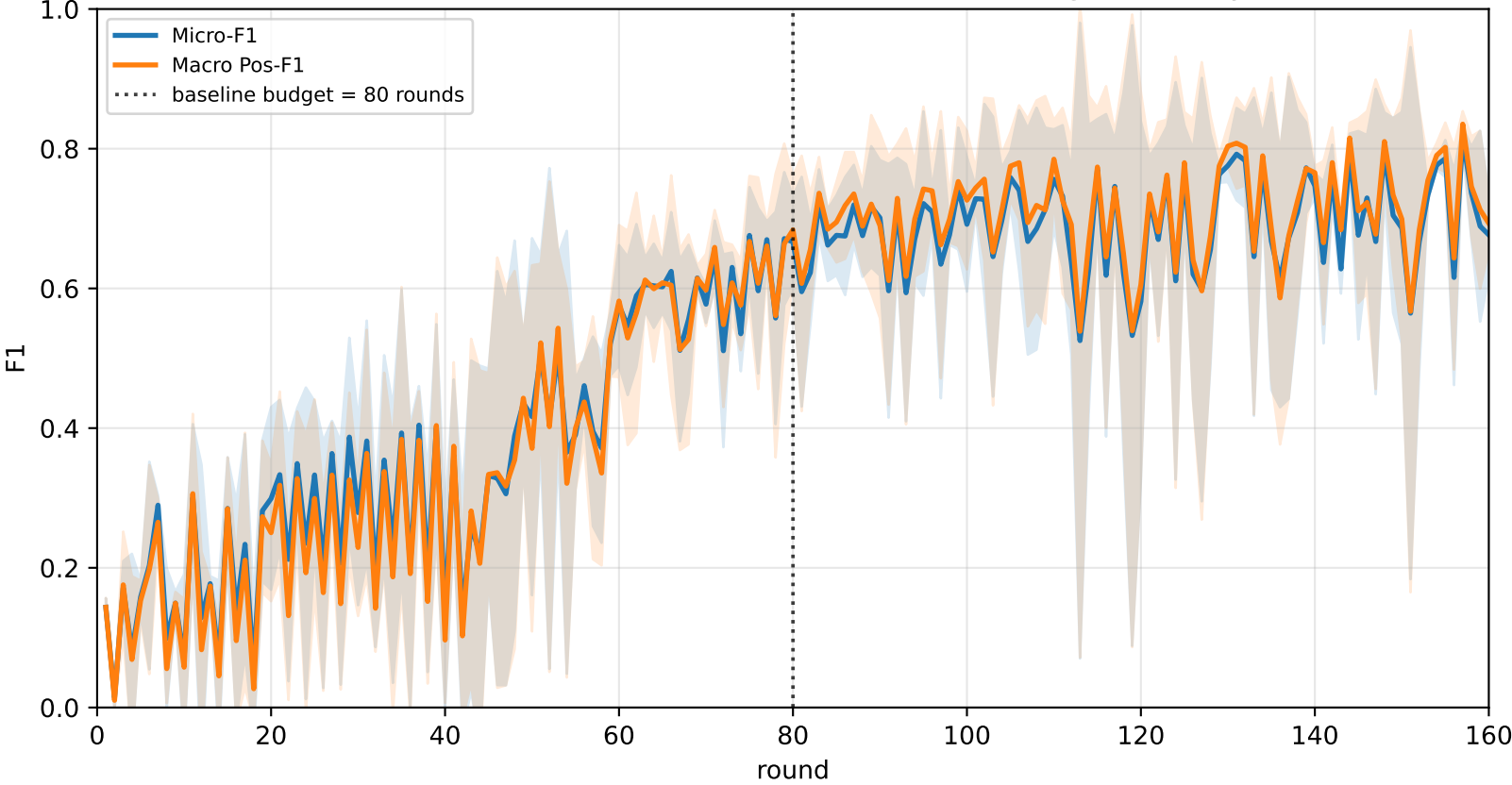
Pooled local centralized micro / macro F1



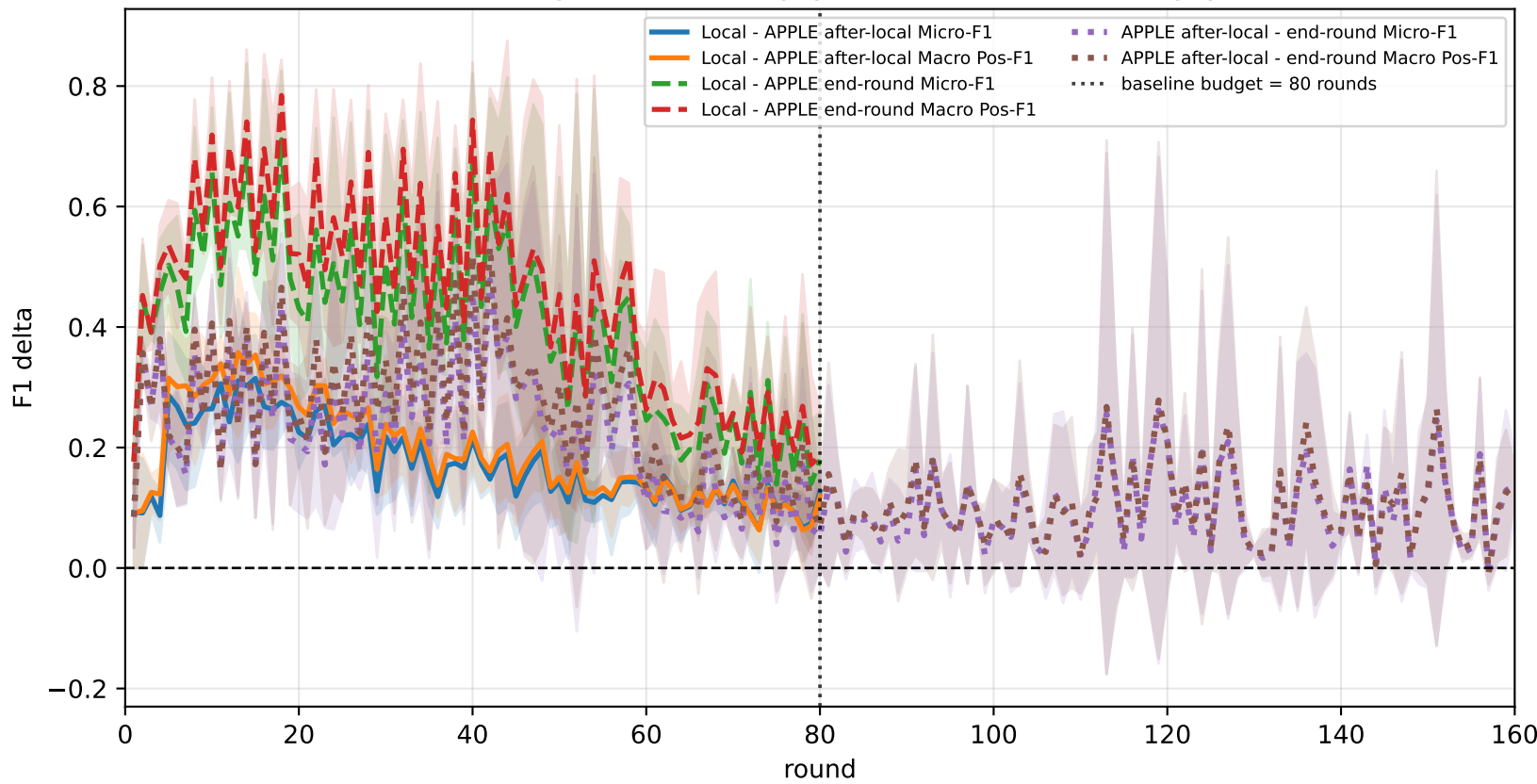
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



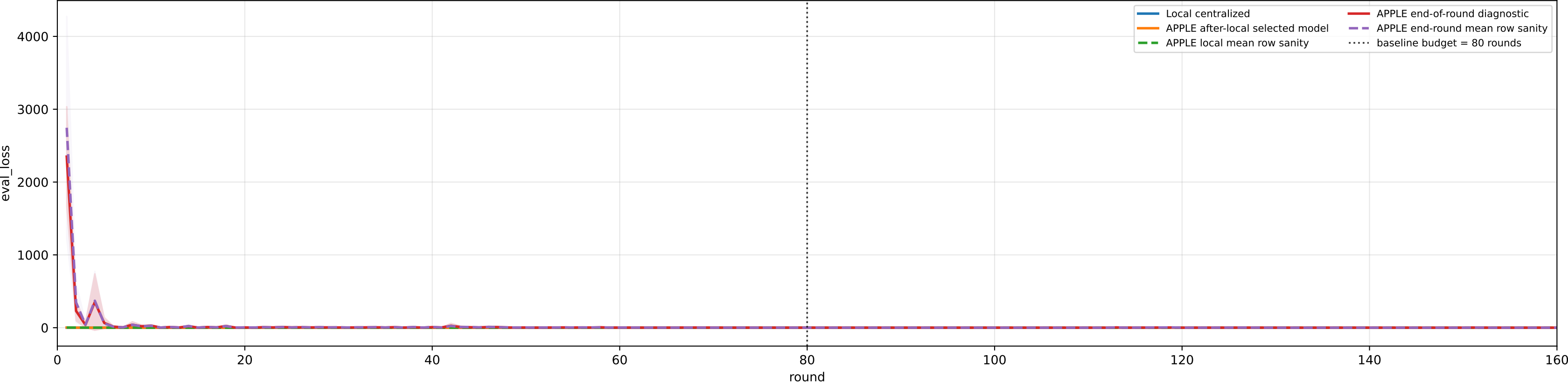
Controlled task-specialization level: Low | APPLE diagnostics | family=specialized_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle6 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3% | specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

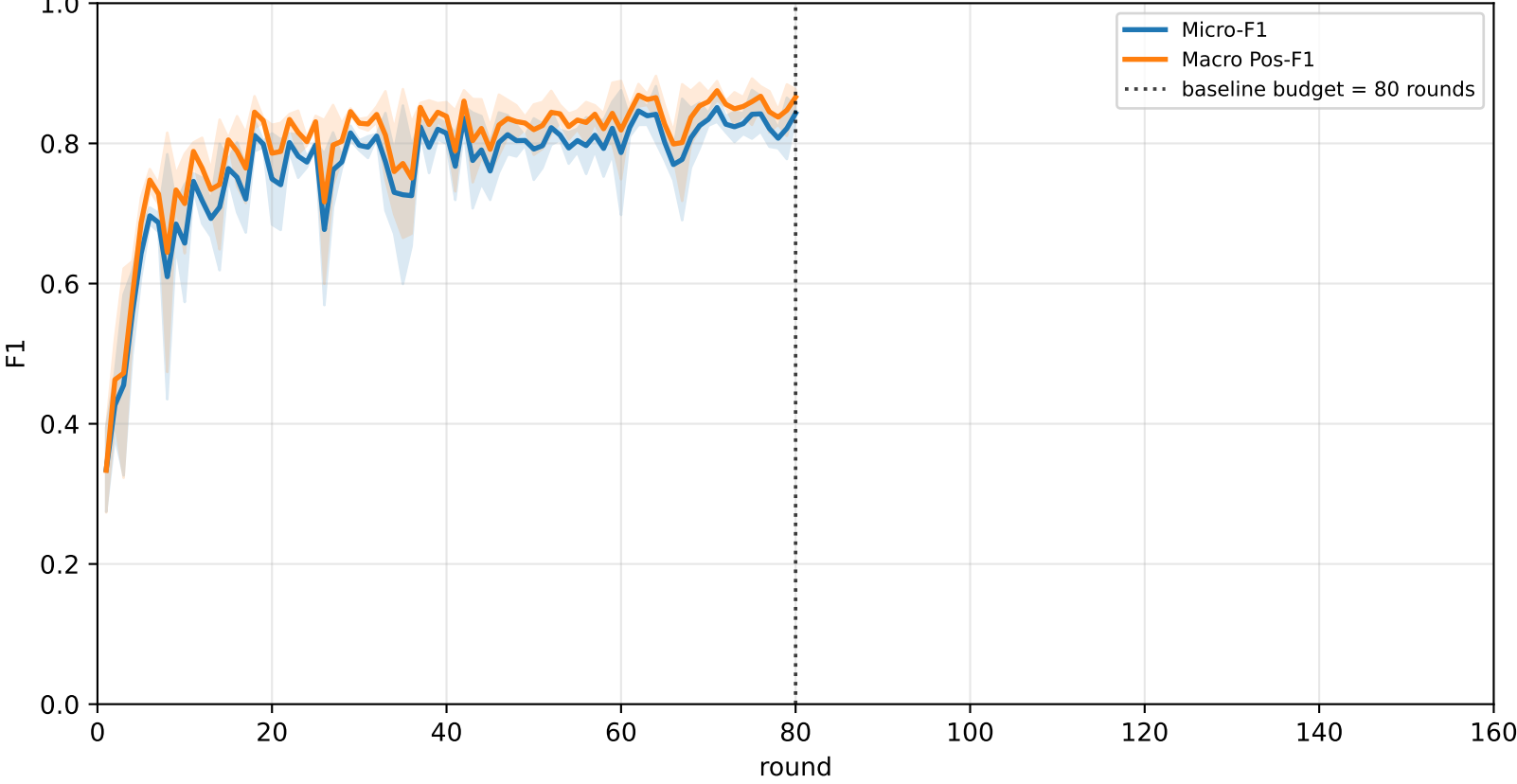
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	84.2% ± 1.5%	86.2% ± 1.0%	97.7% ± 0.7%	93.8% ± 0.8%	88.3% ± 1.2%	79.5% ± 0.8%	71.9% ± 4.5%
APPLE	80.1% ± 1.3%	82.0% ± 1.3%	94.6% ± 1.8%	90.0% ± 1.8%	80.3% ± 2.0%	74.2% ± 0.7%	70.8% ± 1.8%

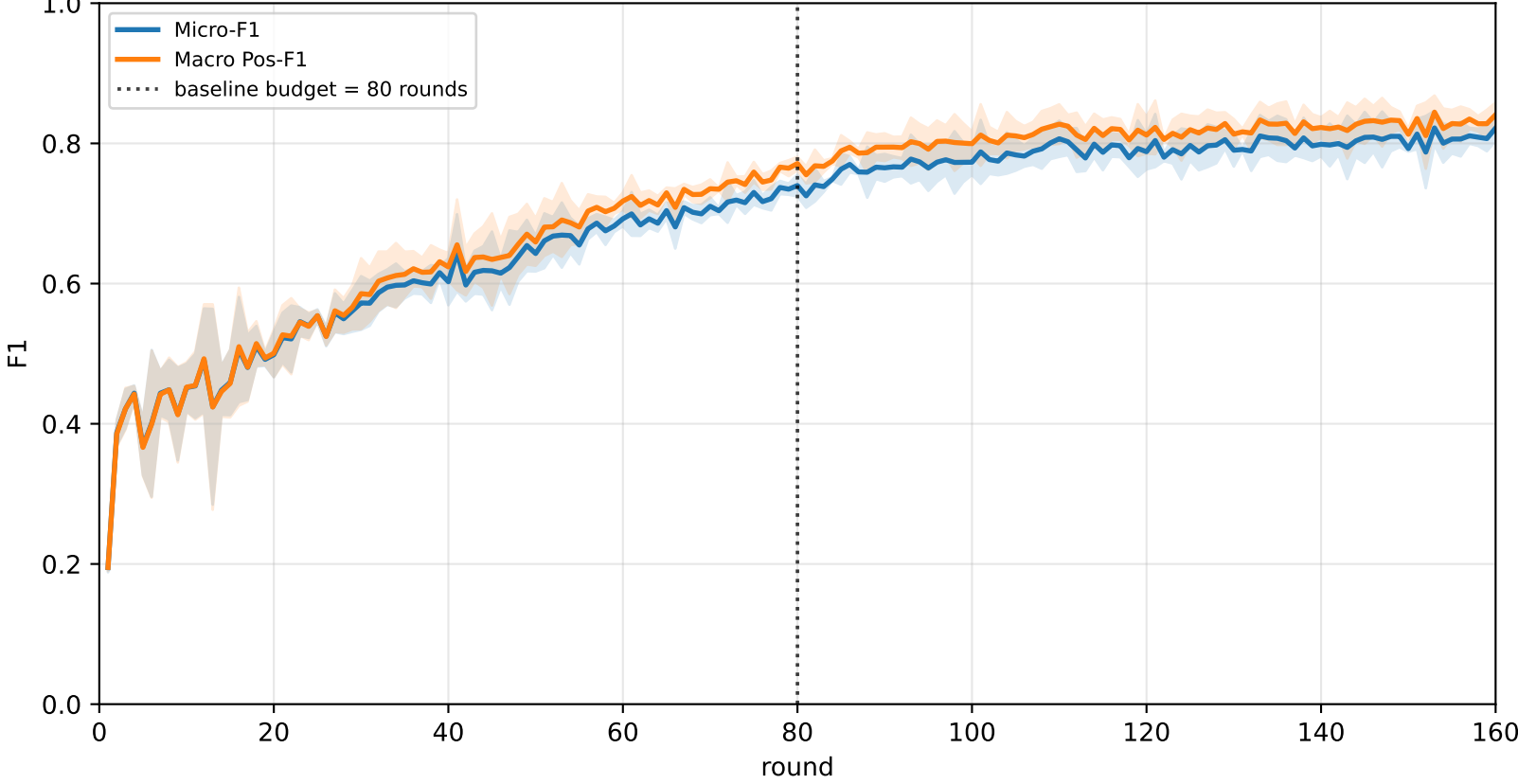
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



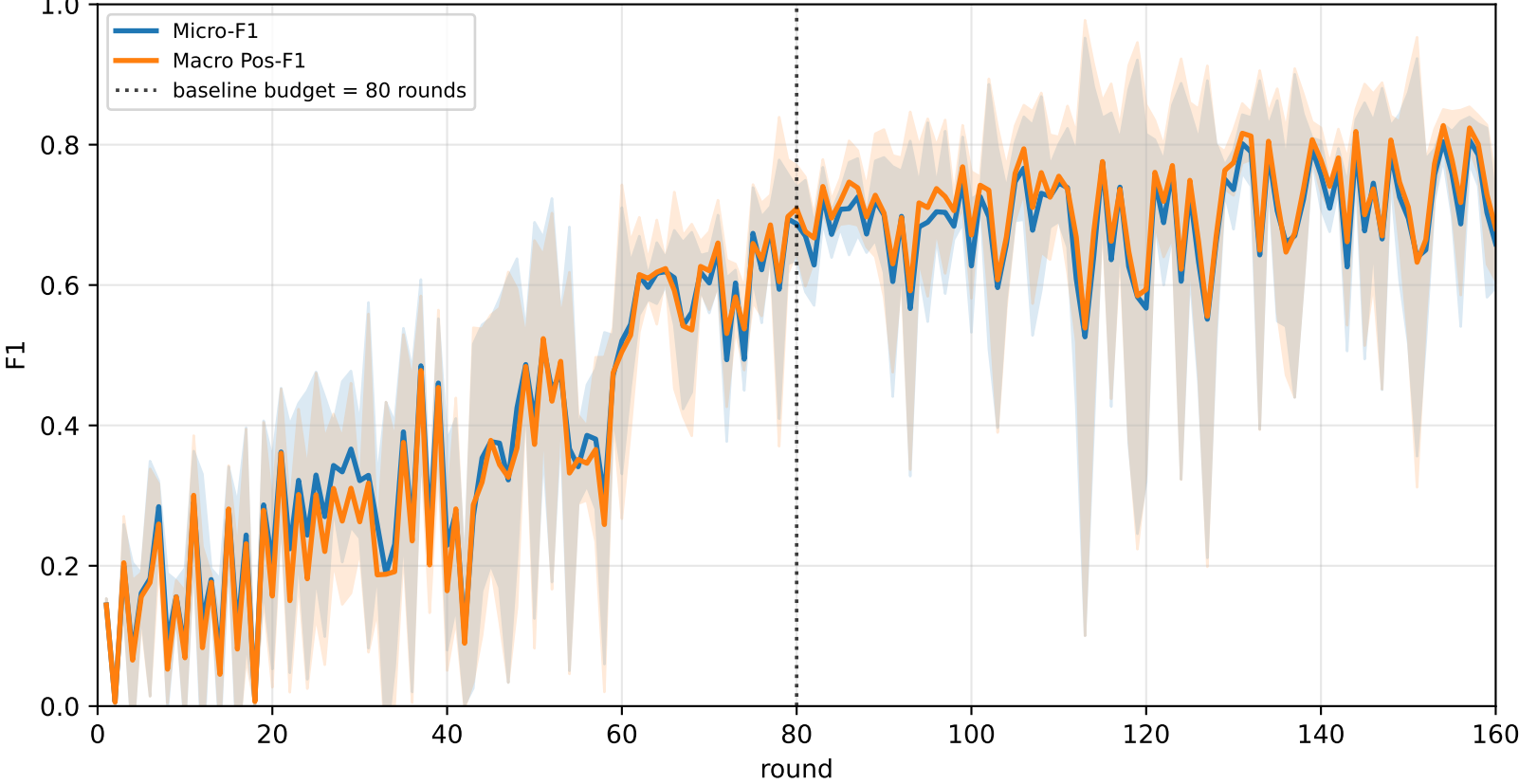
Pooled local centralized micro / macro F1



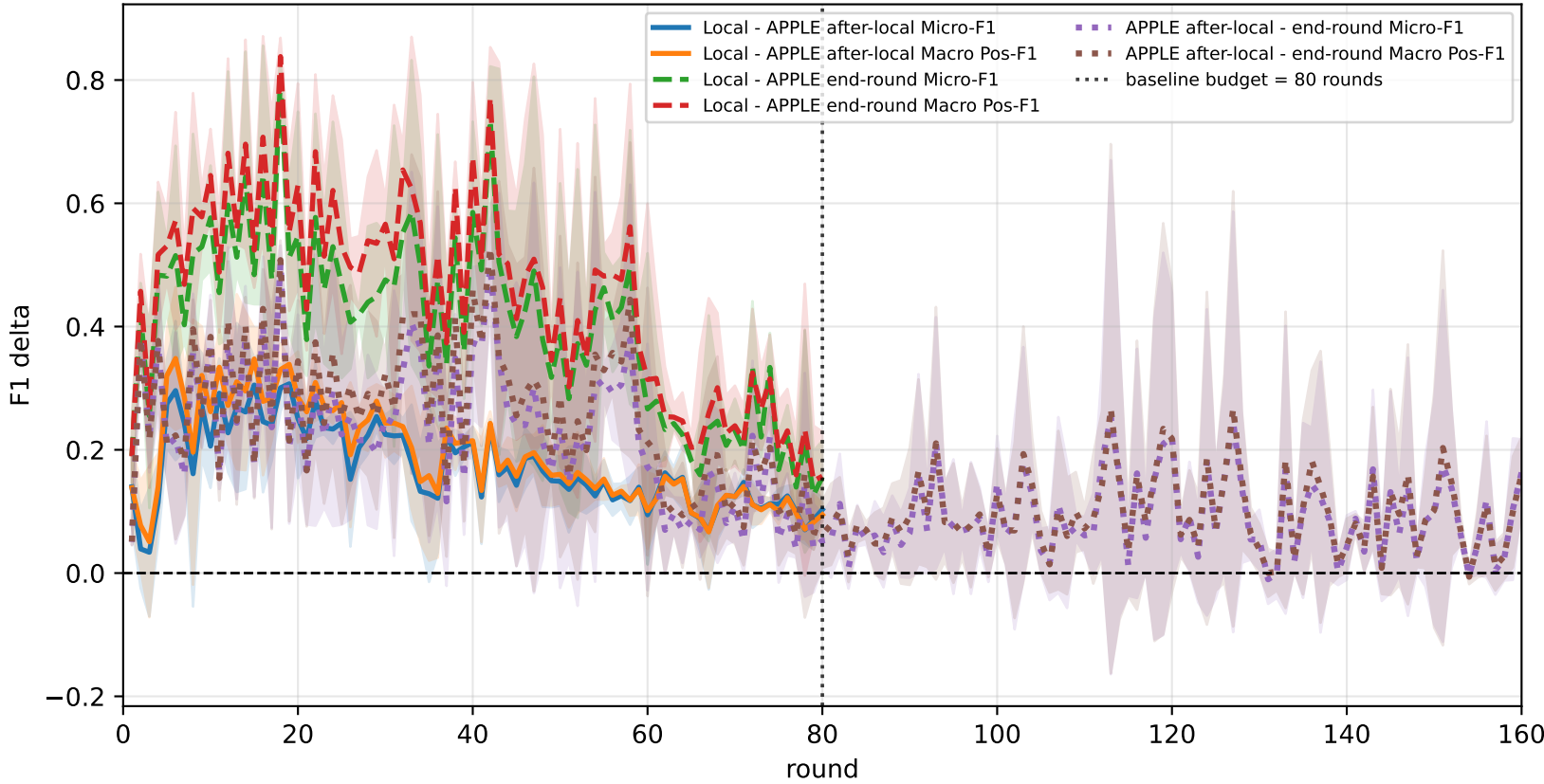
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)

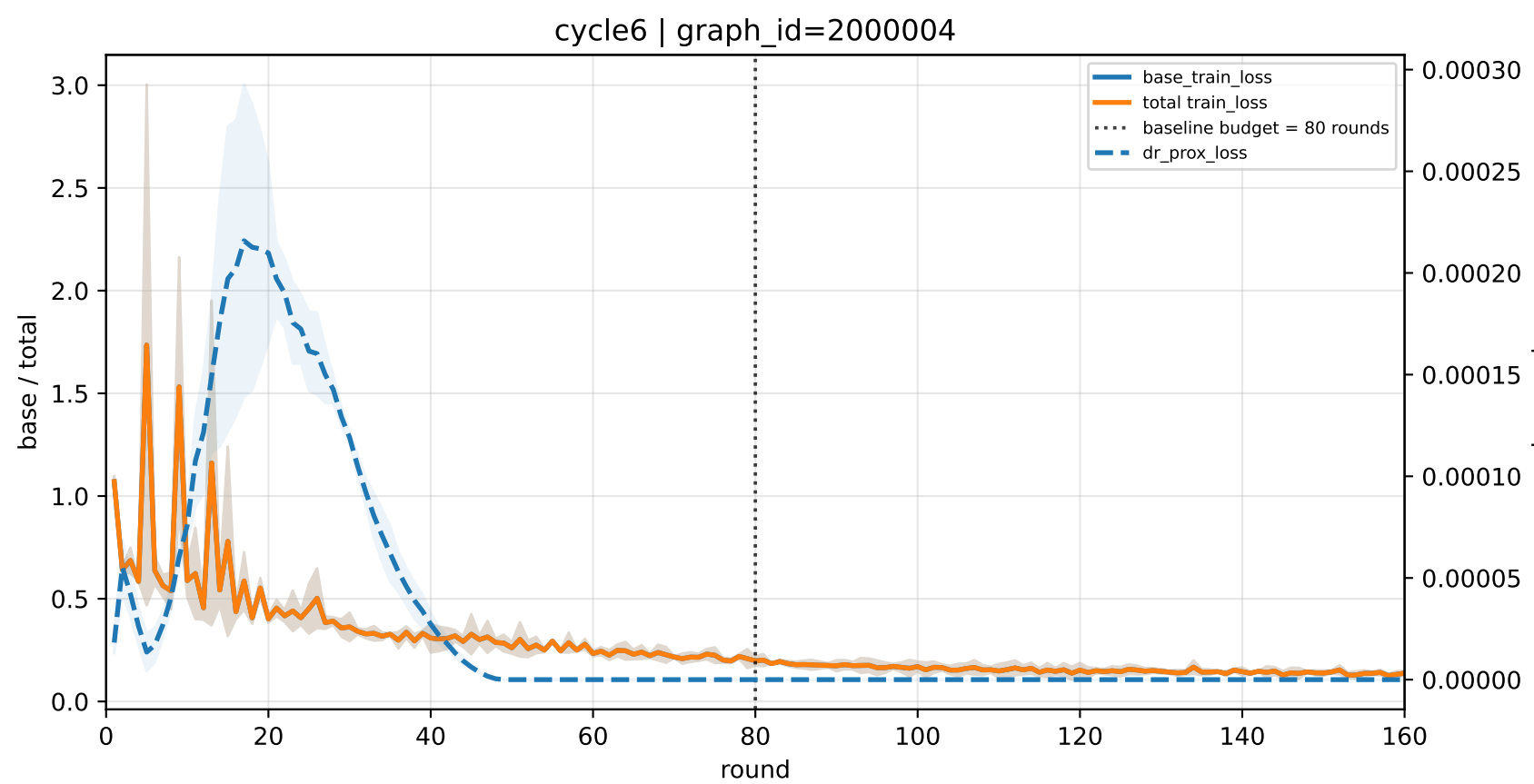
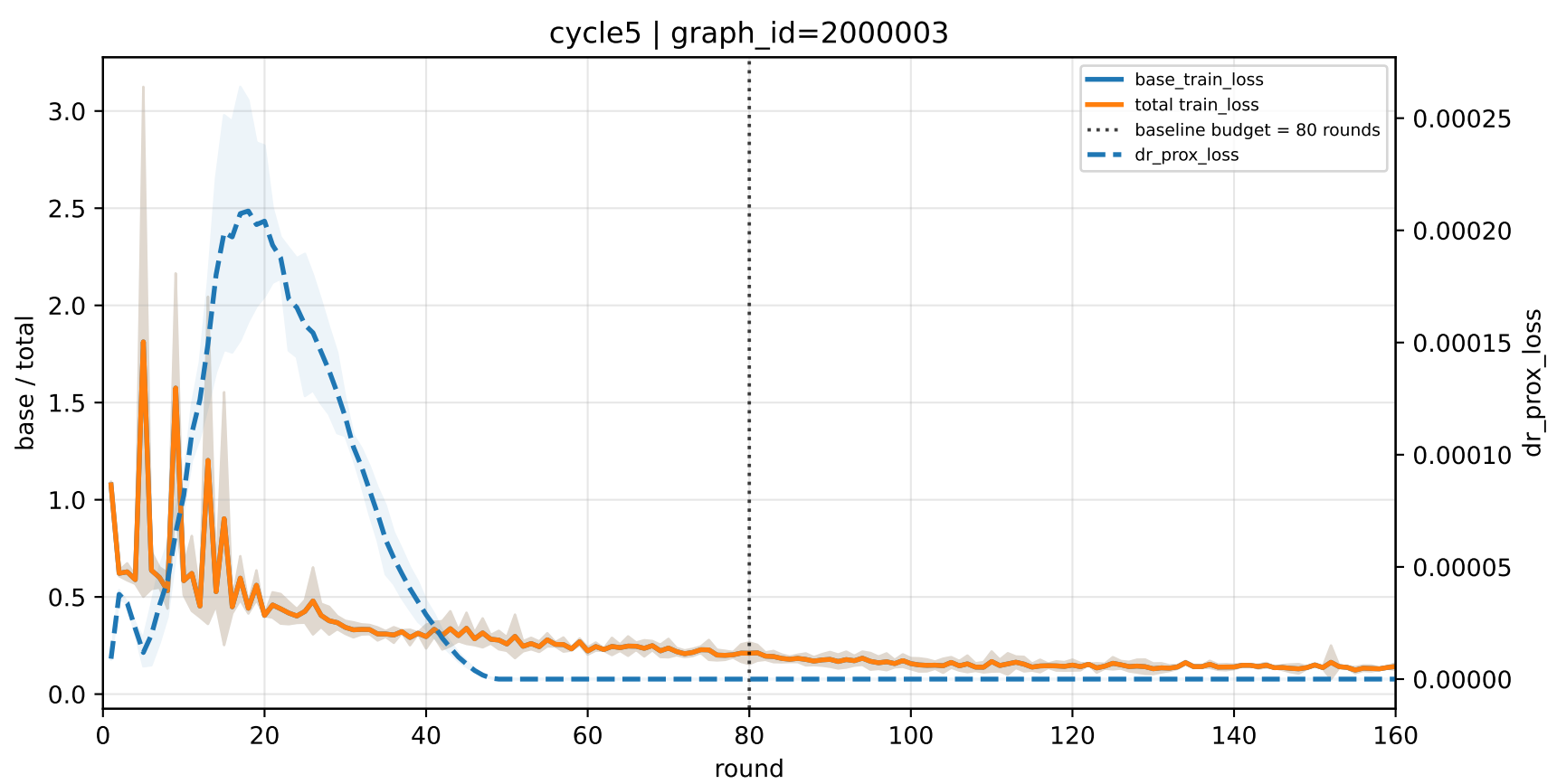
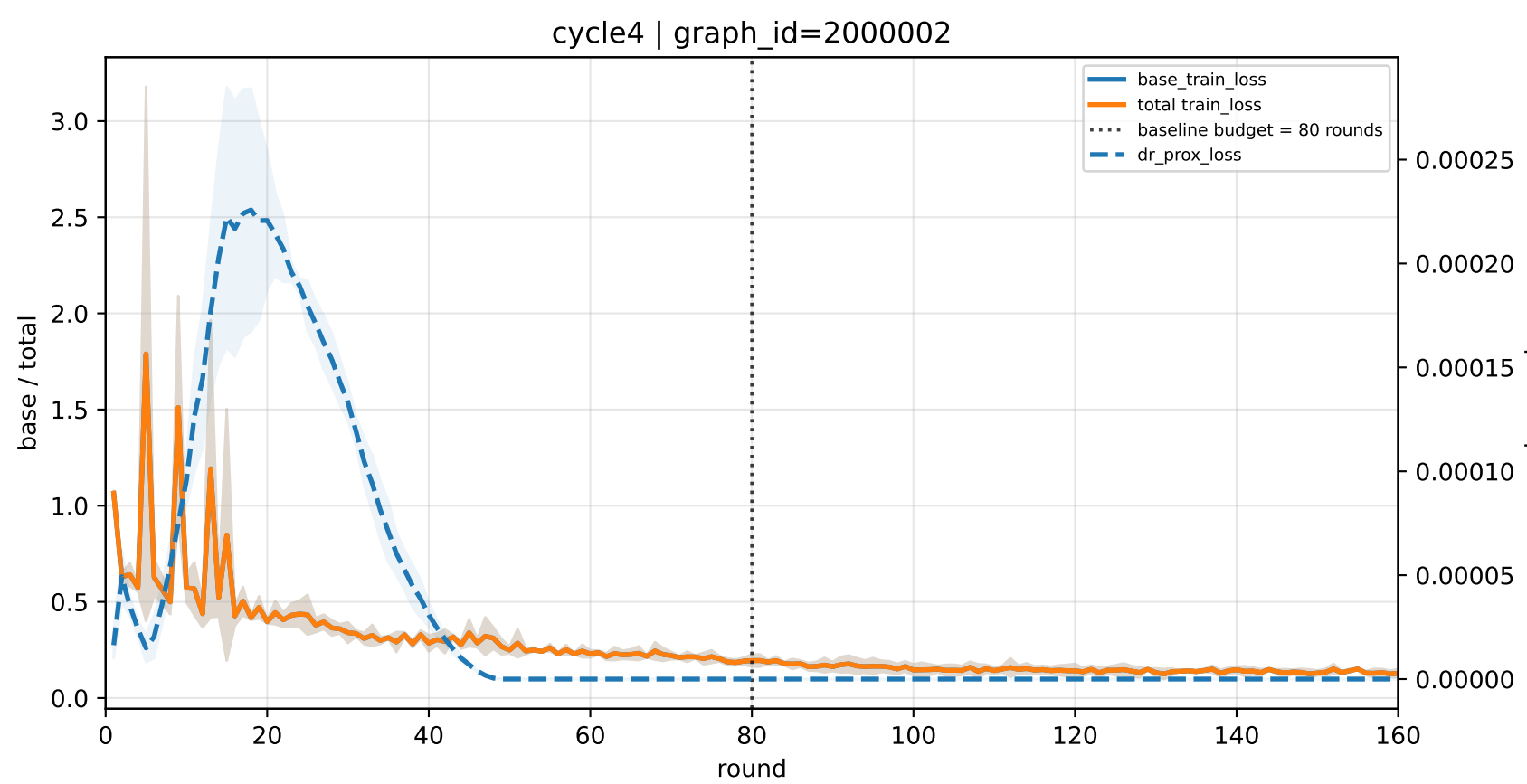
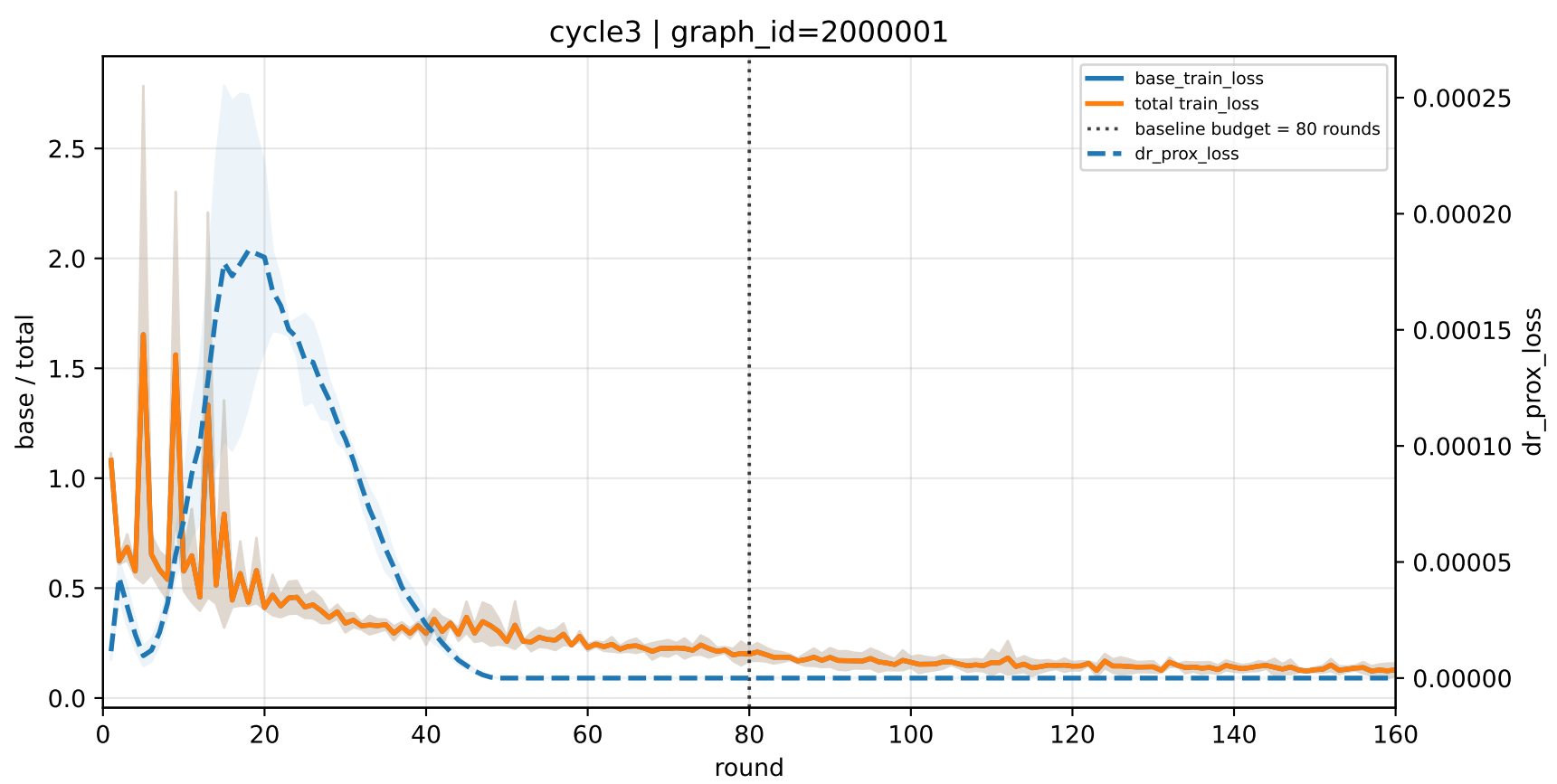
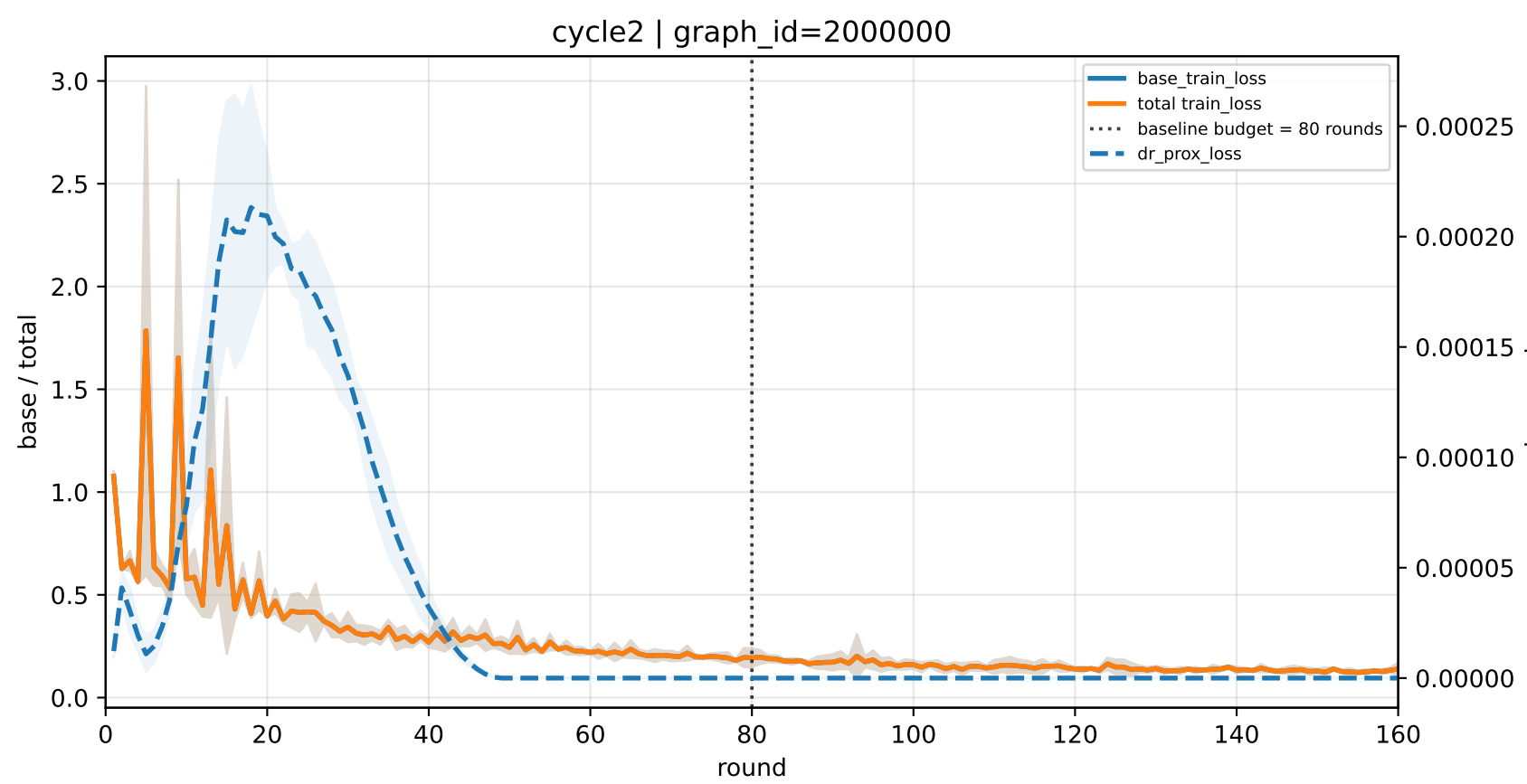
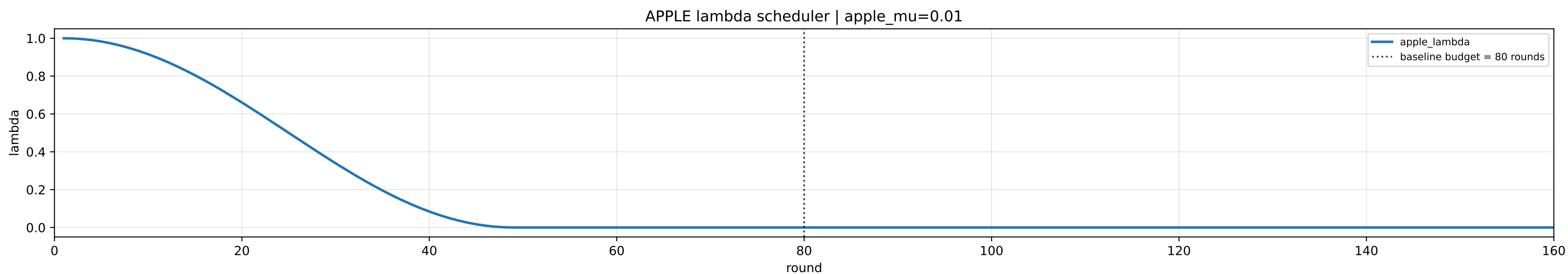


Delta diagnostics: local gap and APPLE mismatch gap



APPLE training-loss decomposition | Low heterogeneity

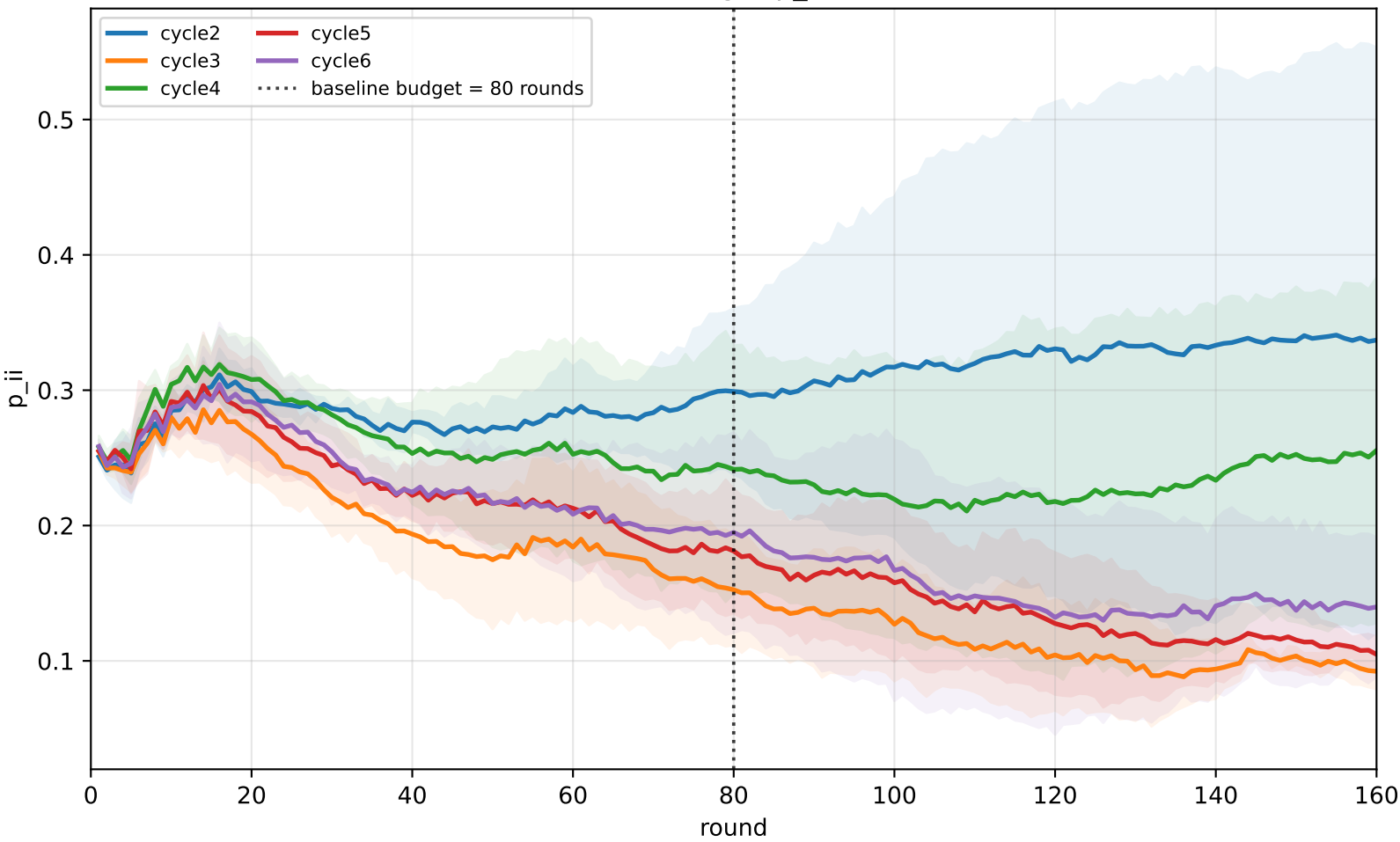
family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003



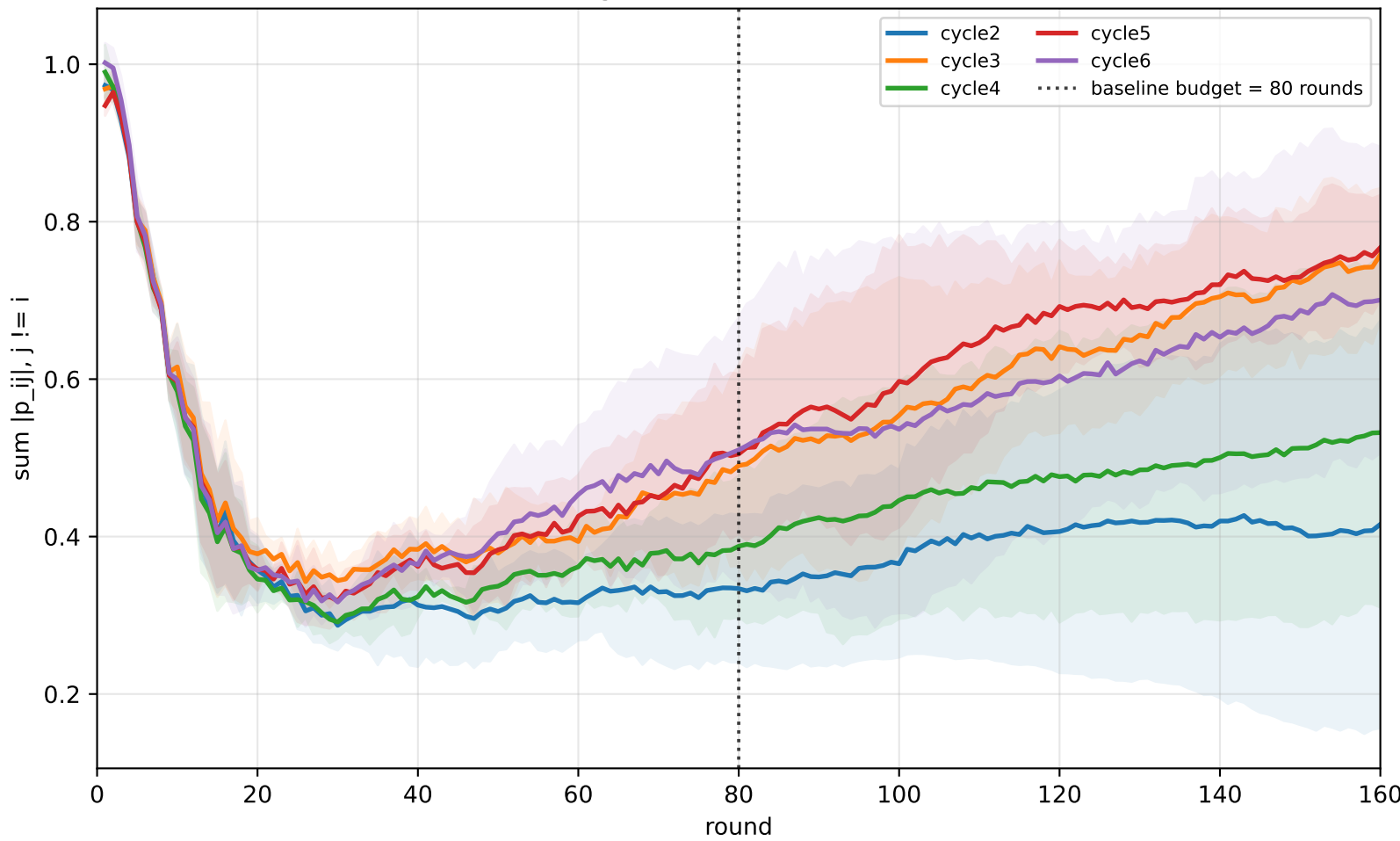
APPLE DR behavior | Low heterogeneity

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 5.3%±0.0% / true 6.3%
specialization=low | specialization_fraction=0.20 | global_visible_support=0.84 | task_jsd_mean=0.003 | task_jsd_median=0.003 | task_jsd_max=0.003 | family_centroid_jsd_mean=0.003

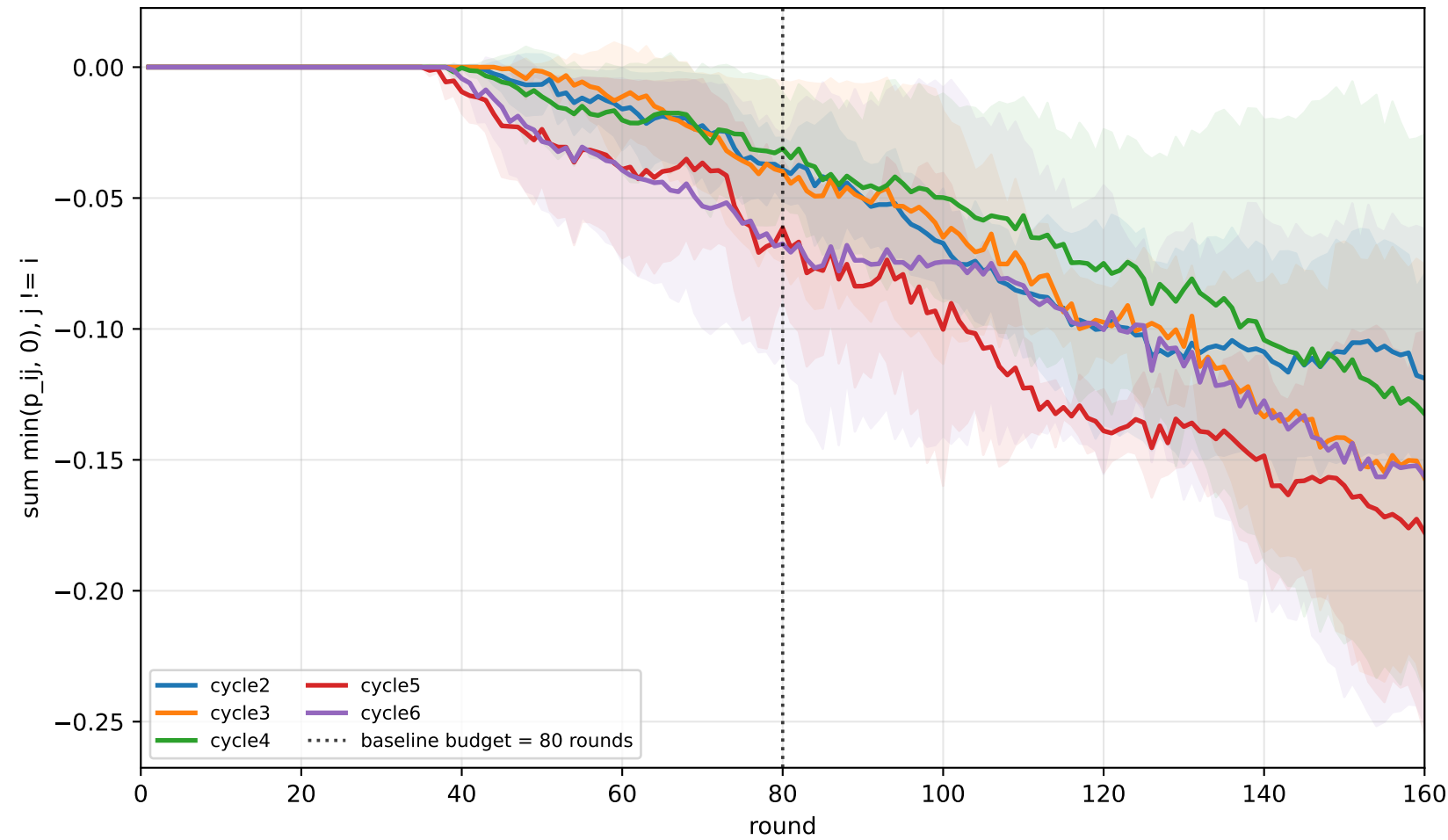
APPLE DR self-weight p_{ii} over rounds



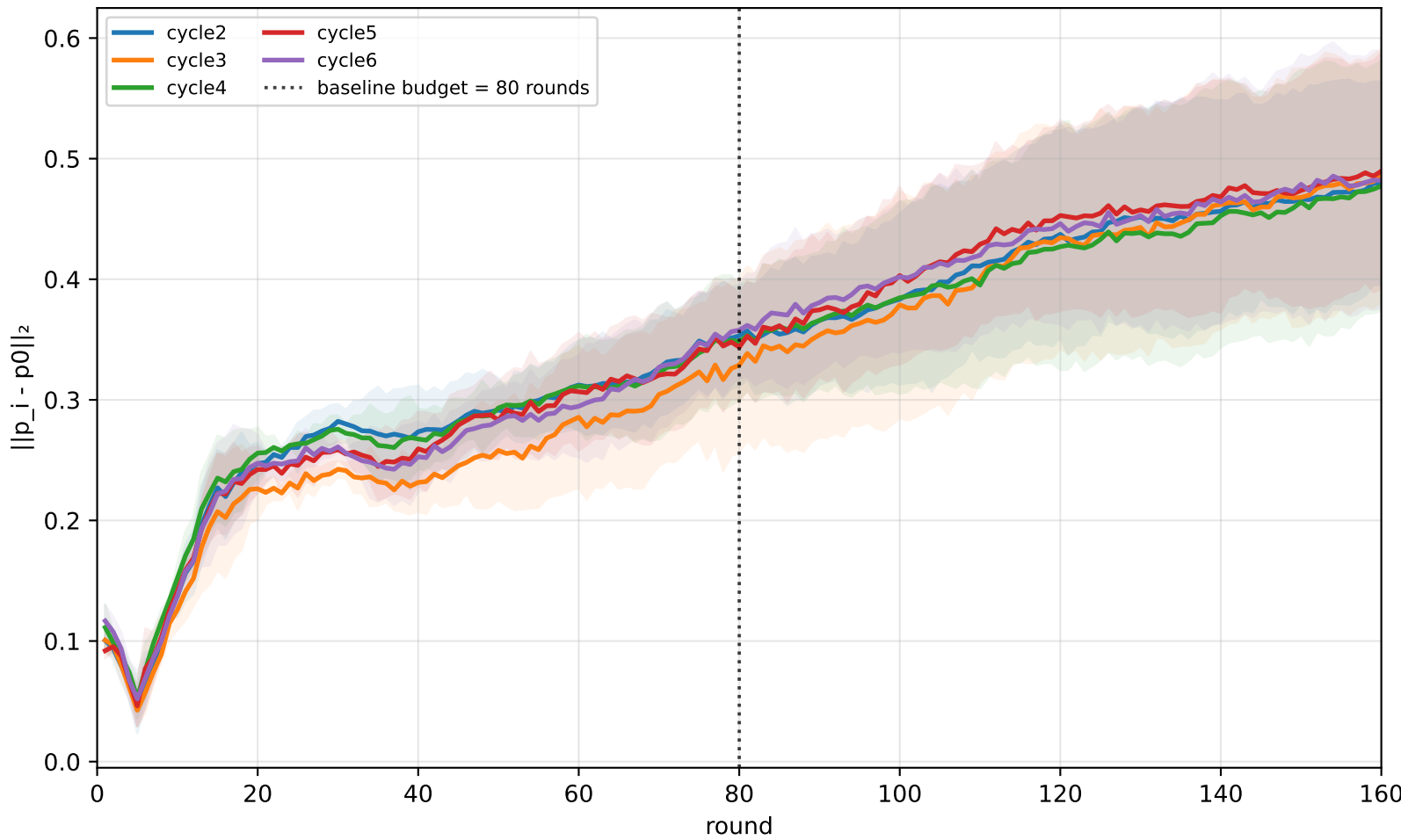
APPLE off-diagonal absolute mass over rounds



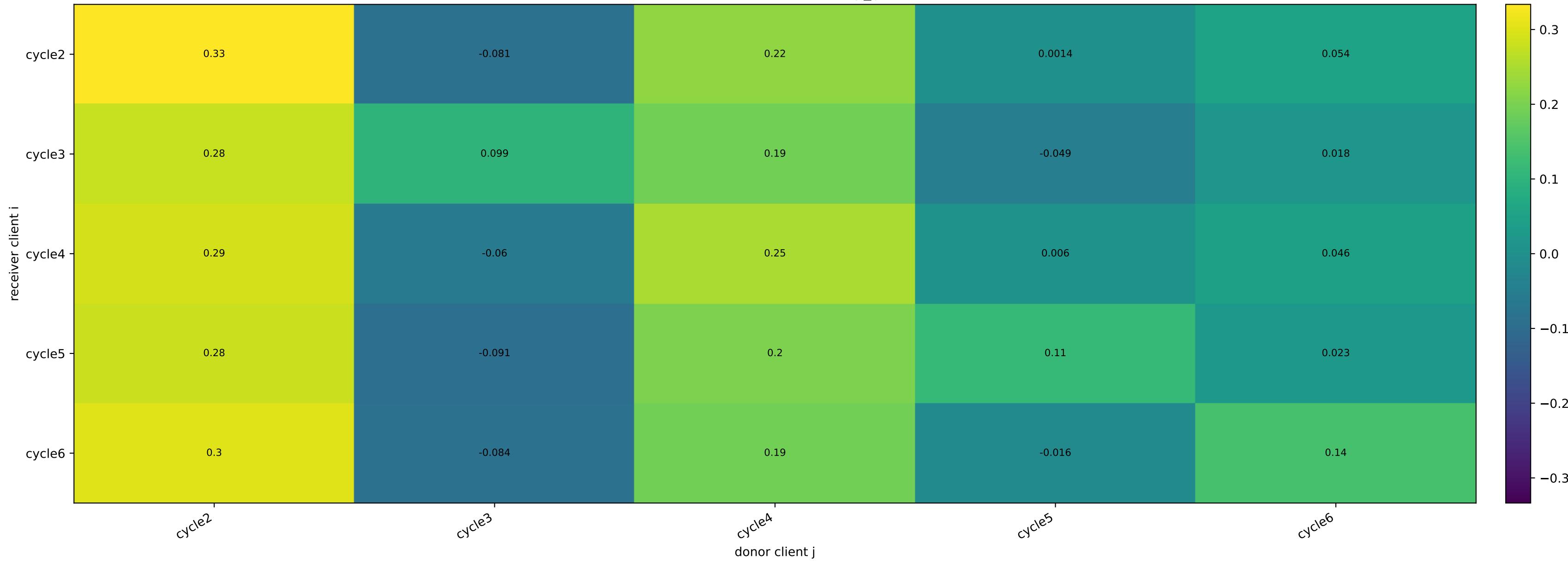
APPLE off-diagonal negative mass over rounds



APPLE DR movement away from p_0



Selected-best-round APPLE DR coefficients p_{ij} , mean over seeds



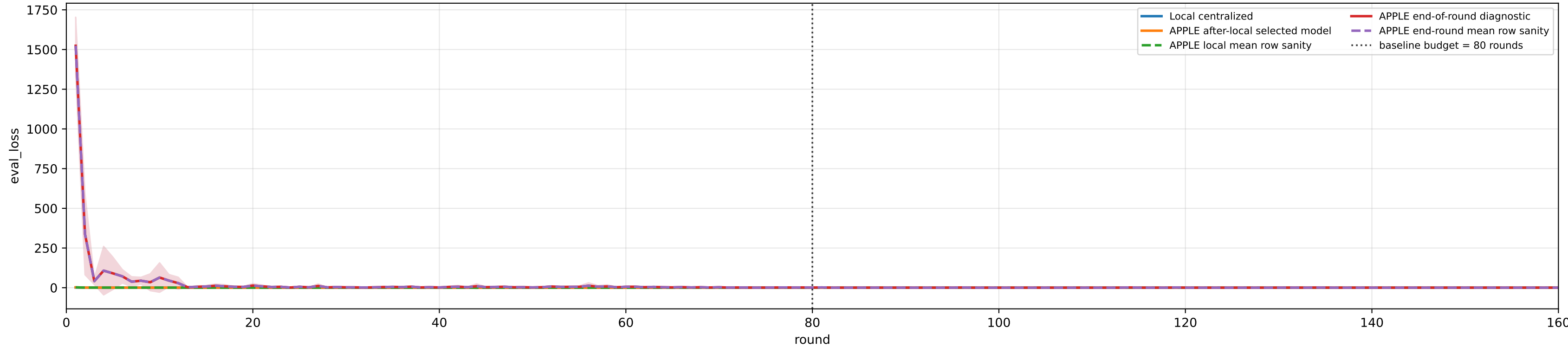
Controlled task-specialization level: Mid | APPLE diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.3%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

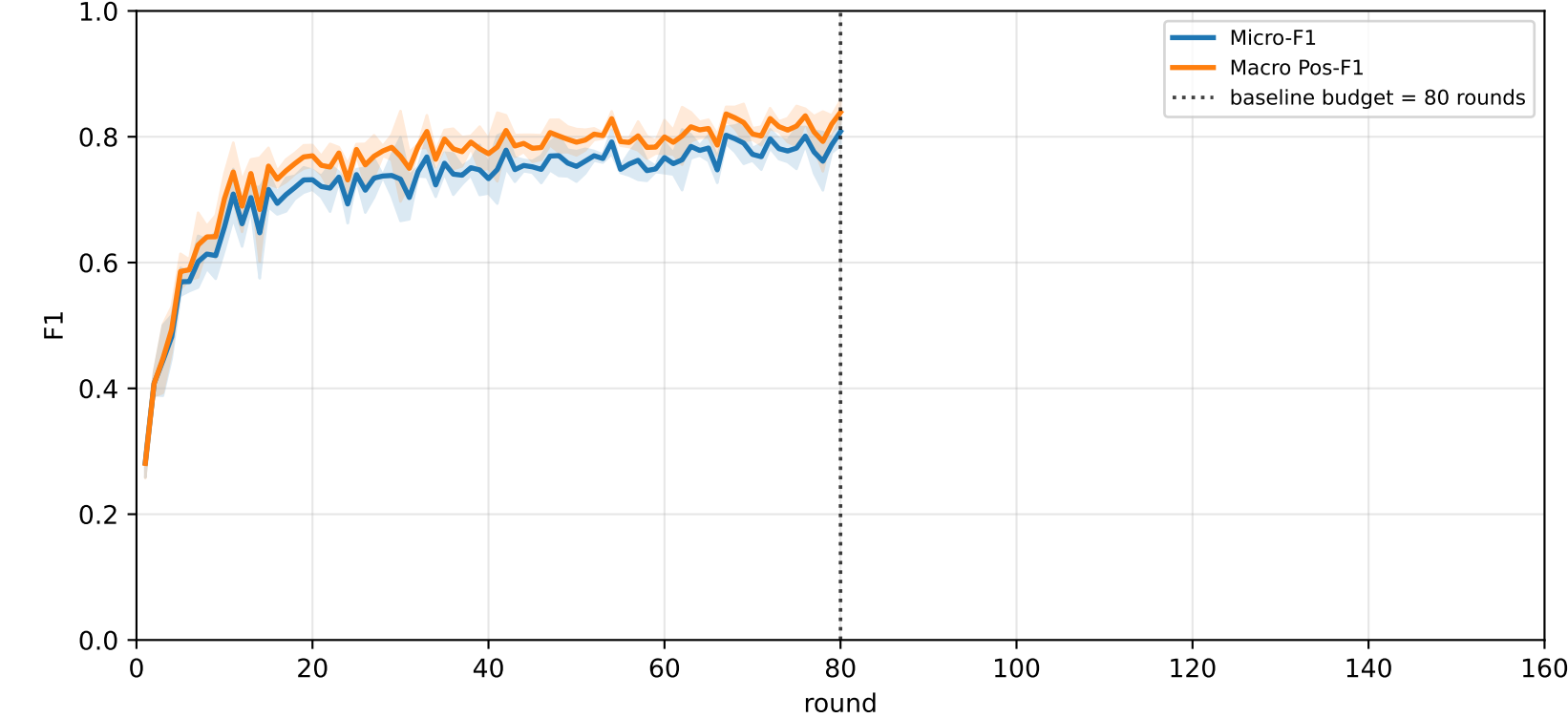
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.6% ± 0.7%	84.6% ± 0.7%	96.9% ± 0.3%	95.3% ± 0.9%	87.5% ± 0.4%	77.3% ± 1.9%	65.6% ± 0.6%
APPLE	80.1% ± 3.1%	82.9% ± 2.5%	96.0% ± 0.9%	92.4% ± 1.6%	82.0% ± 4.2%	77.6% ± 4.1%	66.0% ± 6.1%

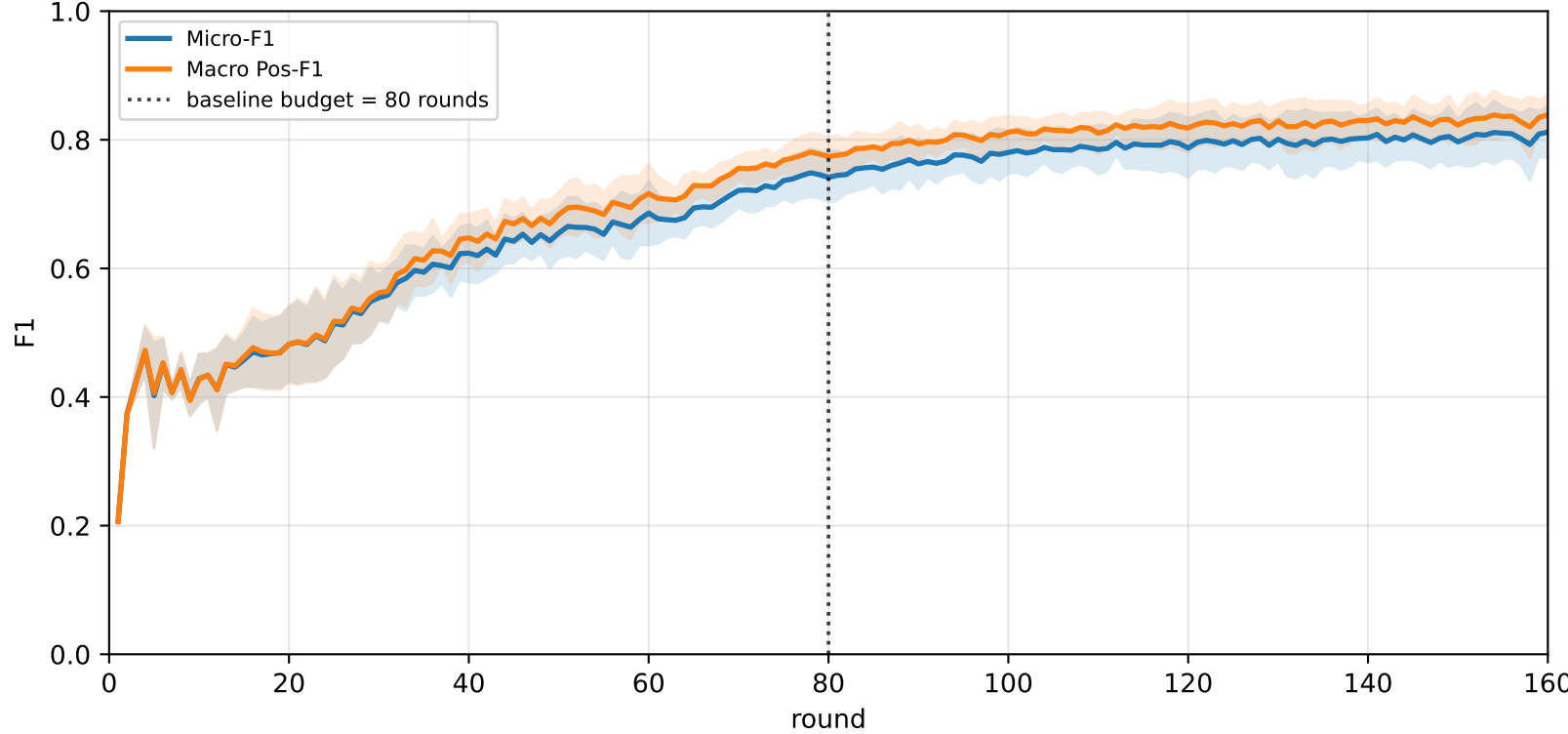
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



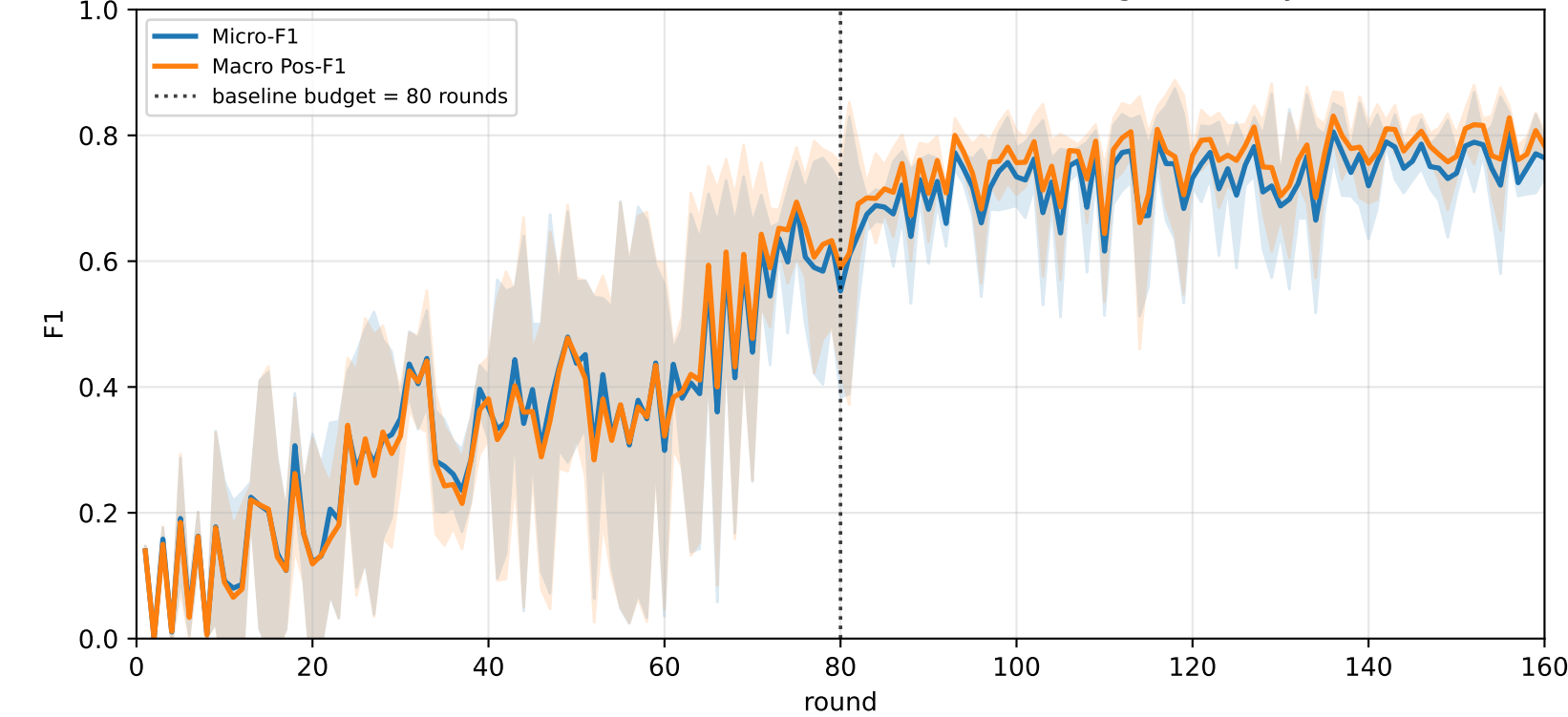
Pooled local centralized micro / macro F1



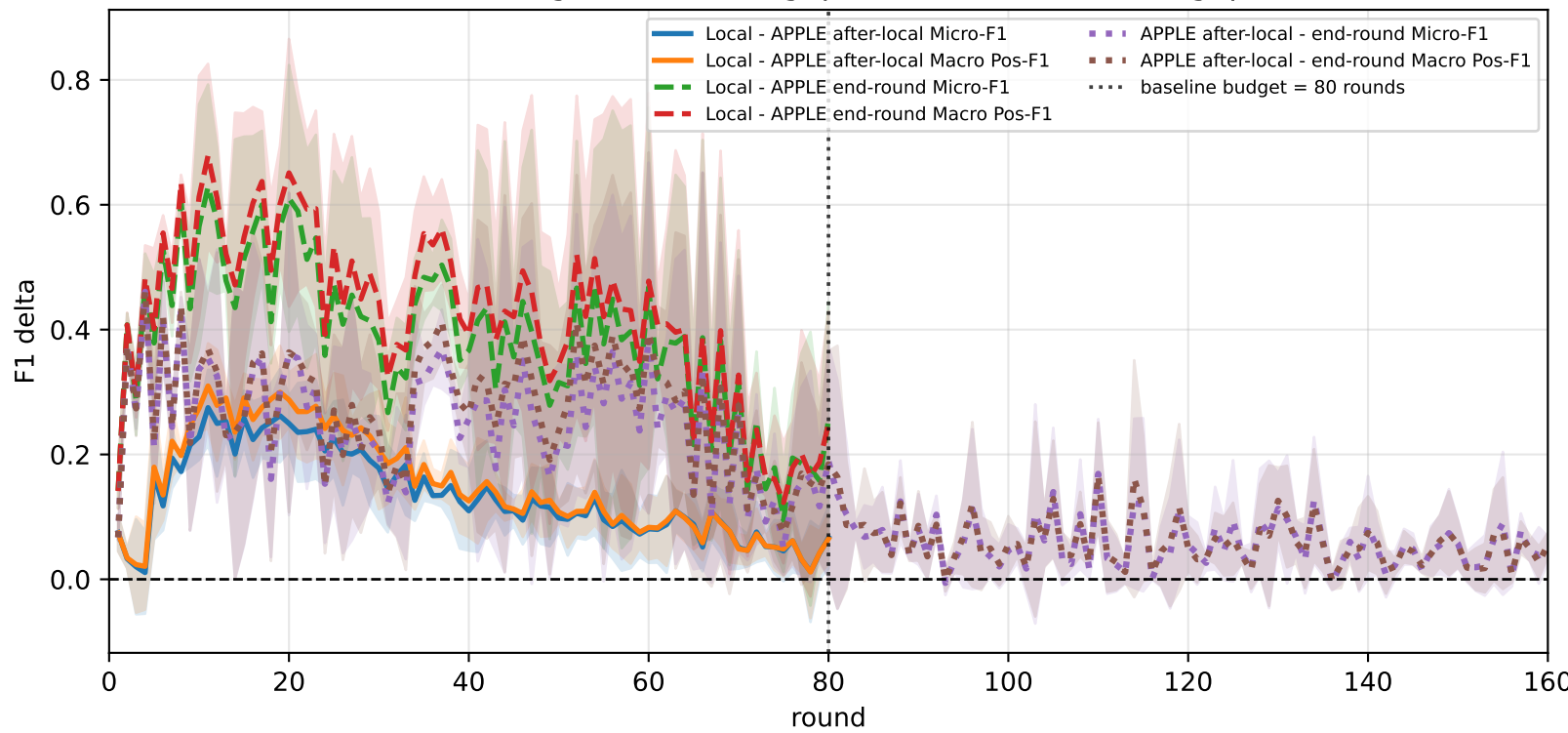
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



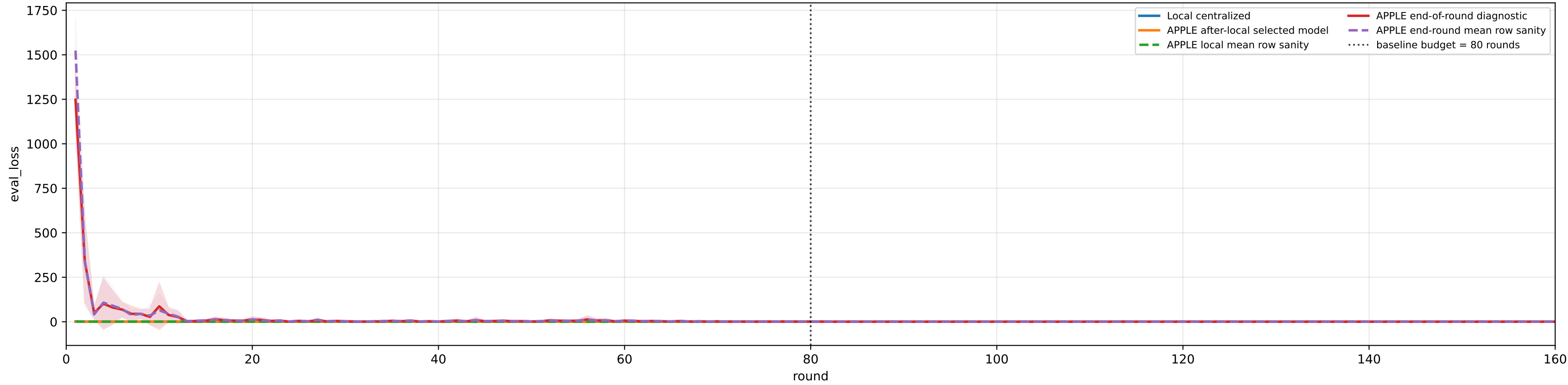
Controlled task-specialization level: Mid | APPLE diagnostics | family=specialized_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

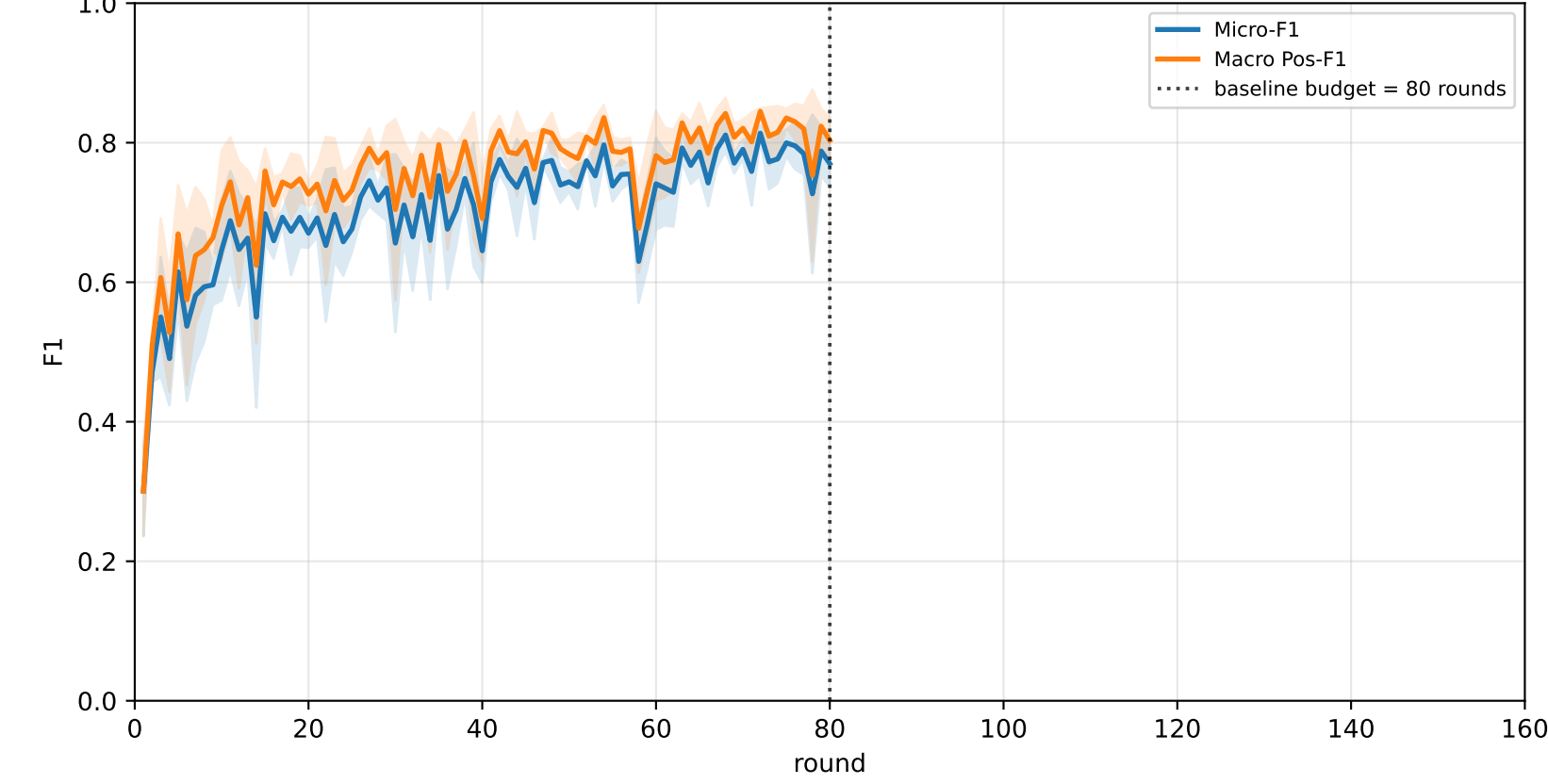
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.3% ± 1.7%	84.2% ± 1.3%	98.2% ± 0.7%	94.8% ± 1.7%	86.7% ± 1.3%	75.0% ± 3.4%	66.4% ± 3.0%
APPLE	80.6% ± 2.9%	83.6% ± 2.6%	98.0% ± 0.3%	93.4% ± 2.5%	83.4% ± 6.1%	78.0% ± 4.0%	65.2% ± 3.7%

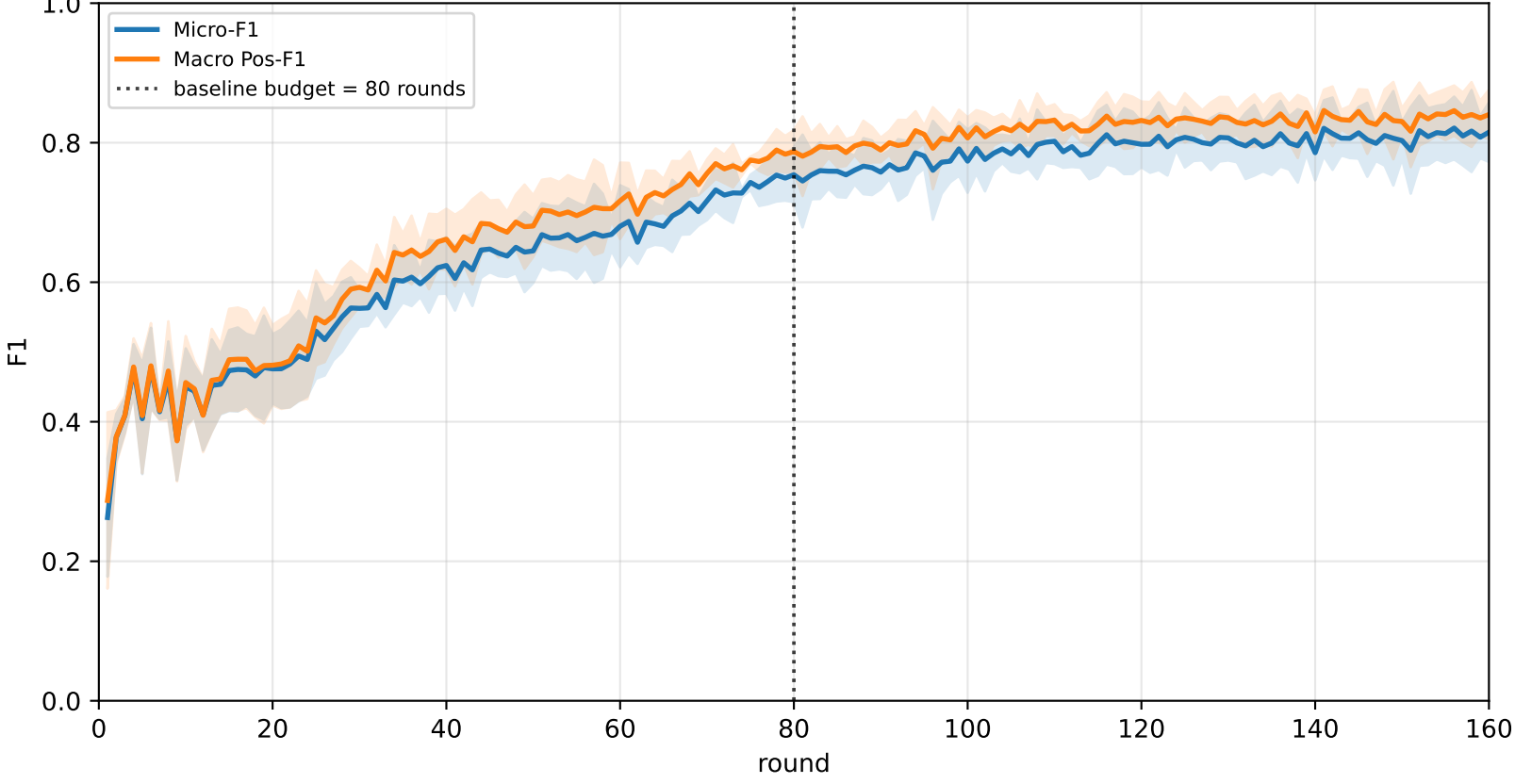
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



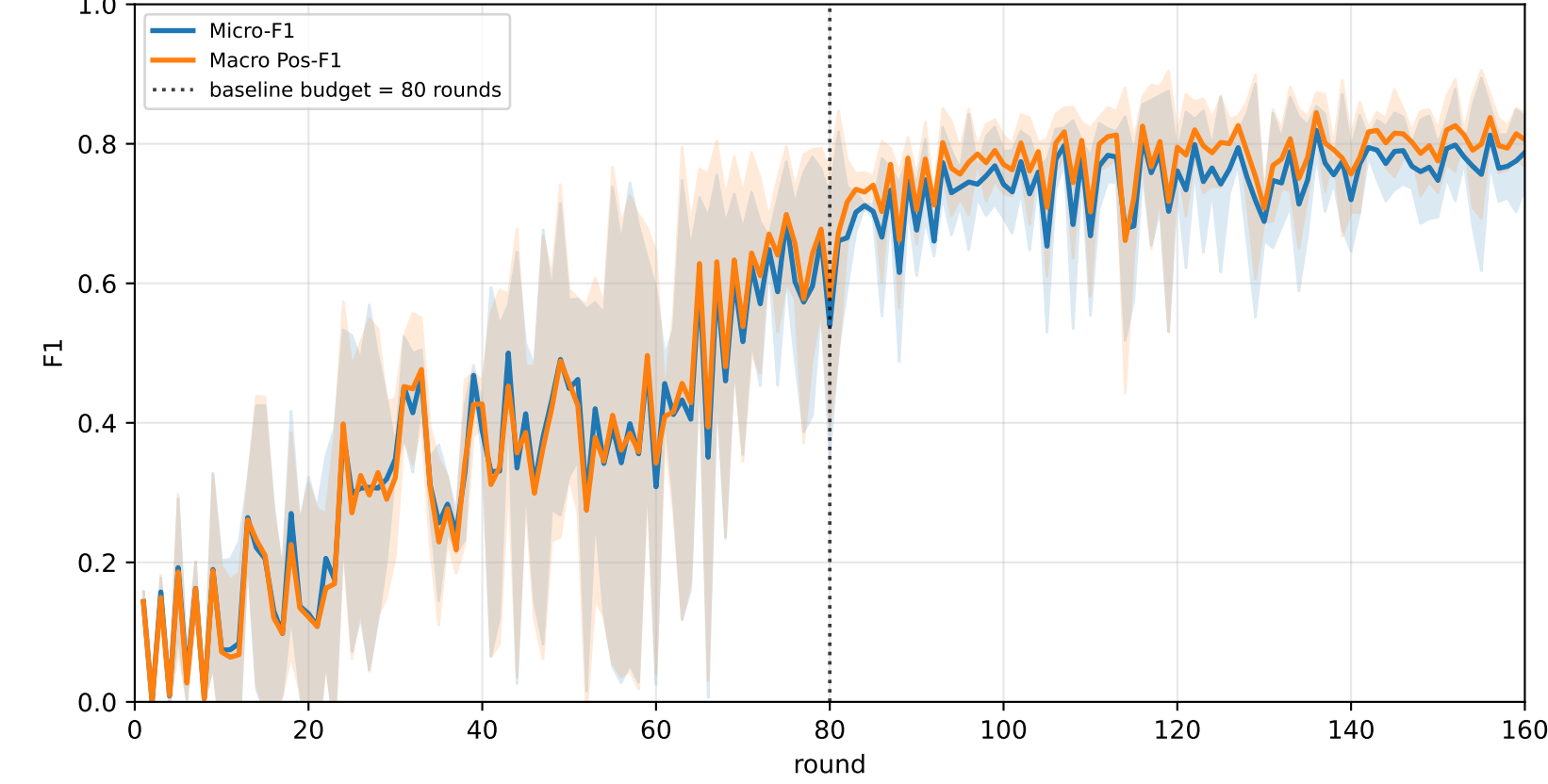
Pooled local centralized micro / macro F1



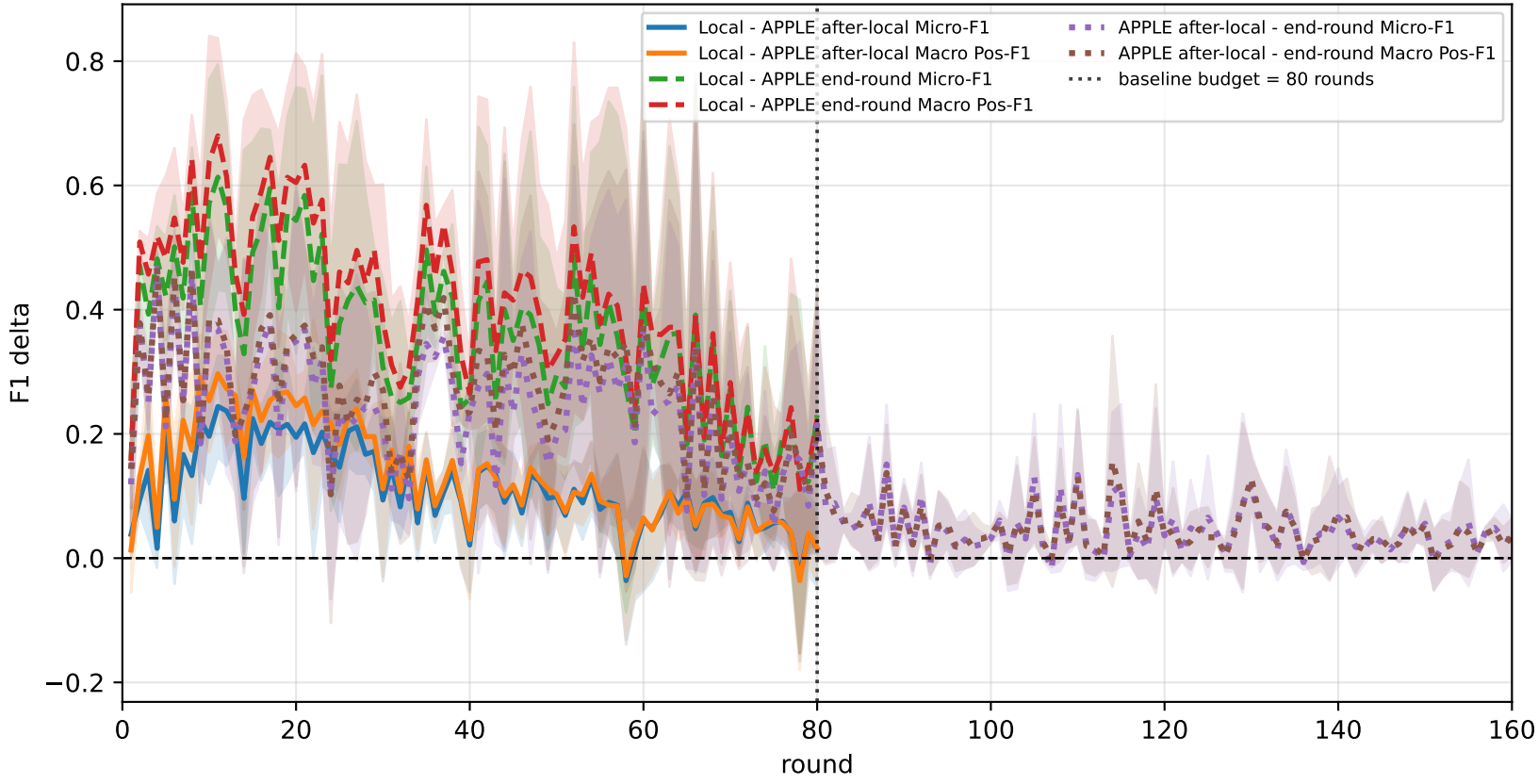
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



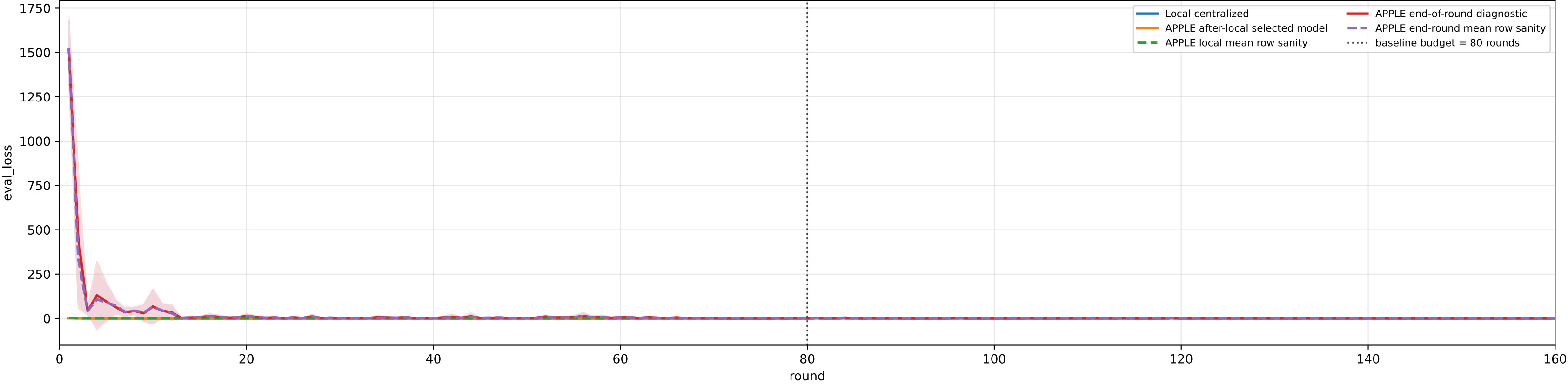
Controlled task-specialization level: Mid | APPLE diagnostics | family=specialized_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

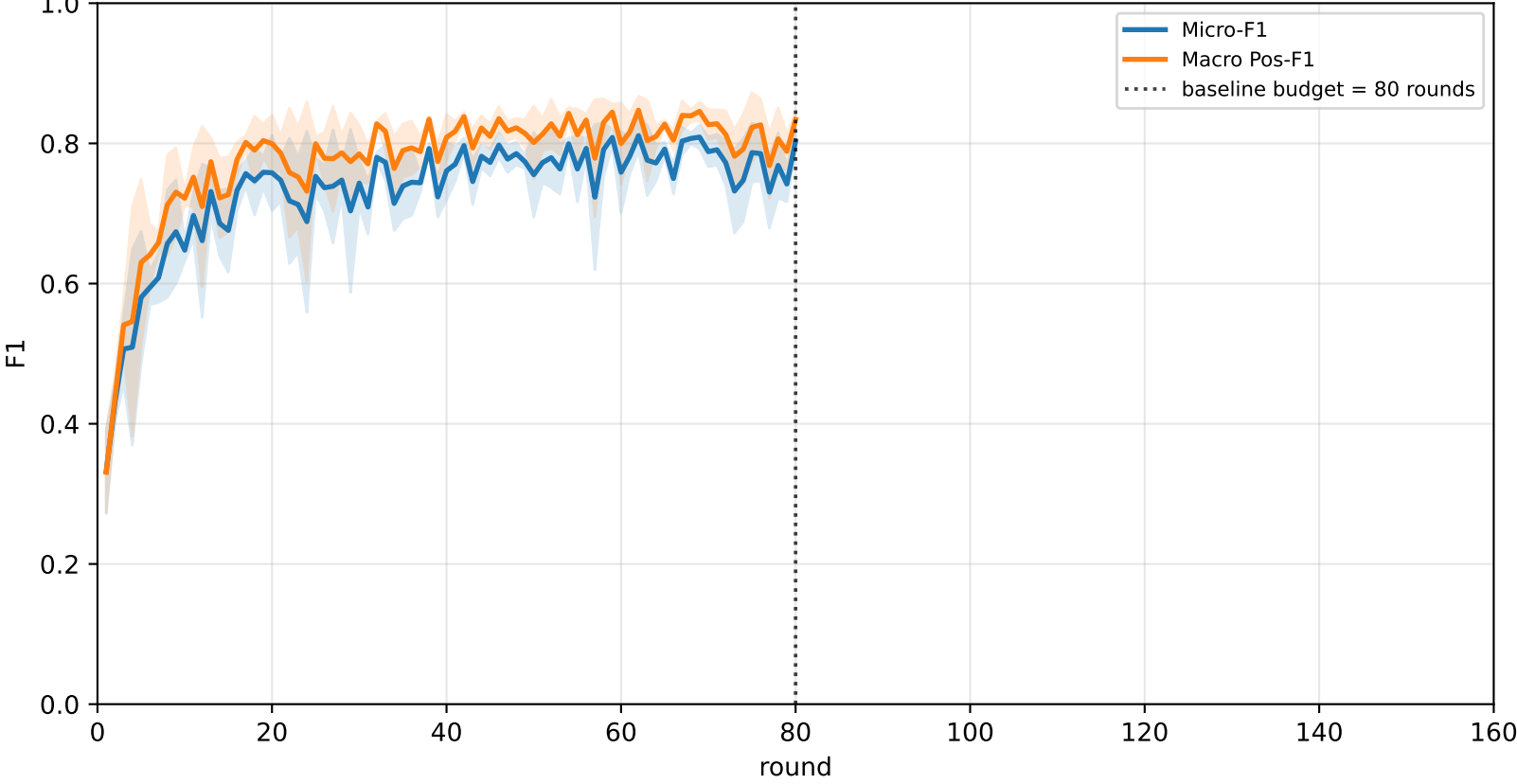
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.1% ± 0.2%	84.7% ± 0.1%	97.7% ± 0.8%	96.9% ± 0.8%	90.8% ± 0.3%	74.1% ± 1.7%	64.0% ± 0.4%
APPLE	79.4% ± 3.6%	82.6% ± 2.4%	96.2% ± 0.5%	94.2% ± 2.2%	80.3% ± 2.8%	77.9% ± 4.2%	64.2% ± 7.9%

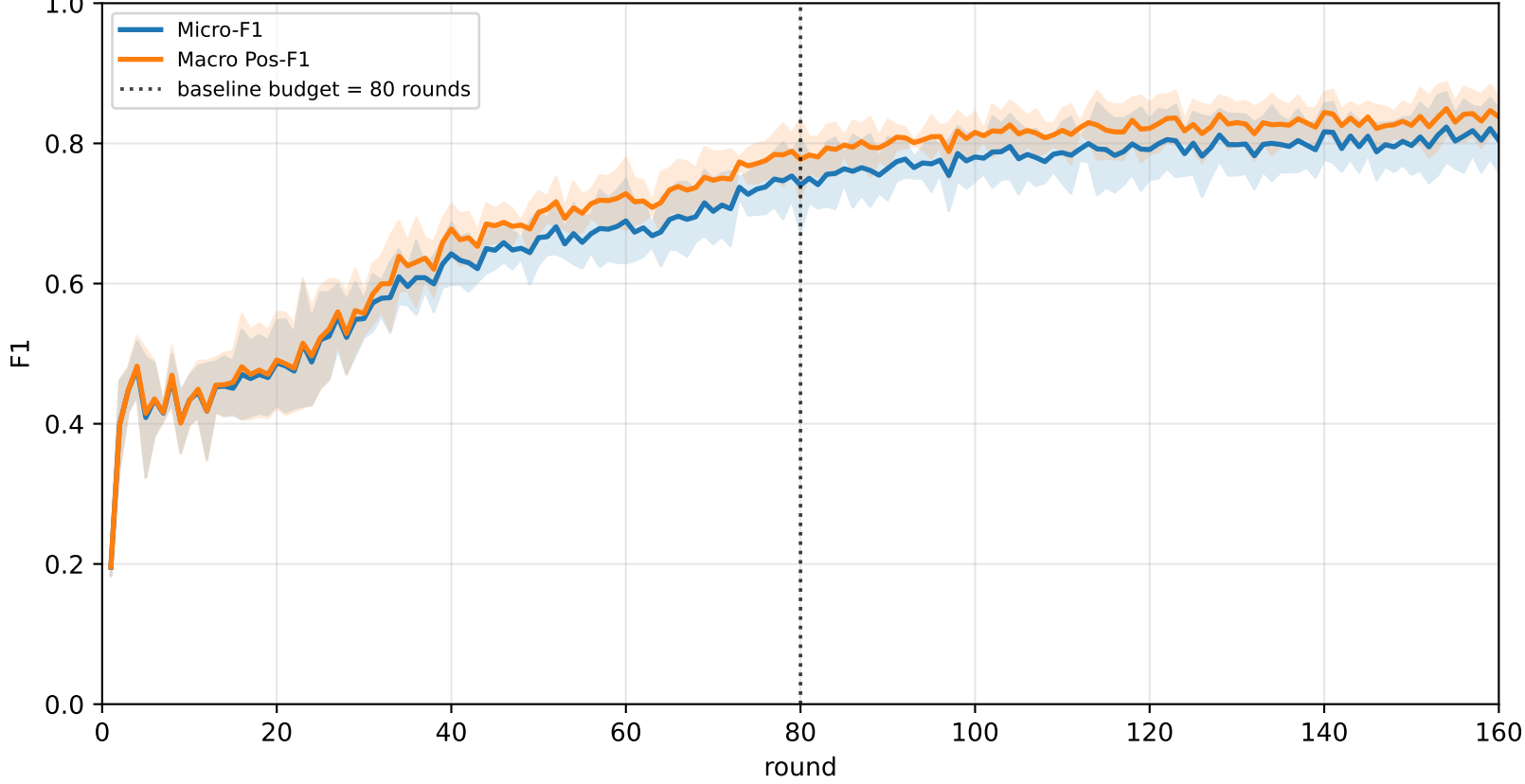
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



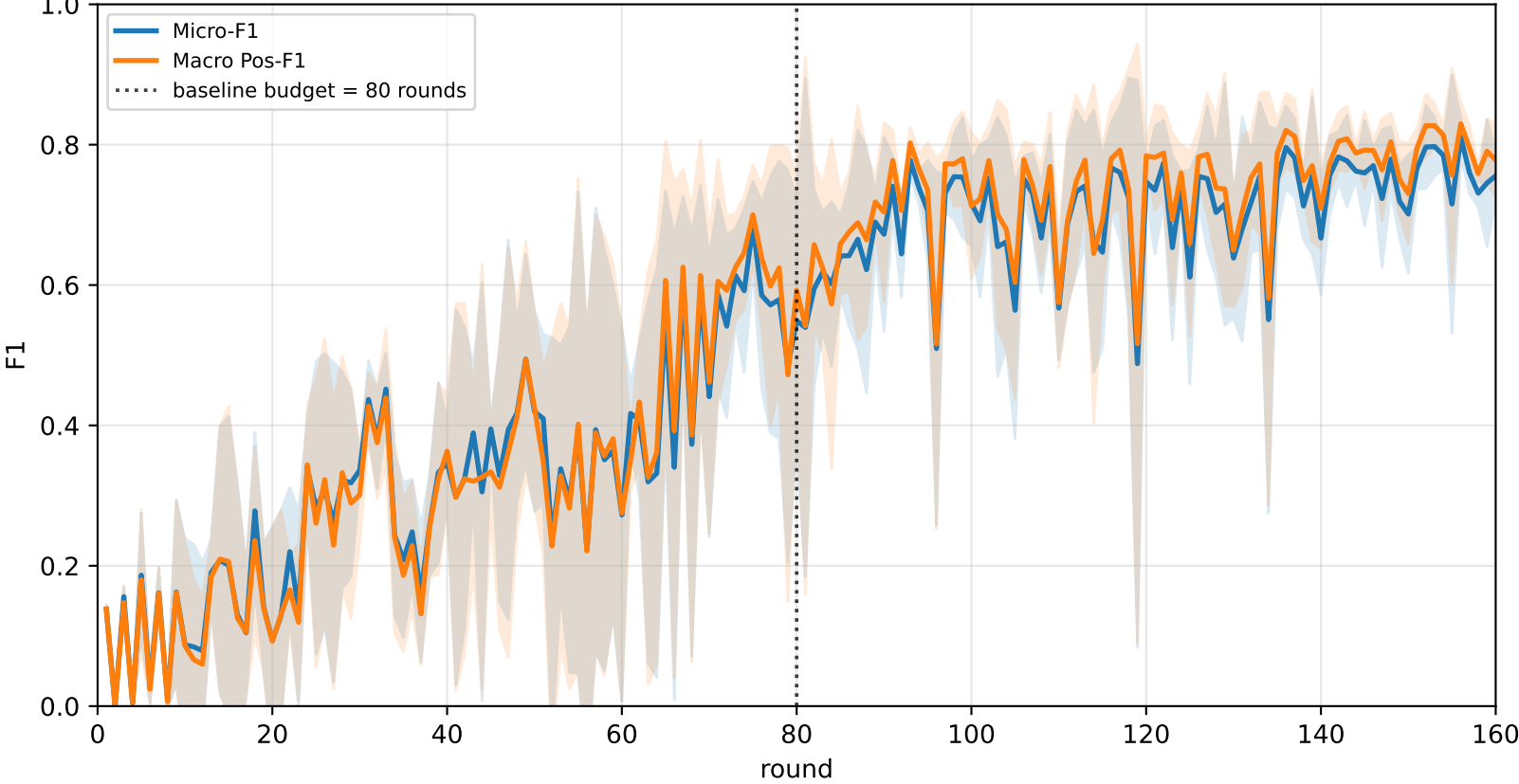
Pooled local centralized micro / macro F1



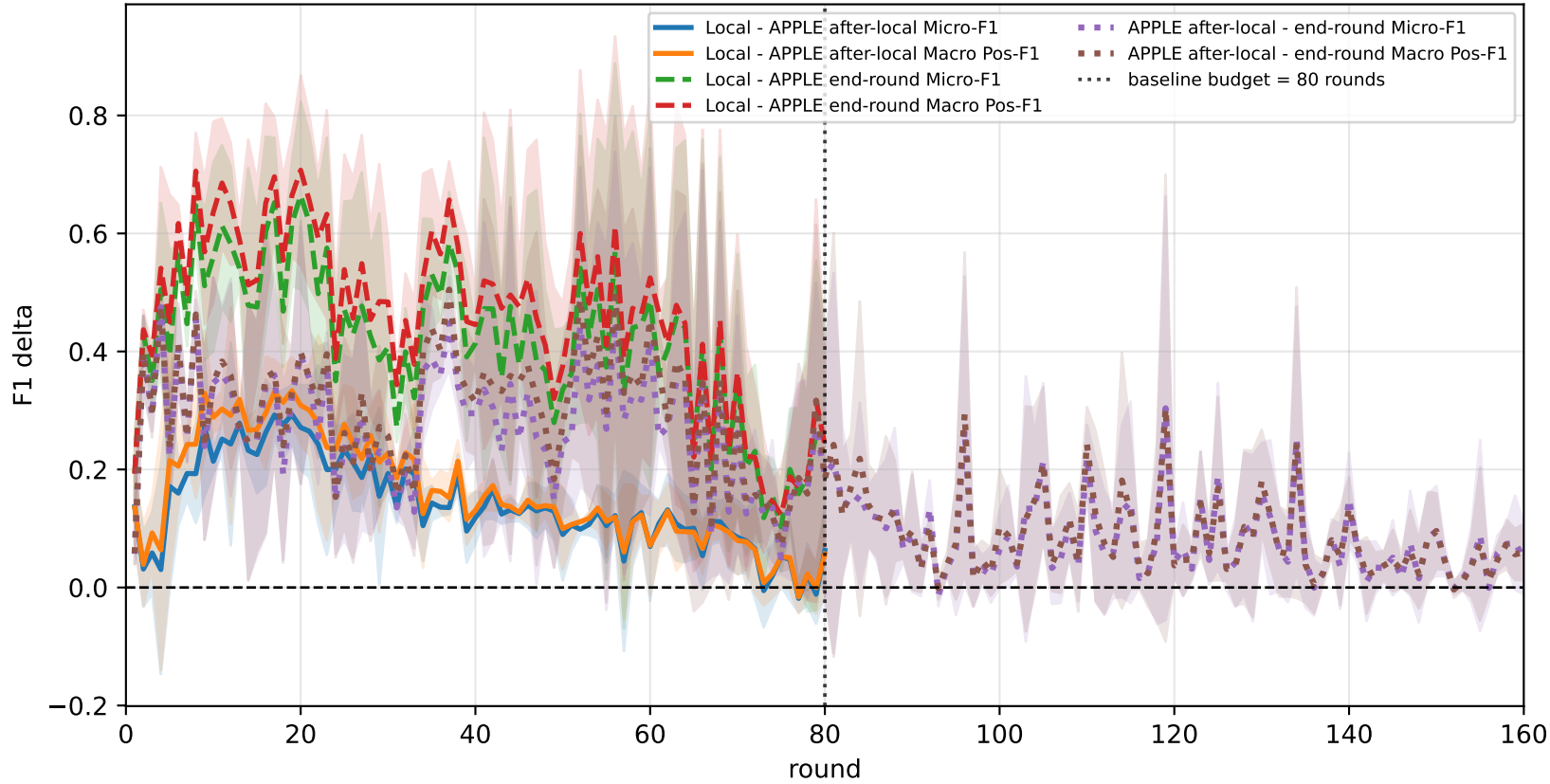
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



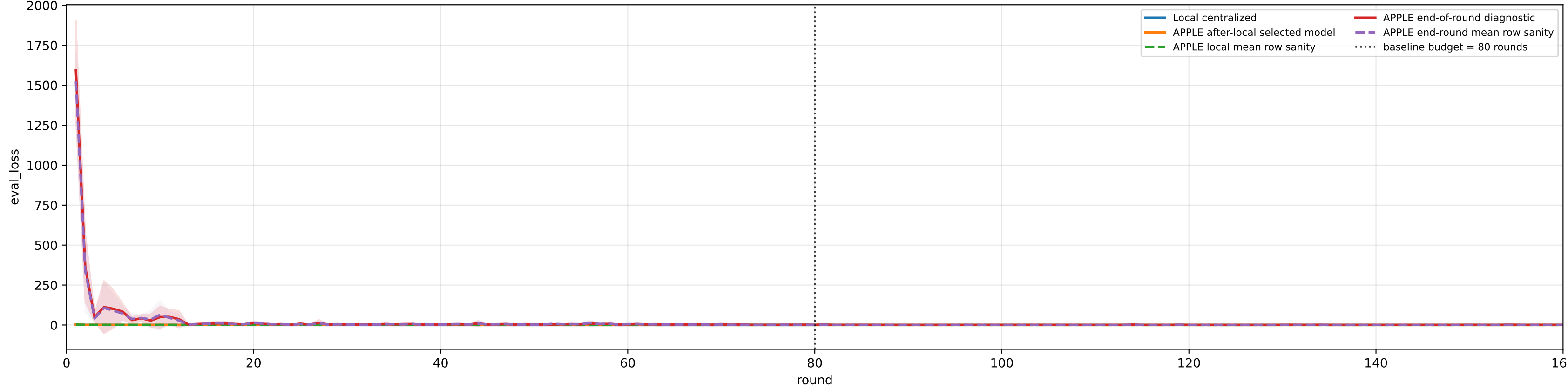
Controlled task-specialization level: Mid | APPLE diagnostics | family=specialized_cycle4 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle4 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

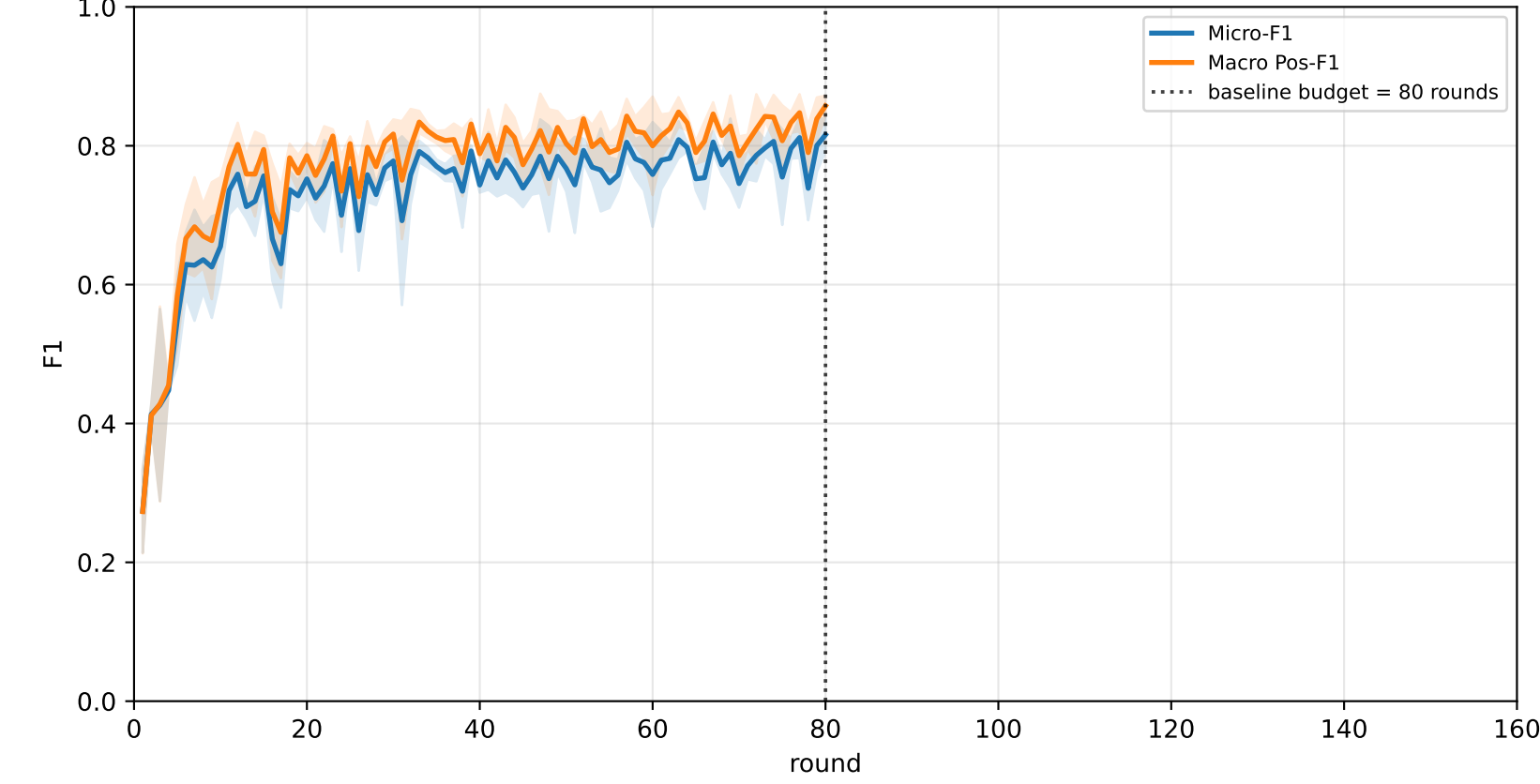
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	82.3% ± 1.4%	85.7% ± 1.2%	97.7% ± 1.1%	96.2% ± 0.1%	89.8% ± 1.6%	80.9% ± 4.0%	63.8% ± 1.5%
APPLE	80.8% ± 3.4%	83.4% ± 2.9%	95.9% ± 1.0%	92.5% ± 2.4%	84.3% ± 5.9%	77.4% ± 4.6%	67.1% ± 6.0%

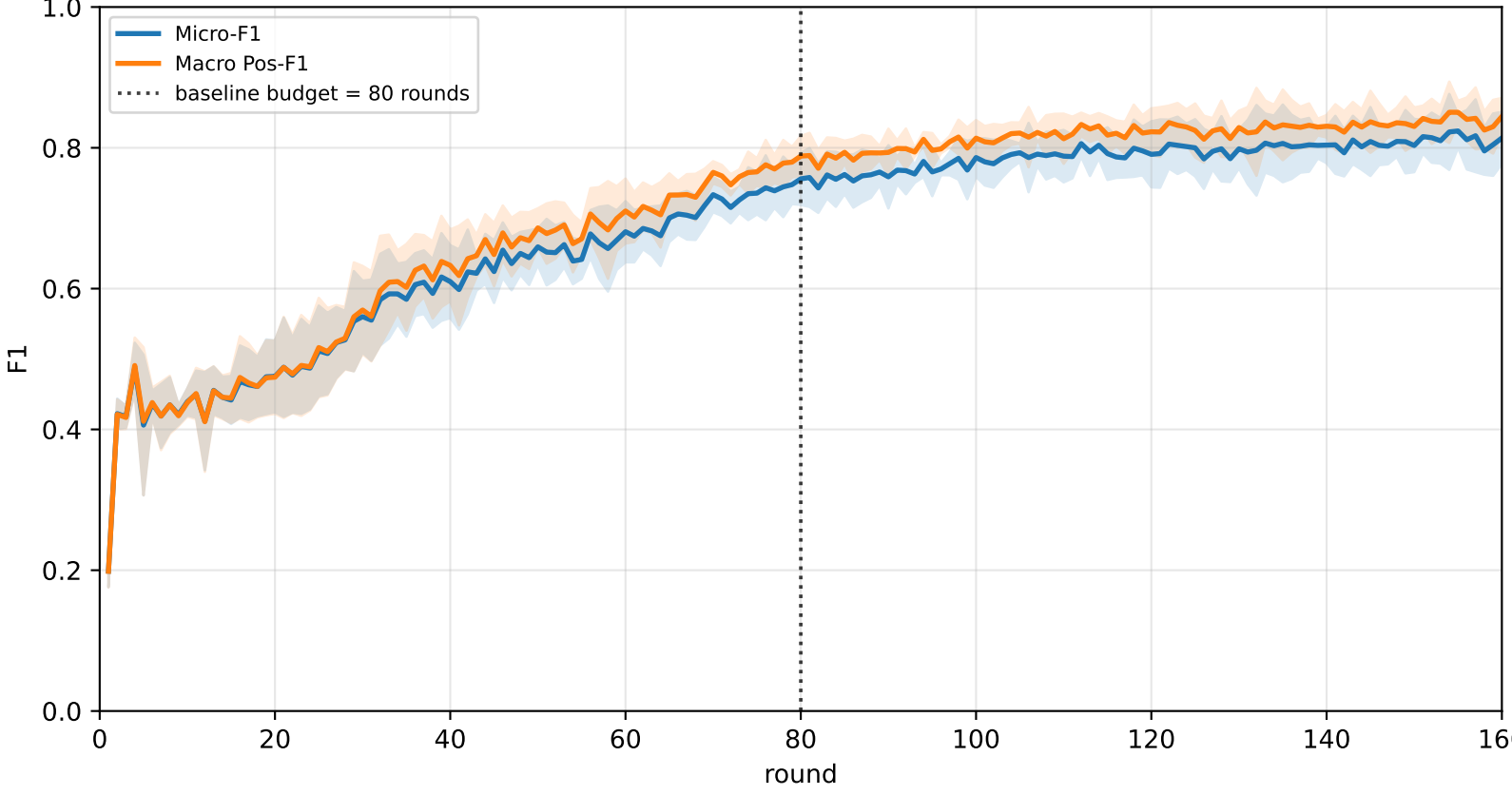
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



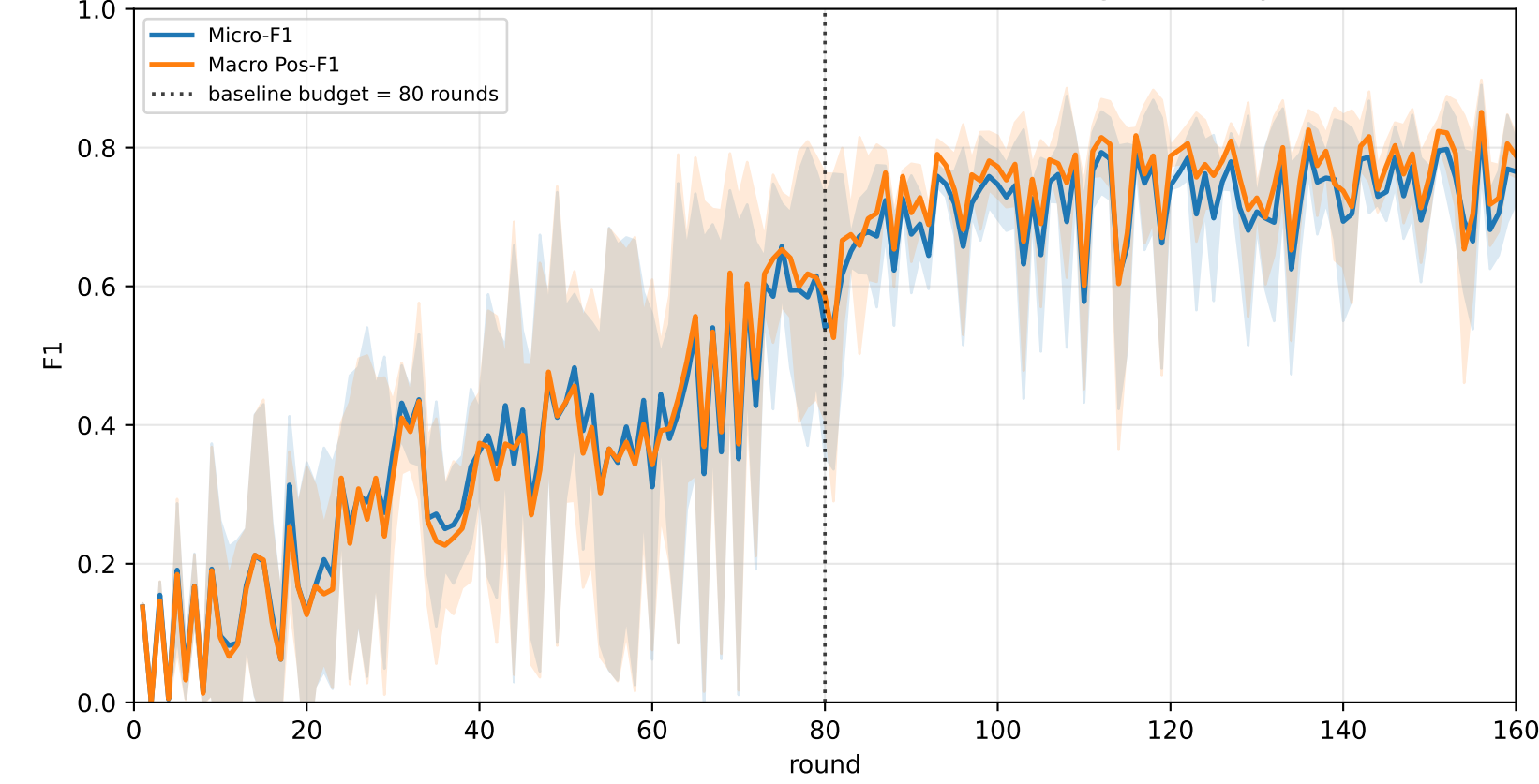
Pooled local centralized micro / macro F1



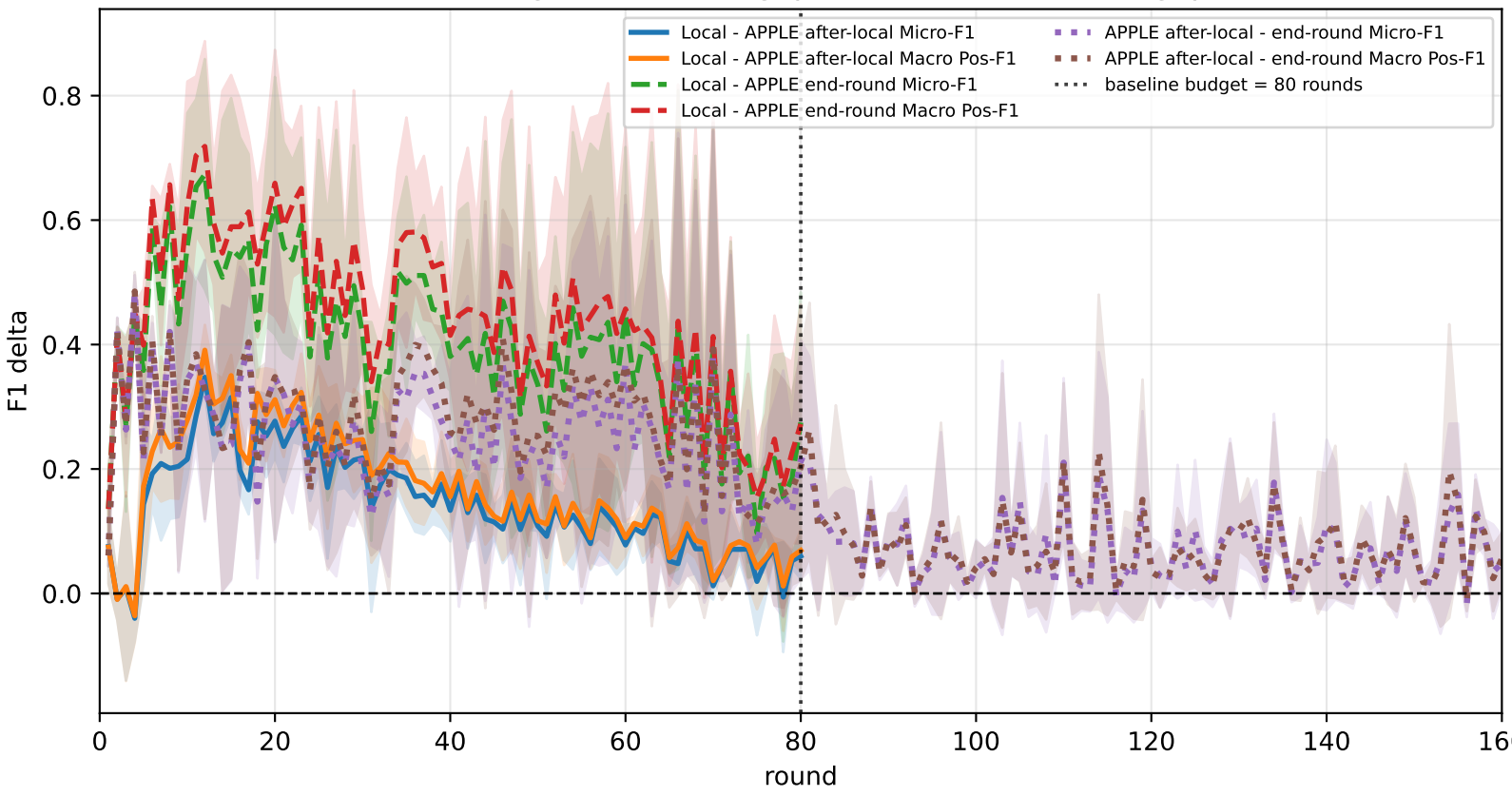
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



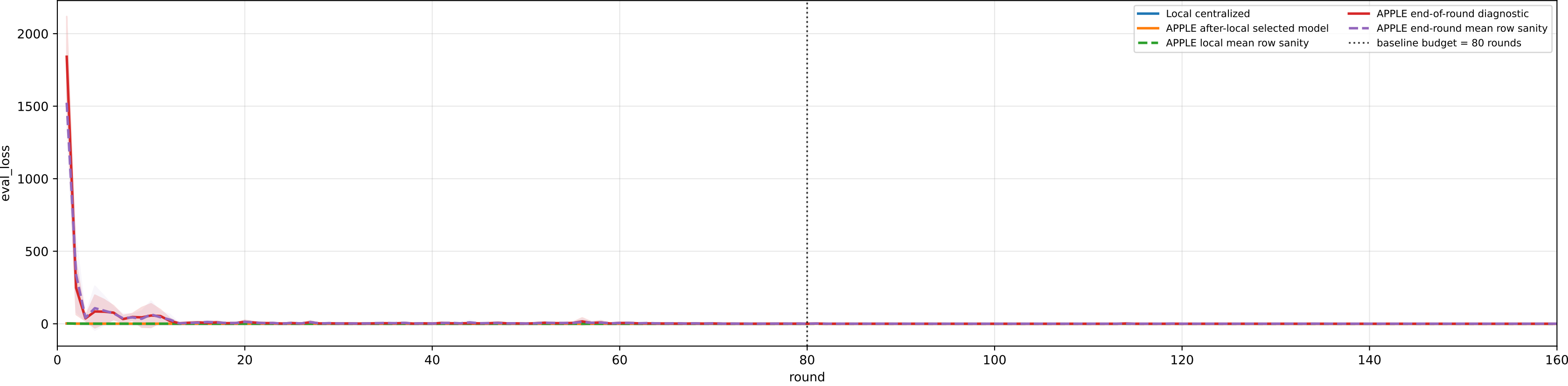
Controlled task-specialization level: Mid | APPLE diagnostics | family=specialized_cycle5 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

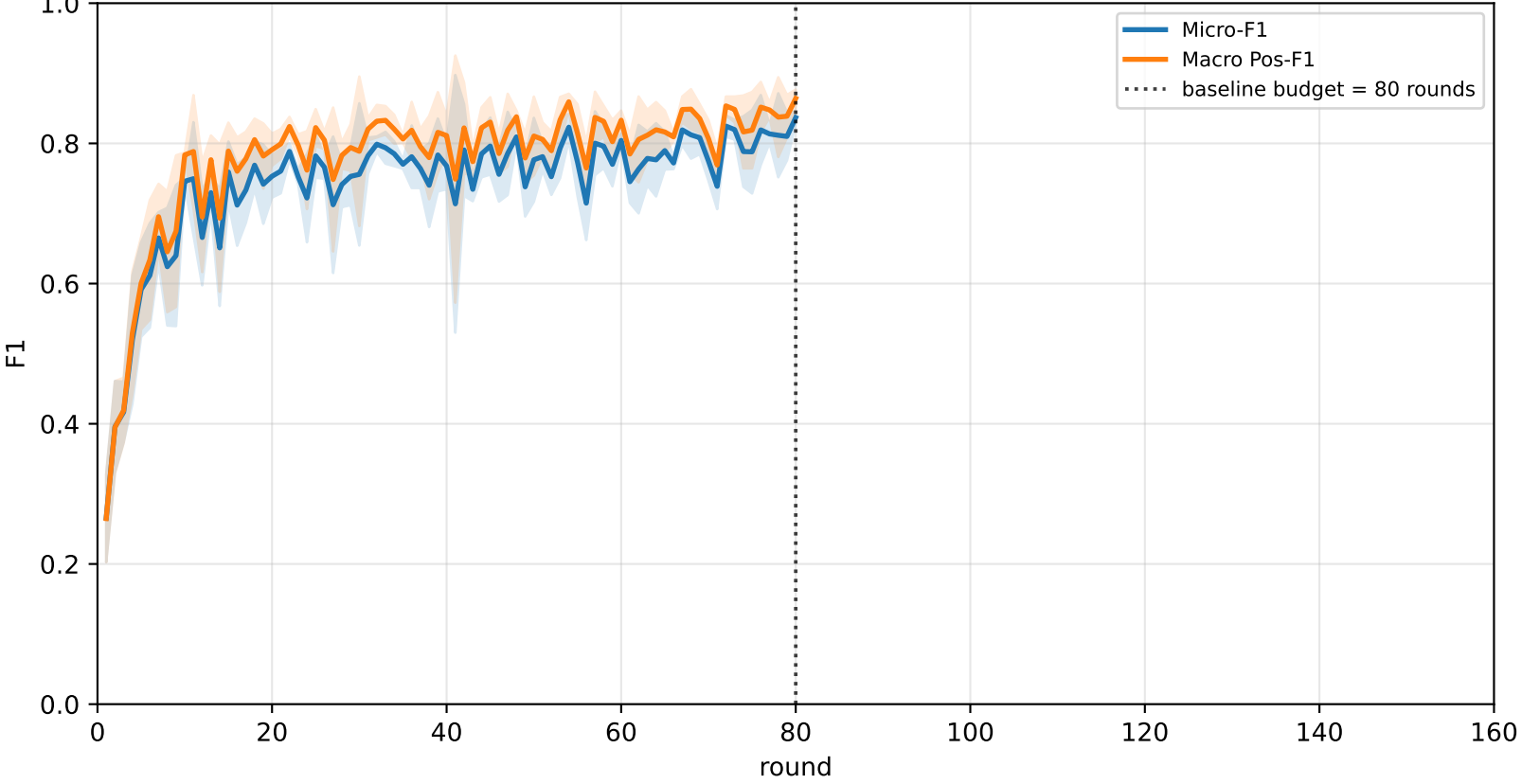
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	82.2% ± 0.8%	84.9% ± 0.9%	95.7% ± 1.8%	95.1% ± 0.8%	86.8% ± 3.3%	80.6% ± 1.1%	66.1% ± 1.3%
APPLE	79.1% ± 3.0%	81.9% ± 2.6%	94.9% ± 1.9%	90.8% ± 1.6%	81.7% ± 3.3%	77.9% ± 4.1%	64.1% ± 5.2%

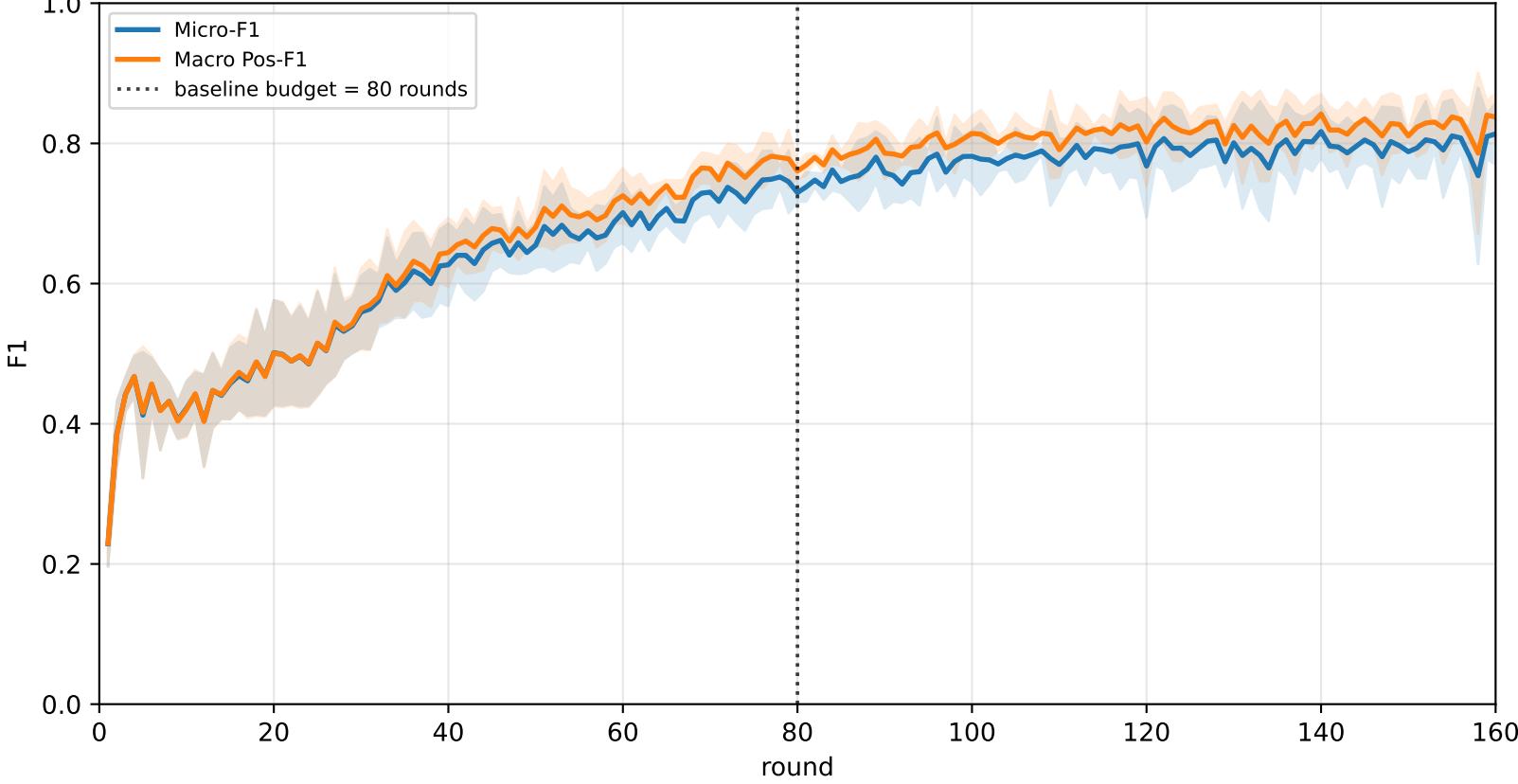
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



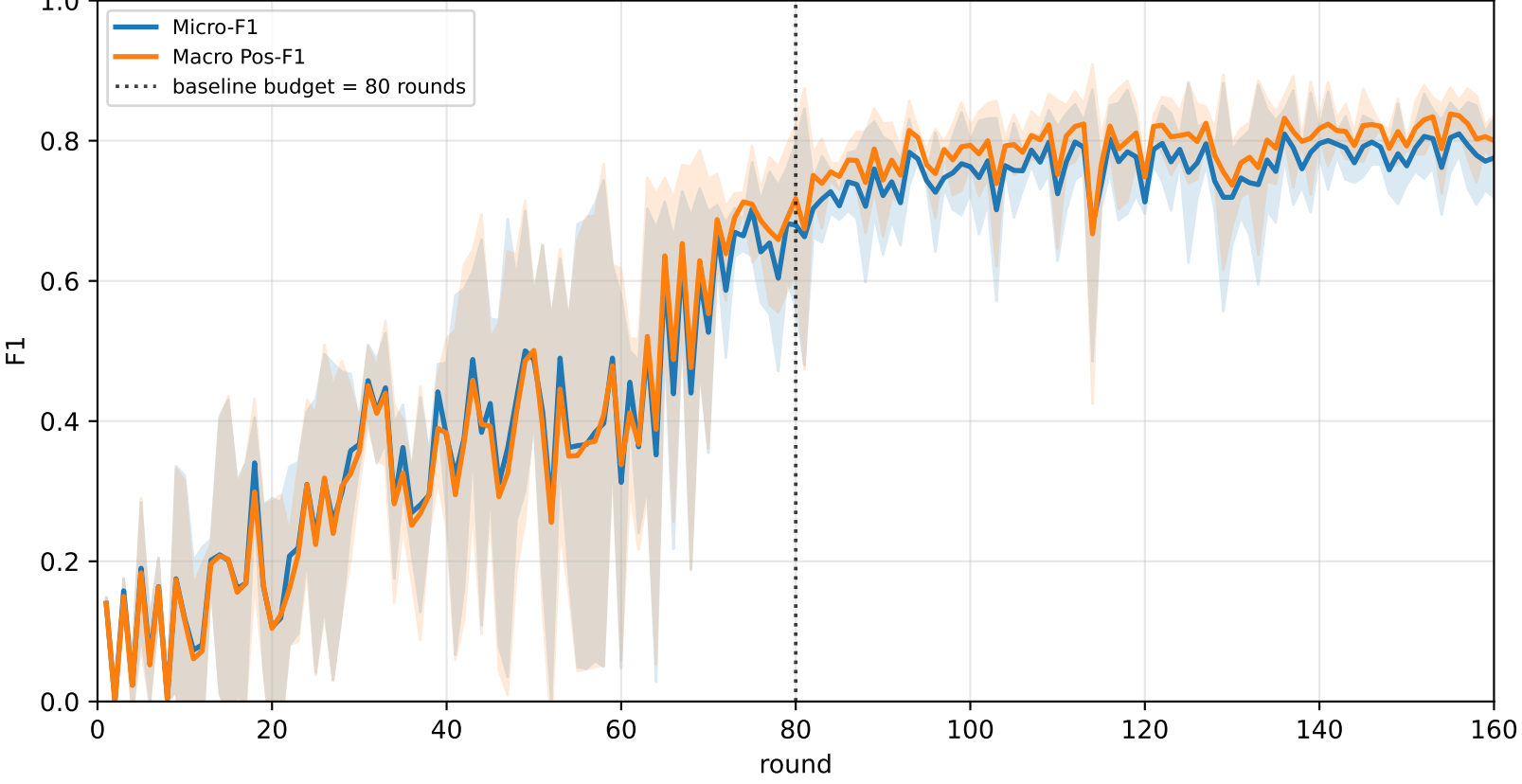
Pooled local centralized micro / macro F1



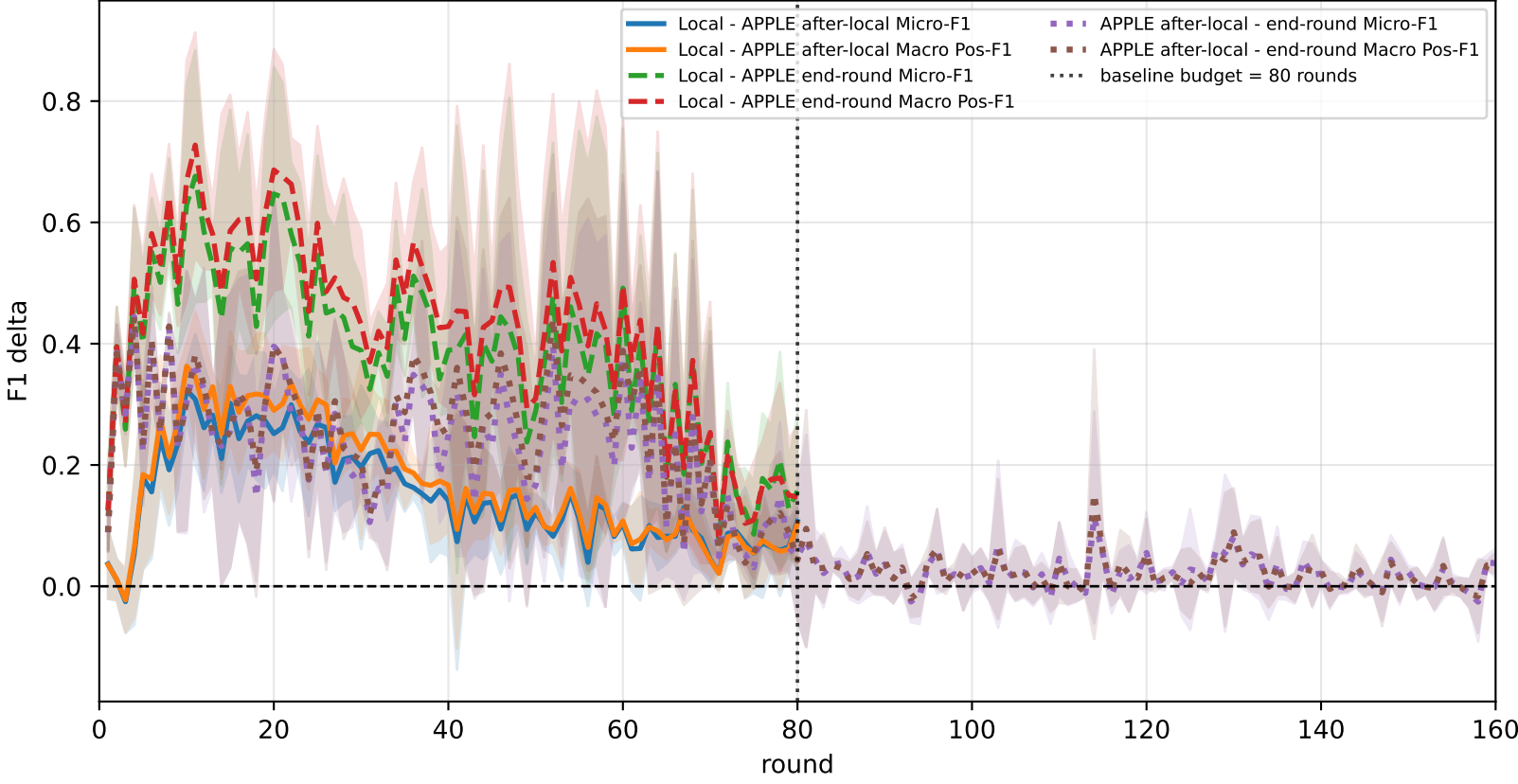
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



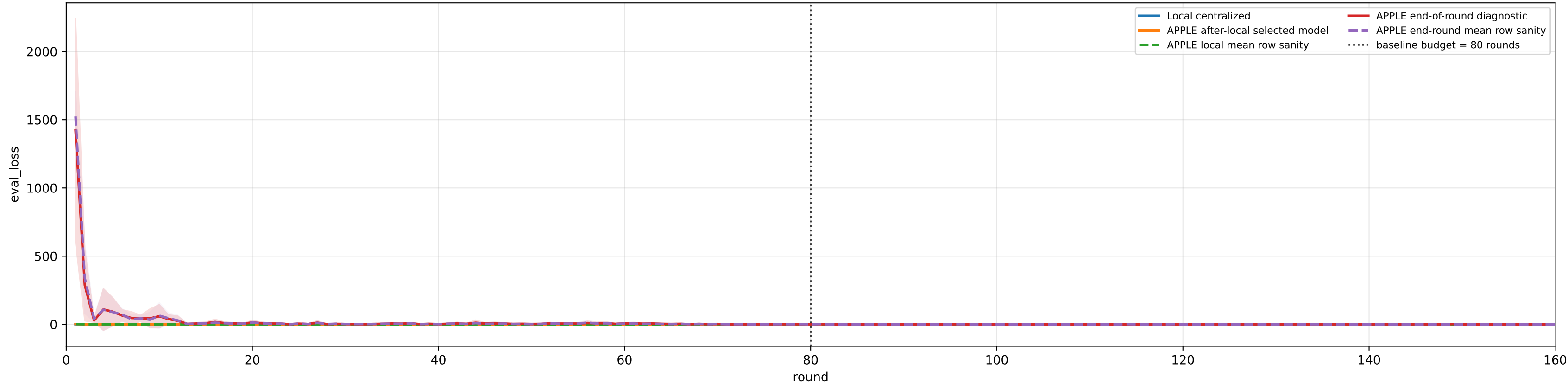
Controlled task-specialization level: Mid | APPLE diagnostics | family=specialized_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle6 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3% | specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

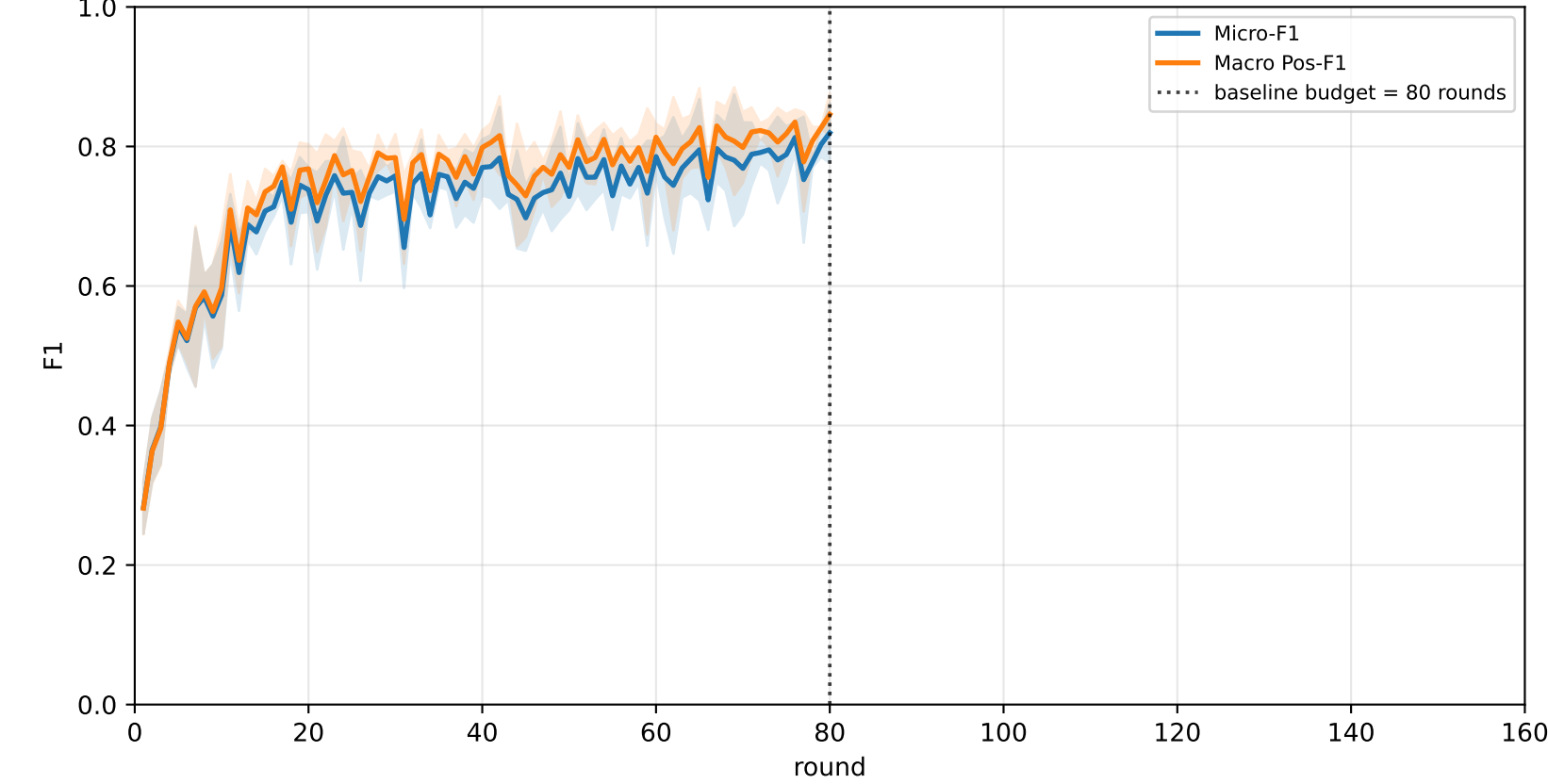
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.3% ± 2.7%	83.7% ± 2.0%	95.6% ± 1.4%	93.8% ± 1.7%	84.2% ± 3.5%	76.6% ± 4.7%	68.5% ± 6.6%
APPLE	80.6% ± 3.2%	82.8% ± 2.4%	95.0% ± 1.8%	91.5% ± 0.3%	80.7% ± 3.4%	77.1% ± 3.9%	69.9% ± 8.2%

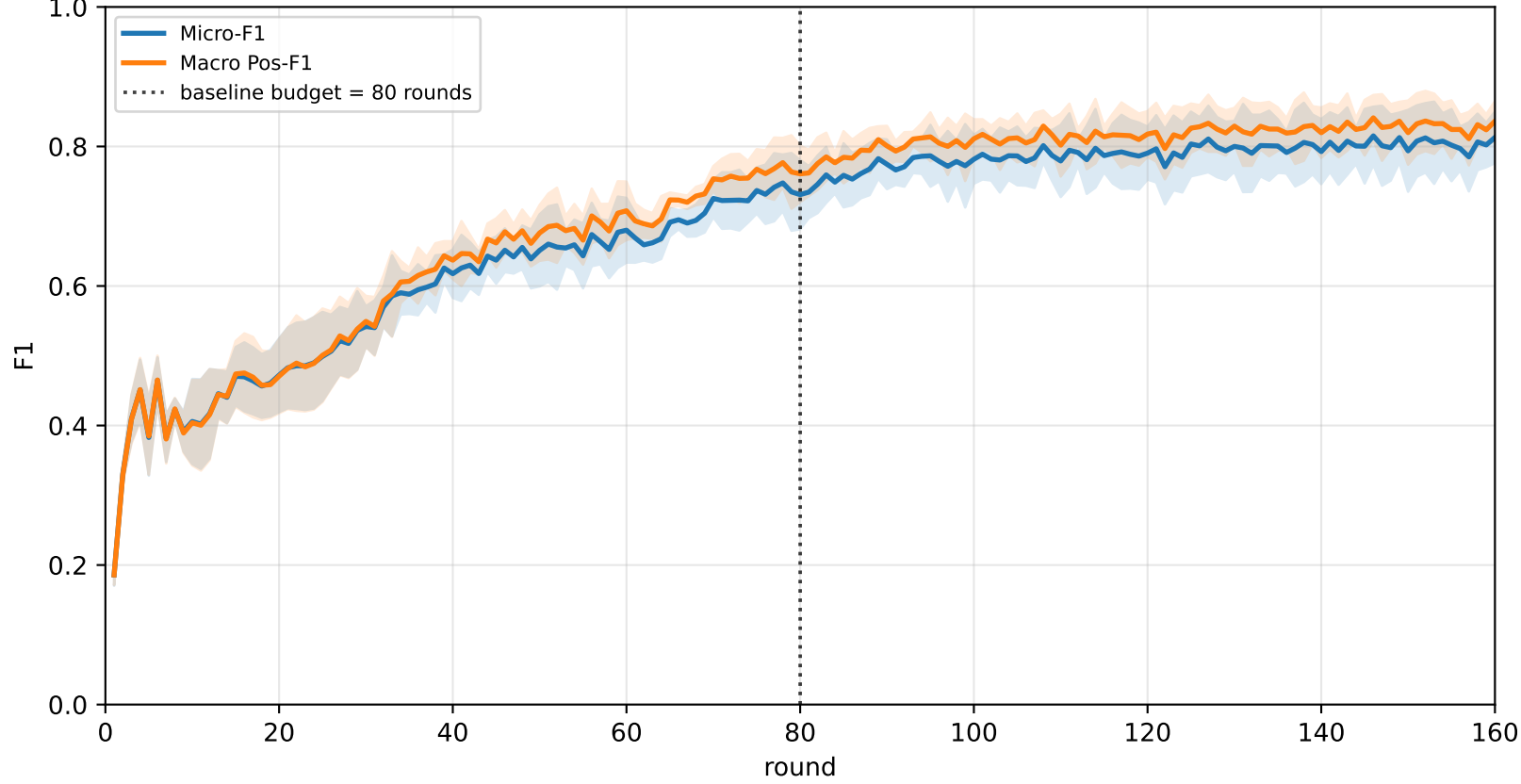
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



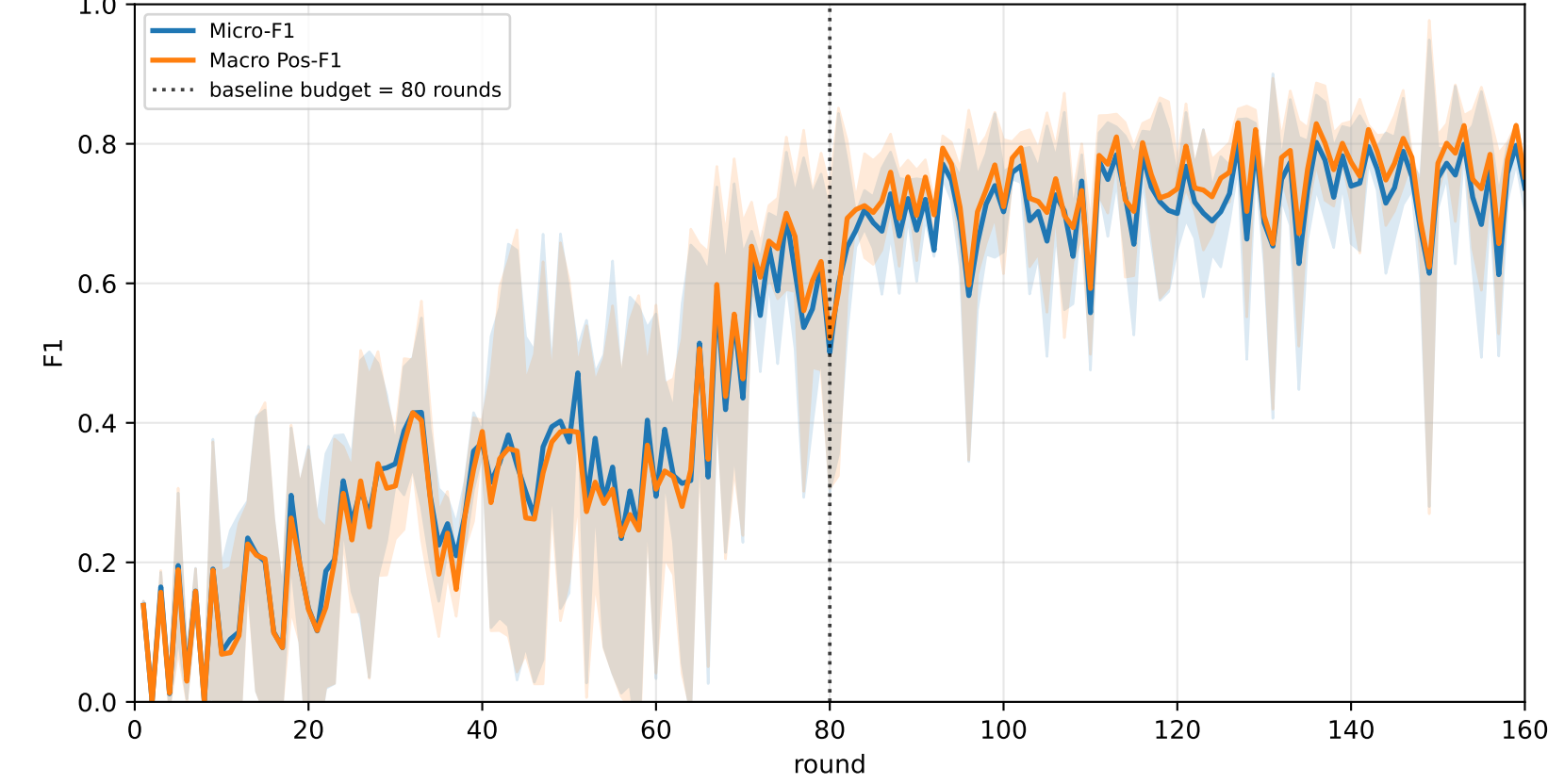
Pooled local centralized micro / macro F1



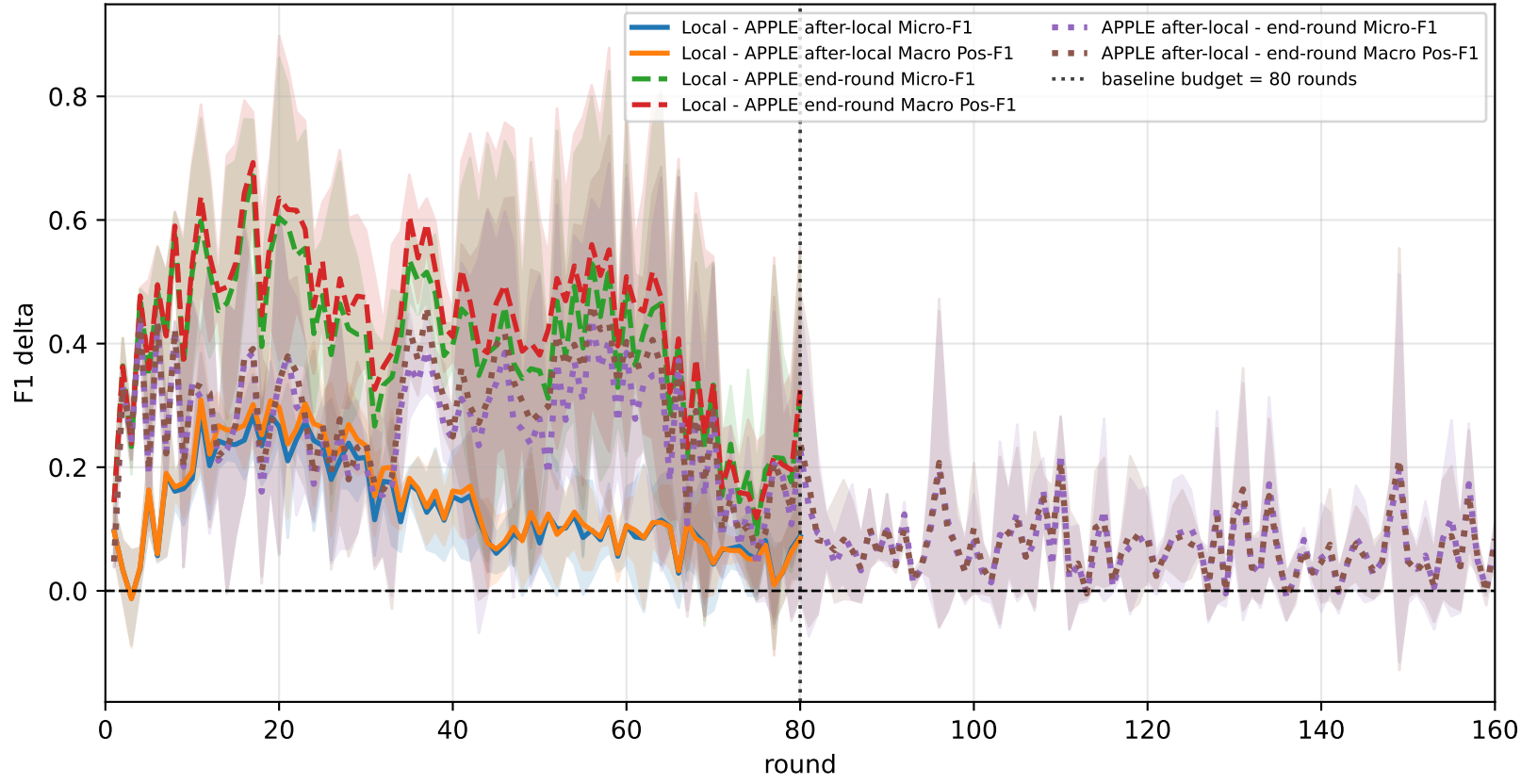
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



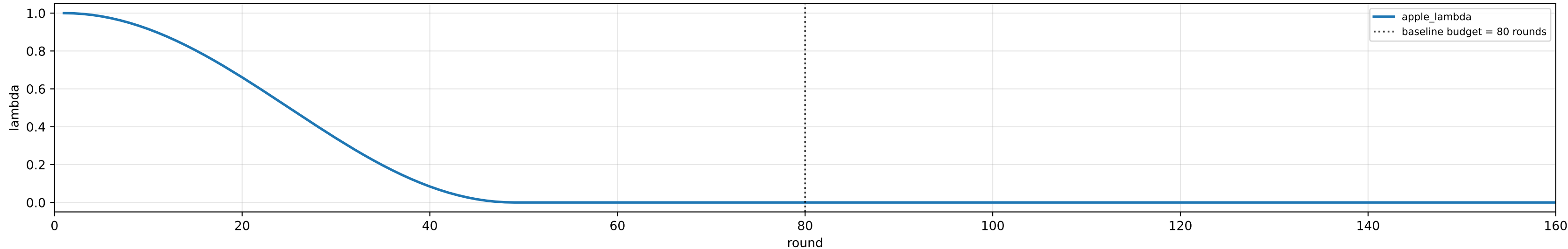
Delta diagnostics: local gap and APPLE mismatch gap



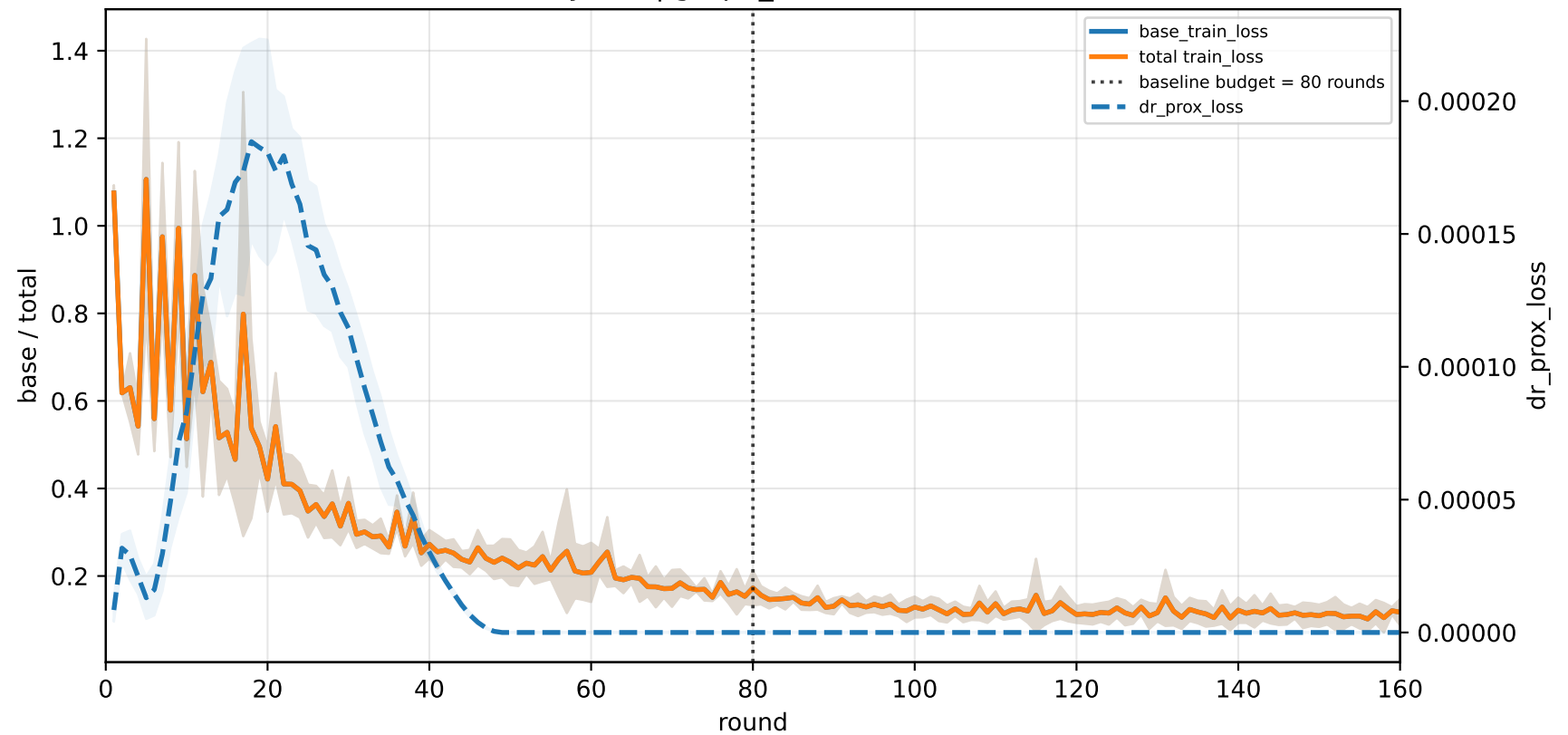
APPLE training-loss decomposition | Mid heterogeneity

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

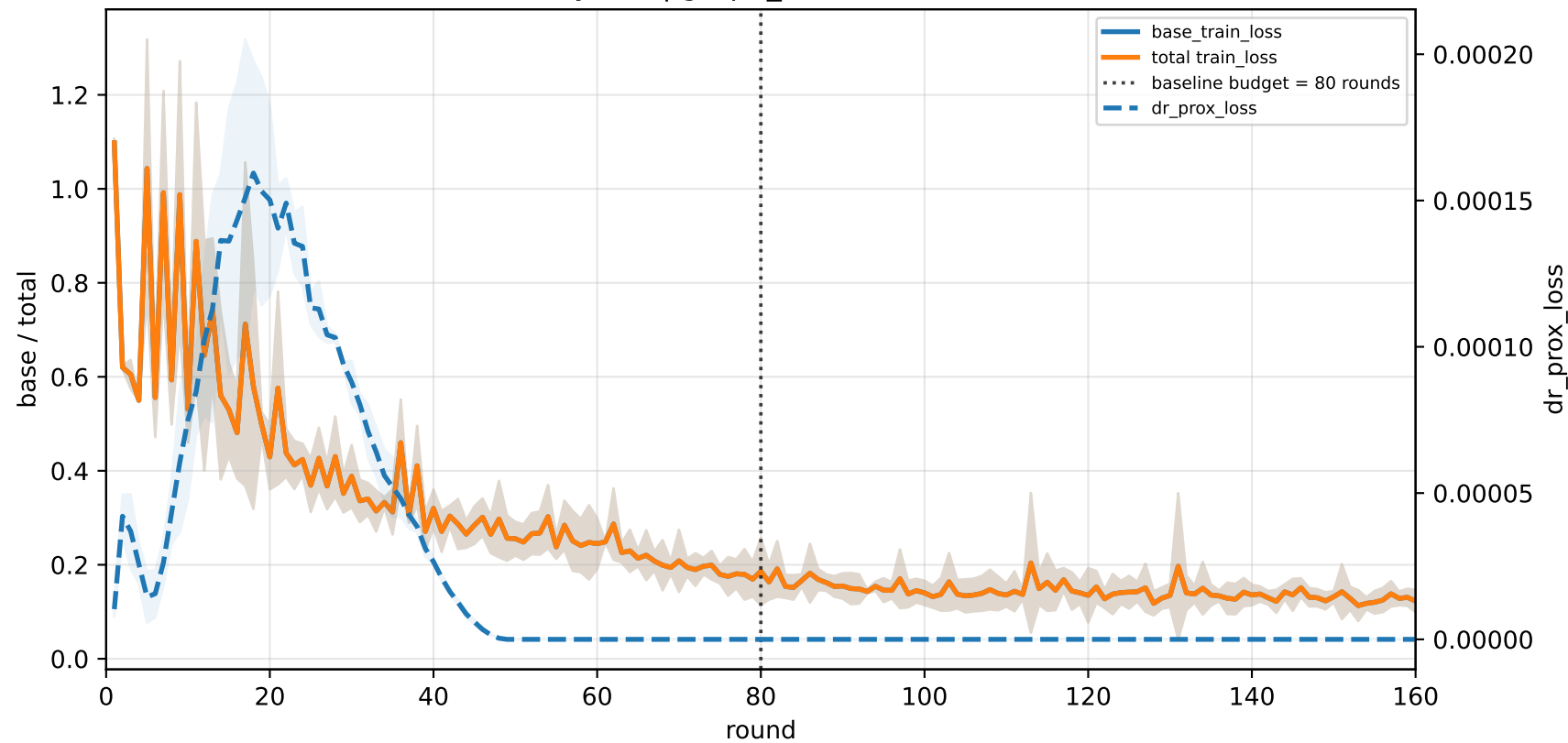
APPLE lambda scheduler | apple_mu=0.01



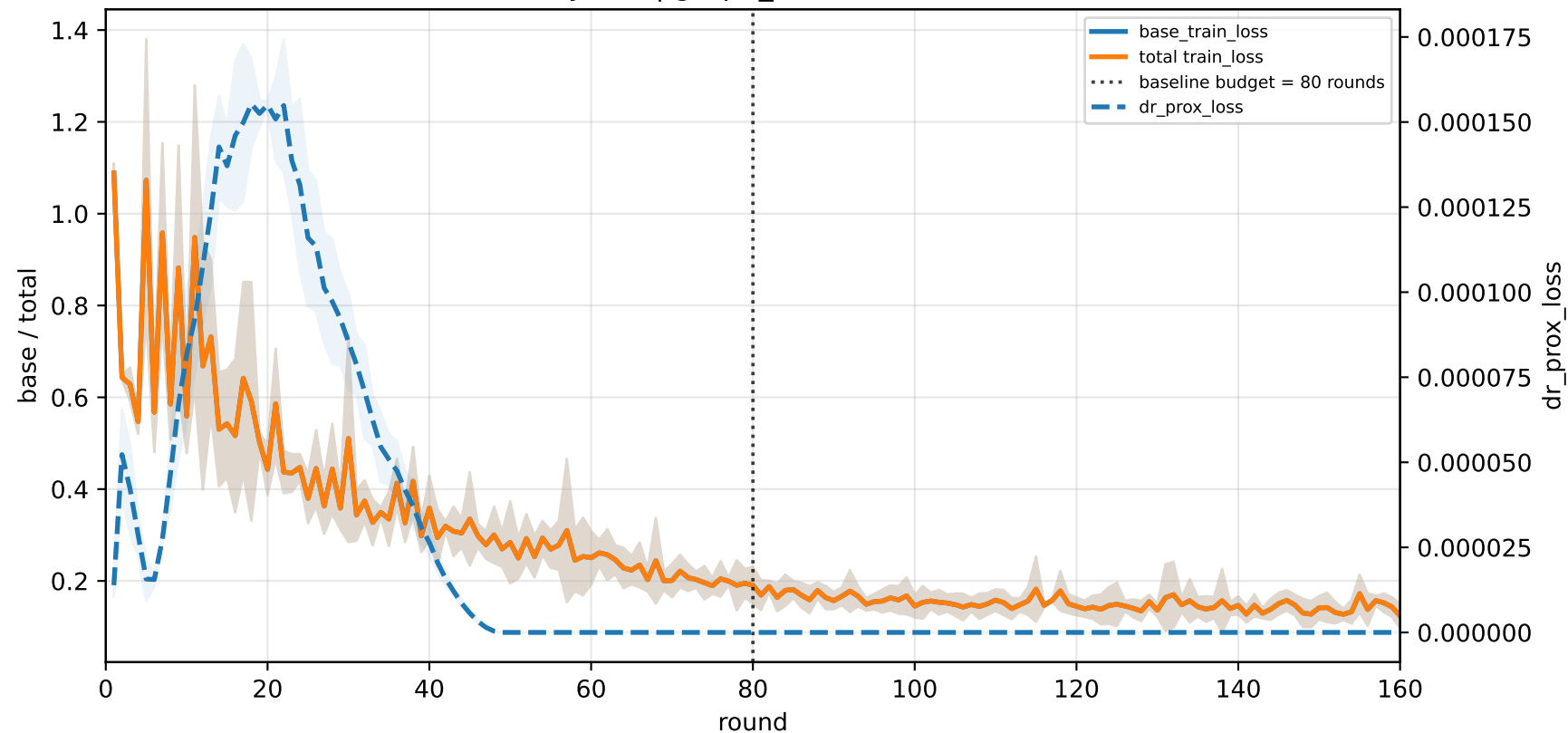
cycle2 | graph_id=2000005



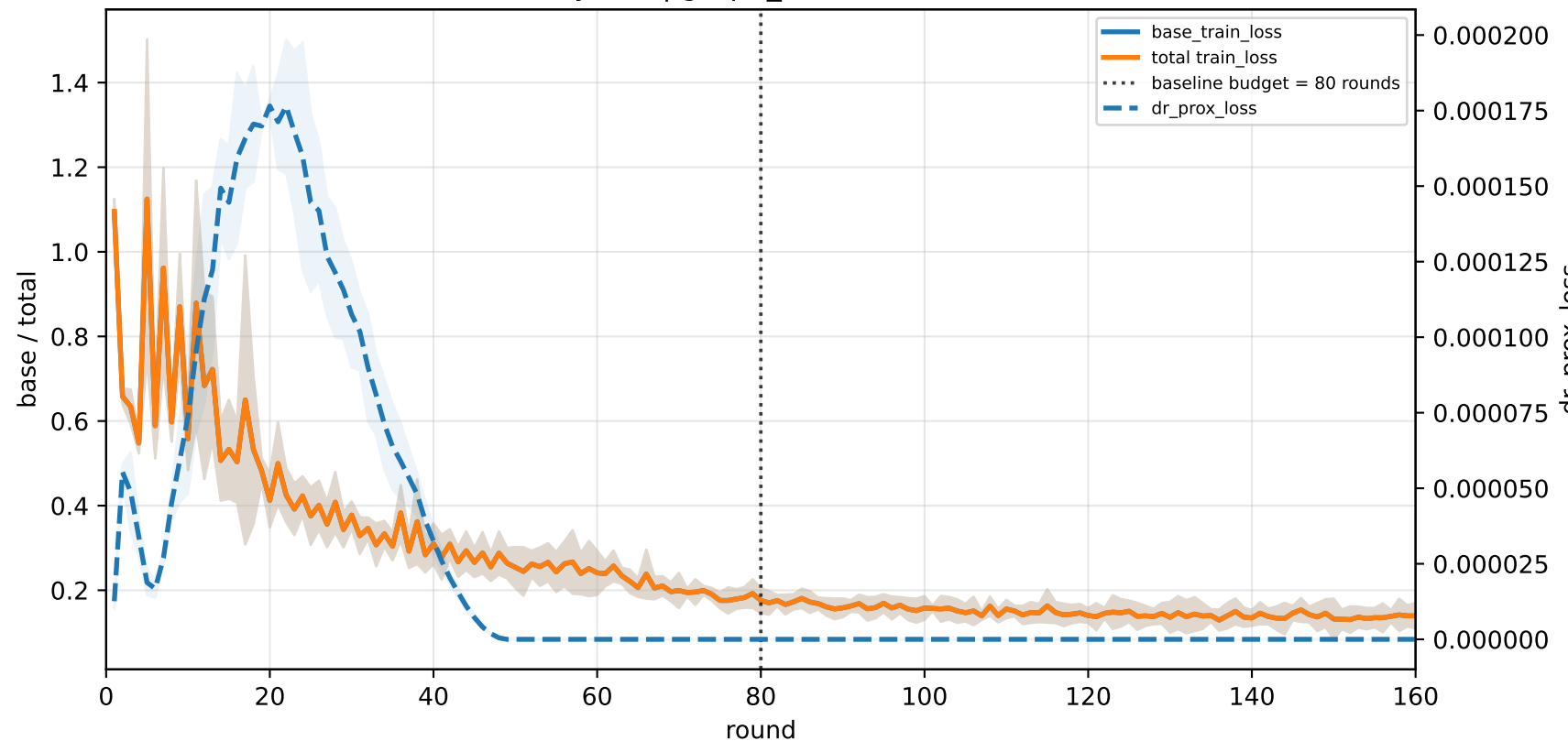
cycle3 | graph_id=2000006



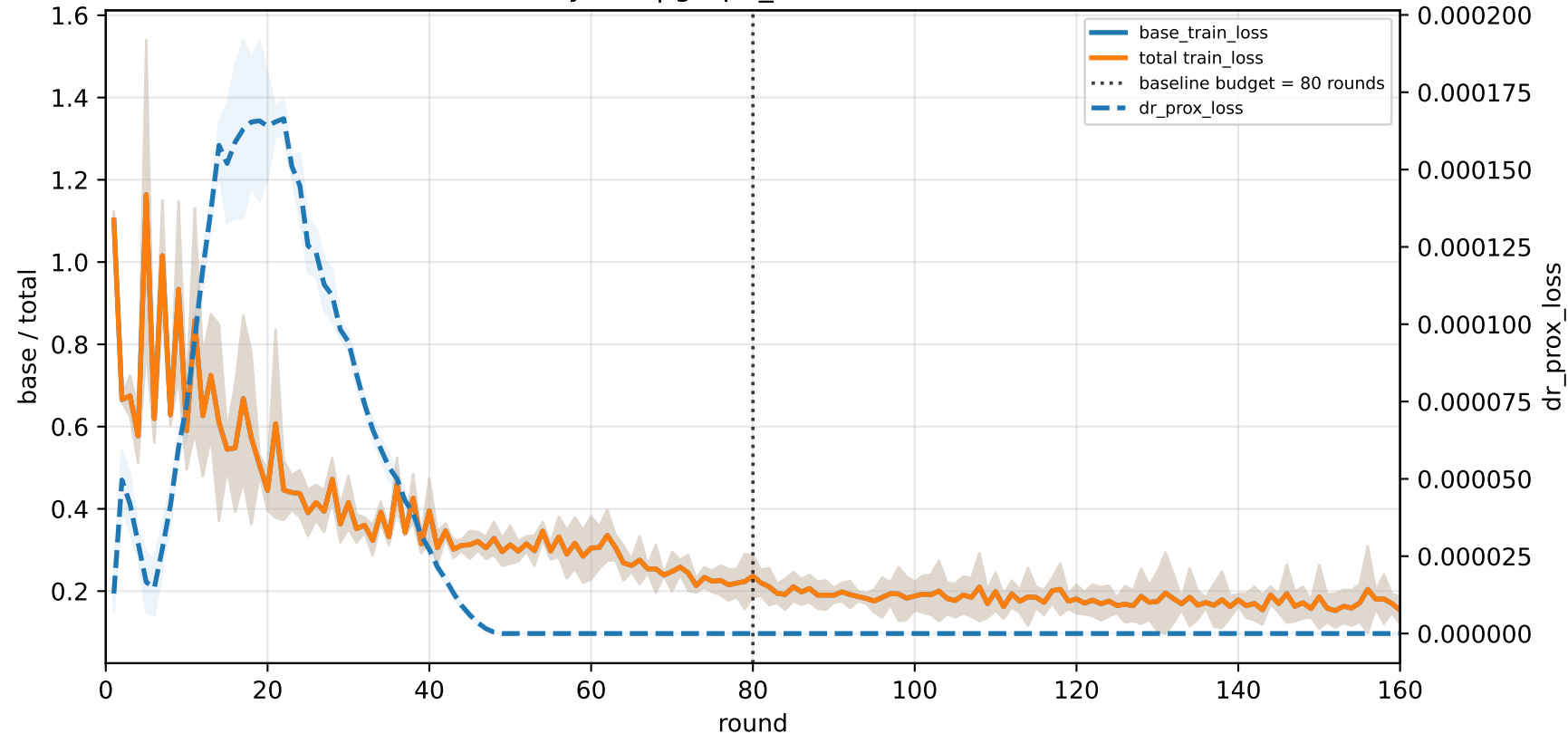
cycle4 | graph_id=2000007



cycle5 | graph_id=2000008



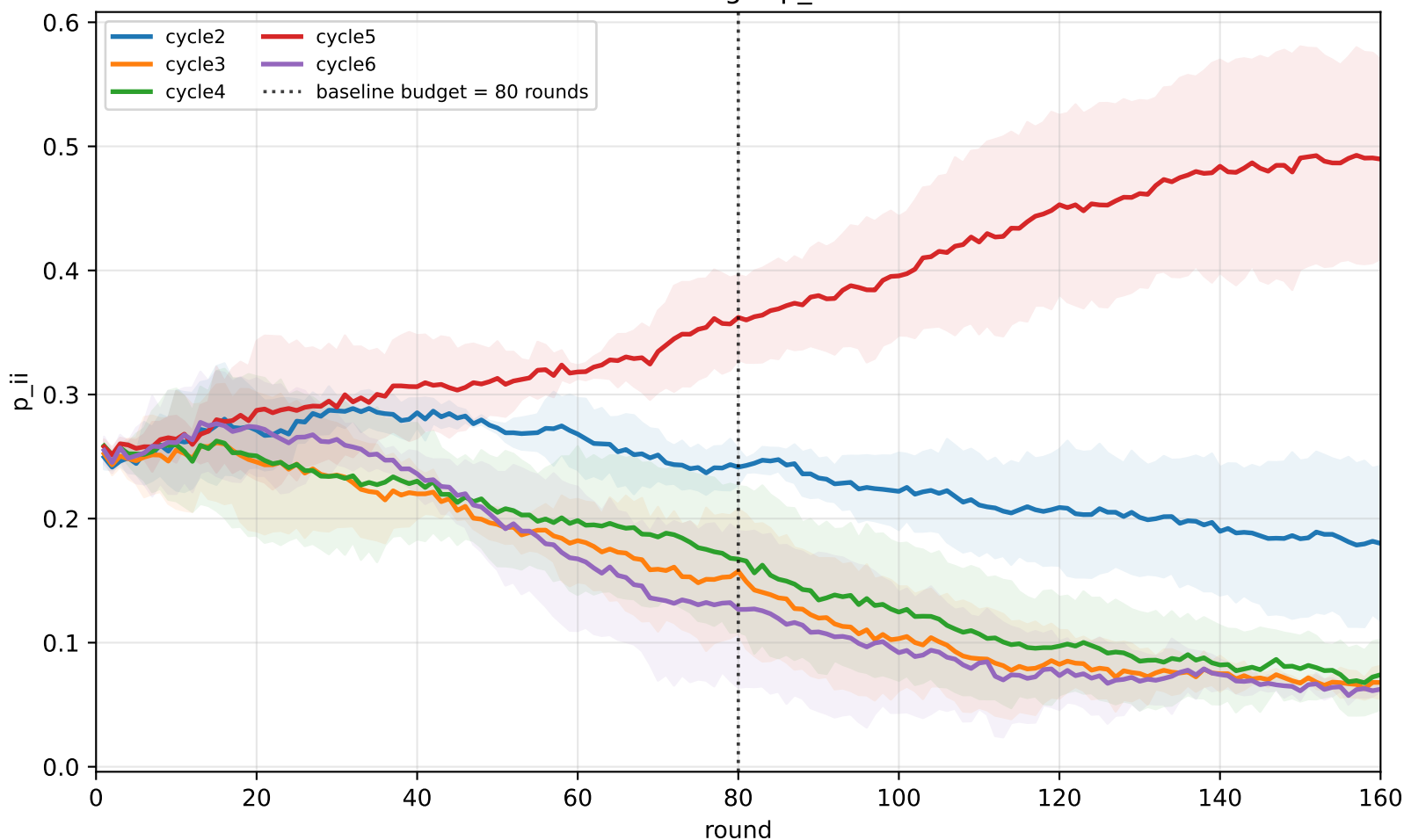
cycle6 | graph_id=2000009



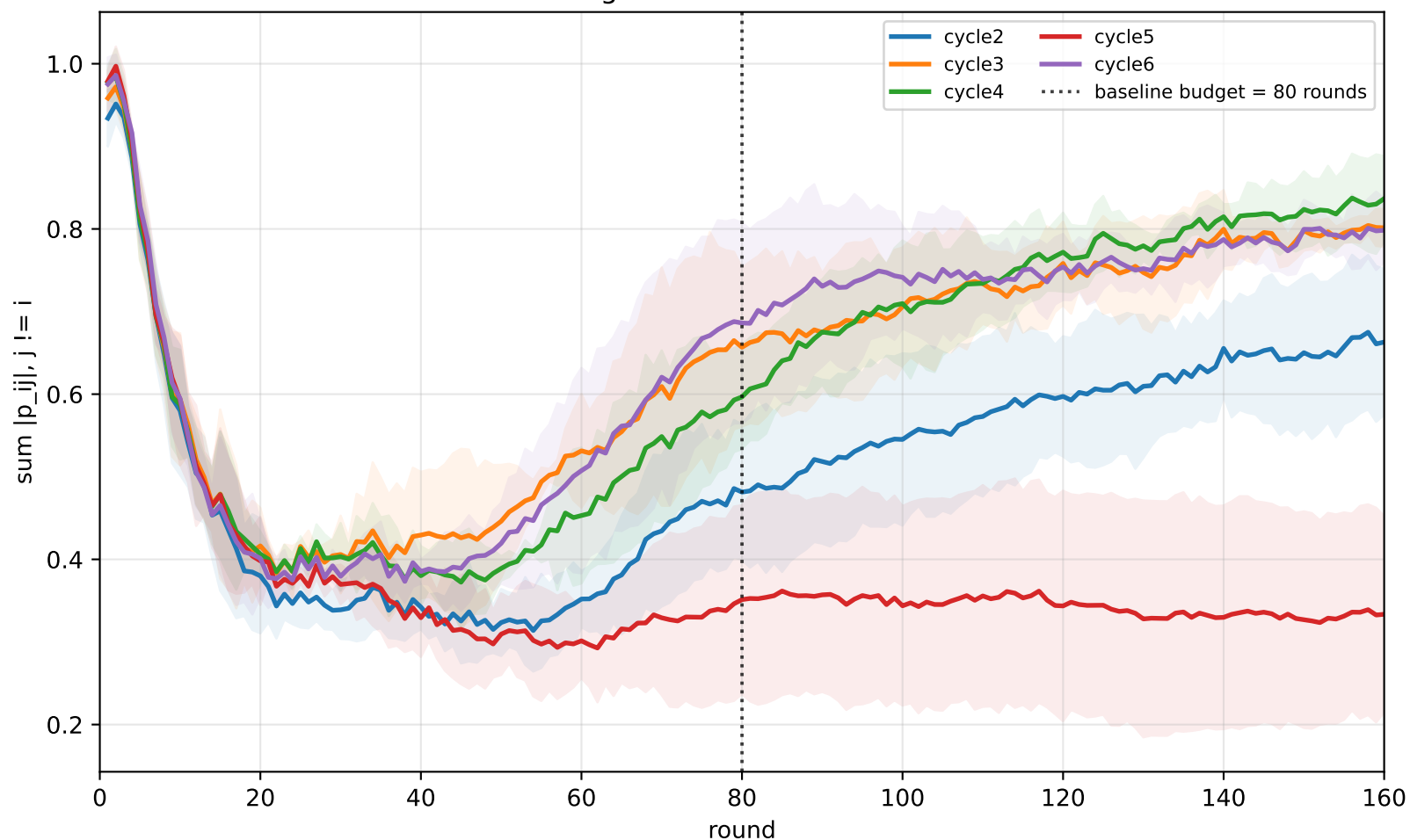
APPLE DR behavior | Mid heterogeneity

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 | family_centroid_jsd_mean=0.028

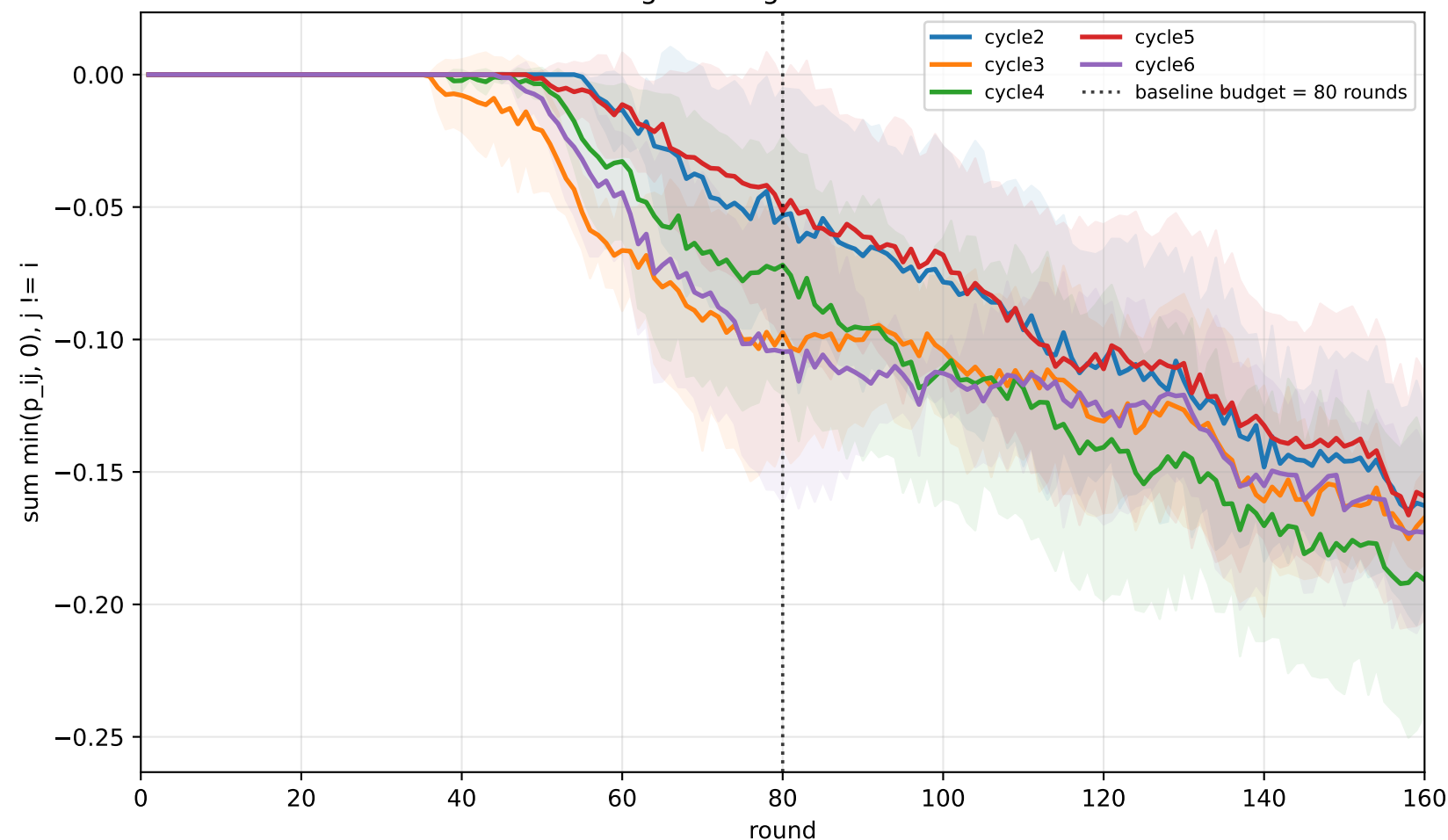
APPLE DR self-weight p_{ii} over rounds



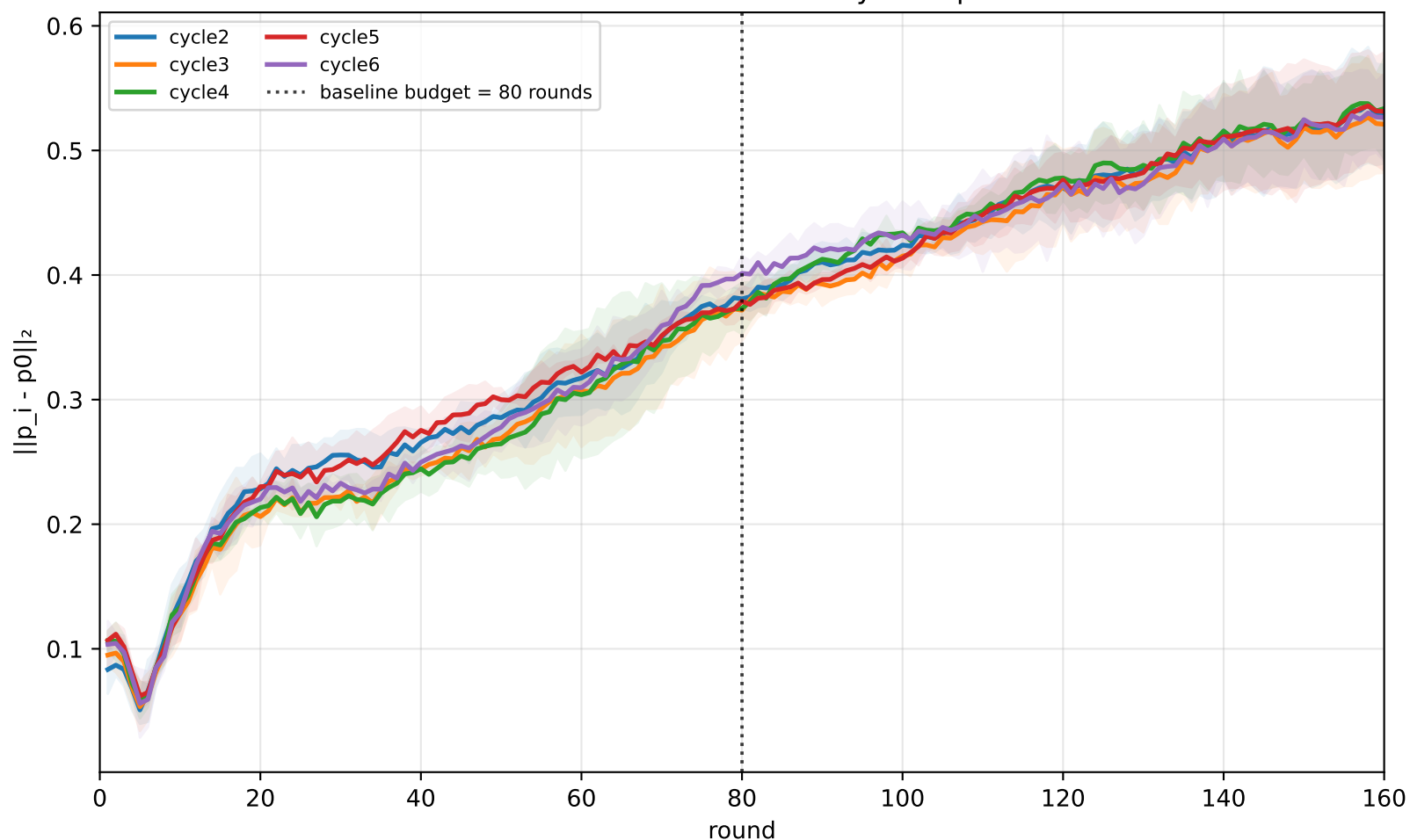
APPLE off-diagonal absolute mass over rounds



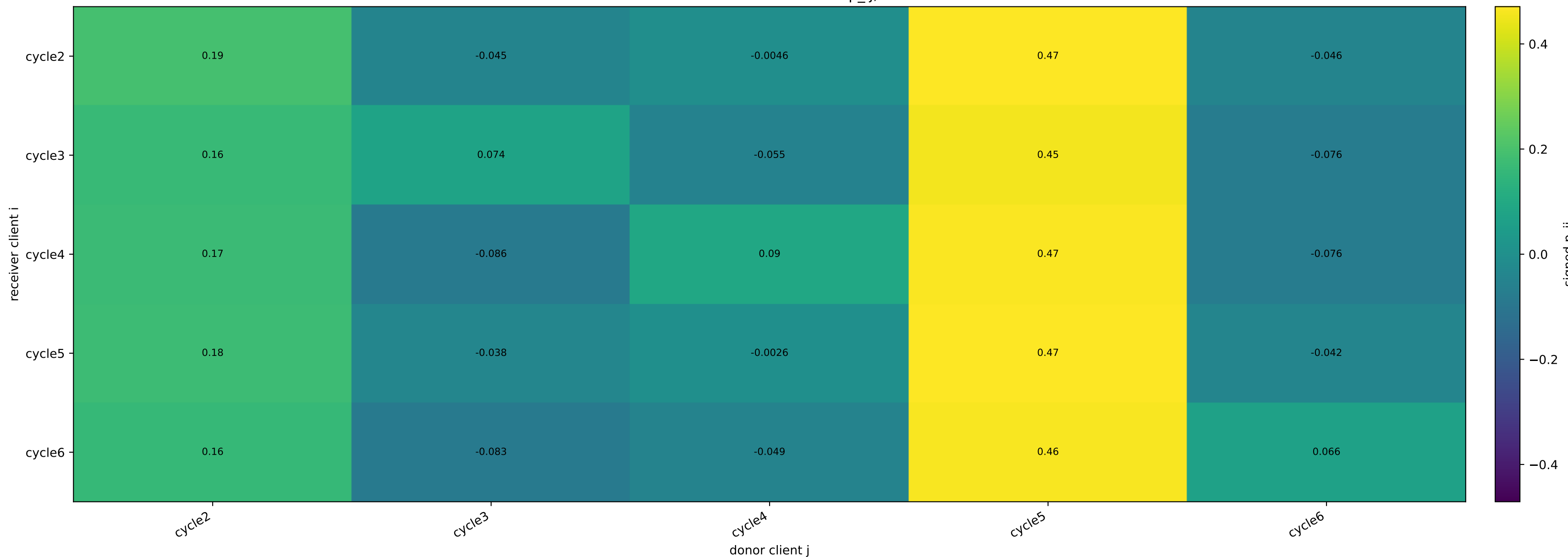
APPLE off-diagonal negative mass over rounds



APPLE DR movement away from p_0



Selected-best-round APPLE DR coefficients p_{ij} , mean over seeds



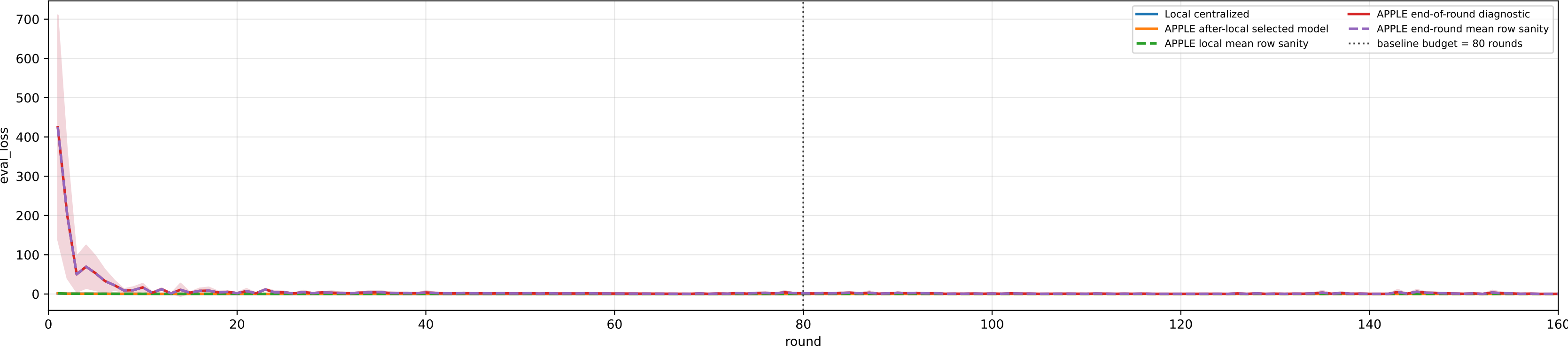
Controlled task-specialization level: High | APPLE diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 | family_centroid_jsd_mean=0.162

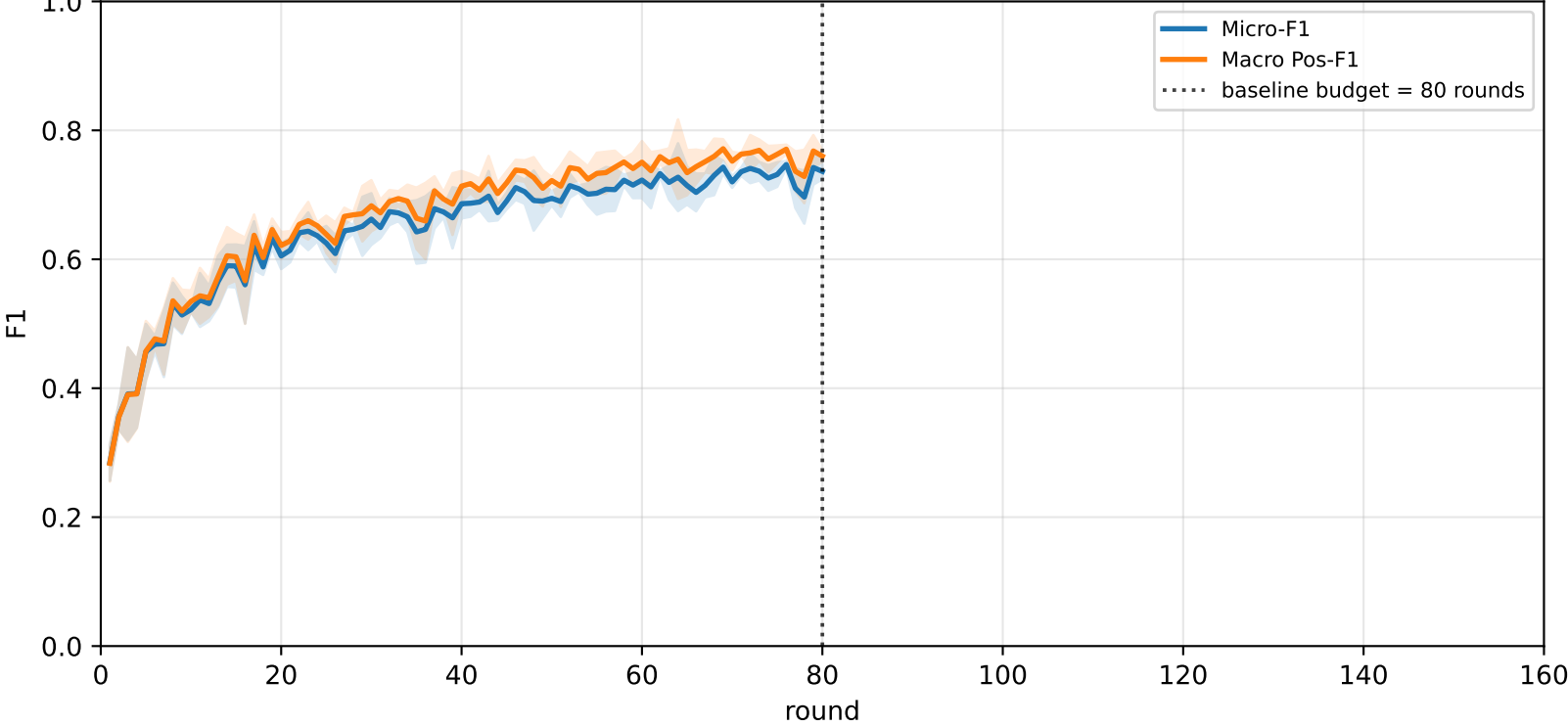
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	74.7% ± 1.4%	78.7% ± 1.0%	91.5% ± 2.6%	89.1% ± 0.9%	80.9% ± 1.9%	68.6% ± 3.9%	58.8% ± 1.7%
APPLE	76.0% ± 3.2%	79.0% ± 2.8%	92.5% ± 1.3%	87.2% ± 2.2%	78.7% ± 3.0%	73.5% ± 3.2%	61.3% ± 4.5%

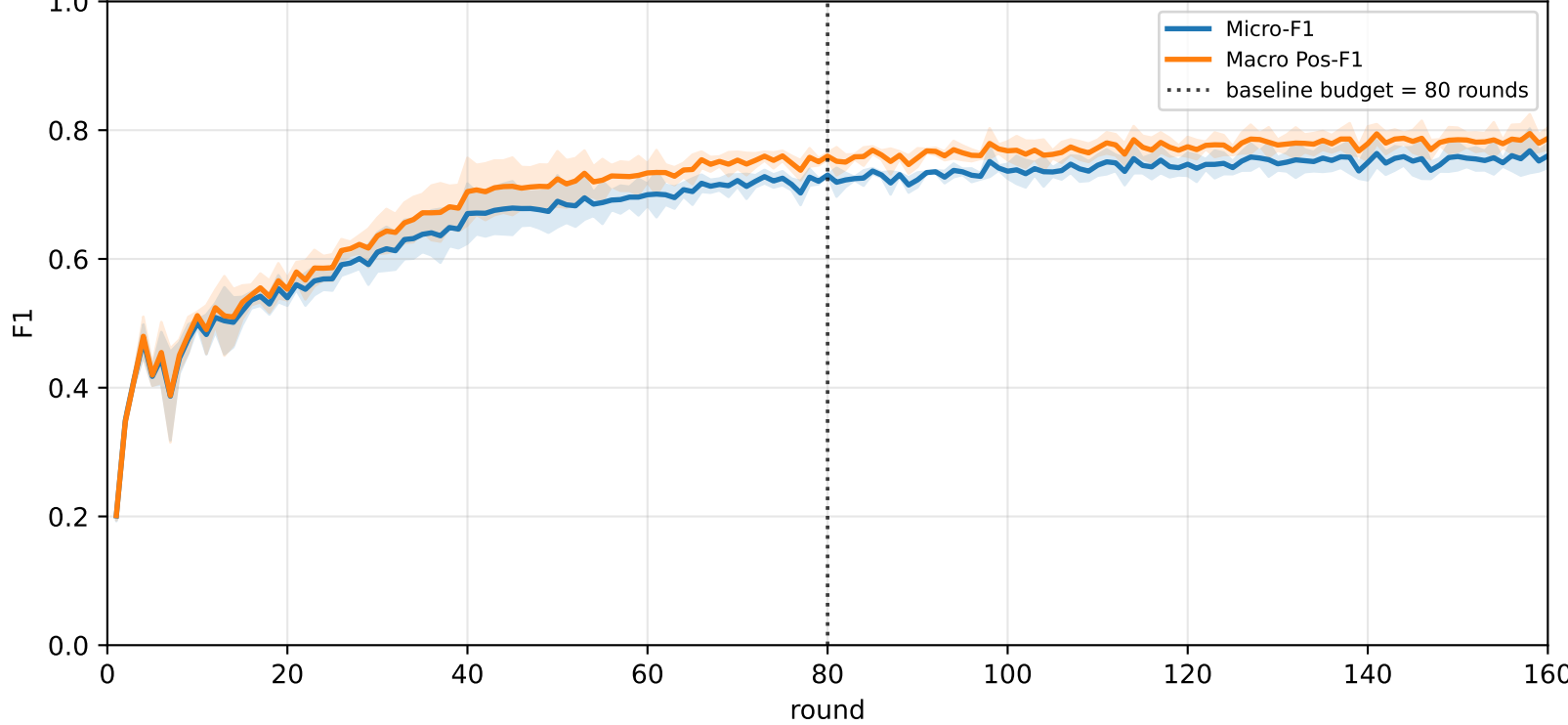
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



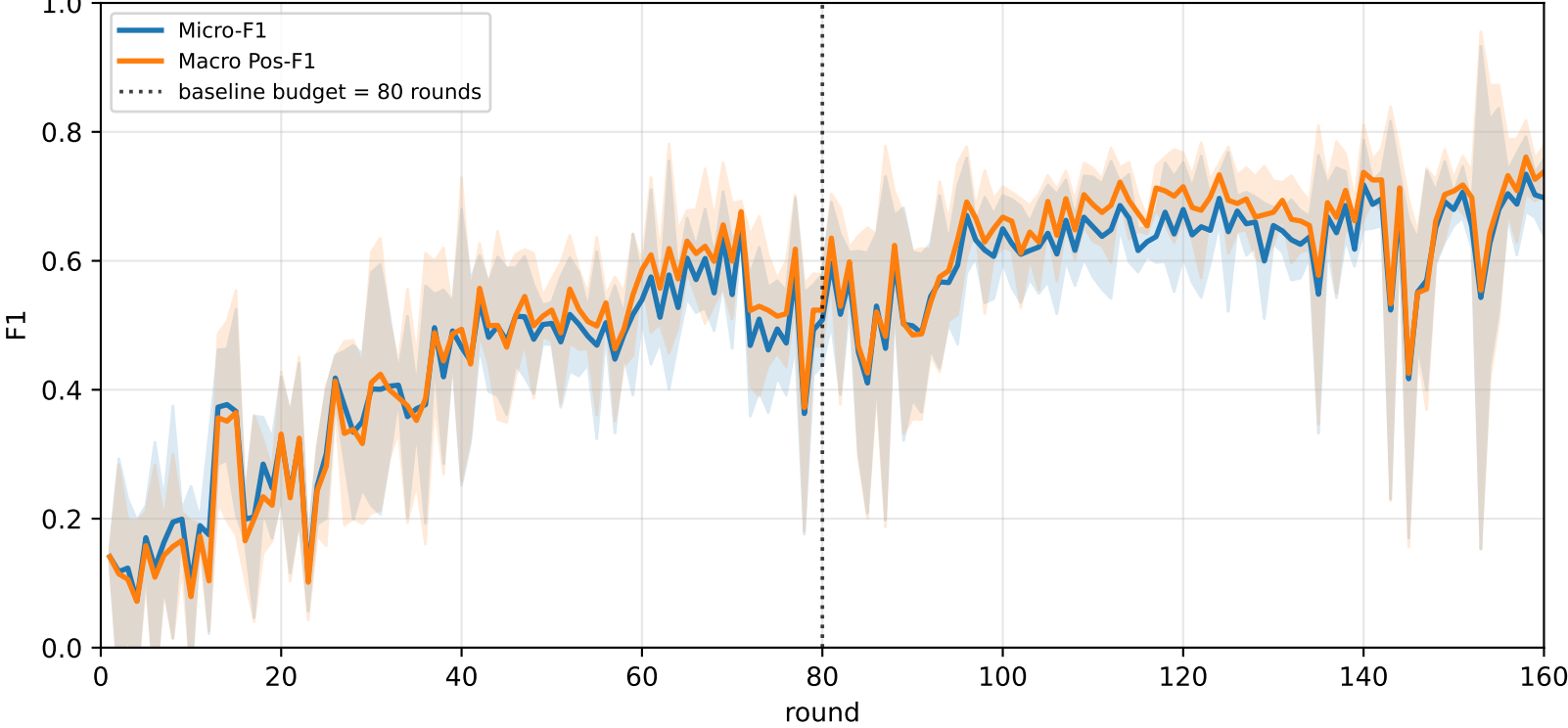
Pooled local centralized micro / macro F1



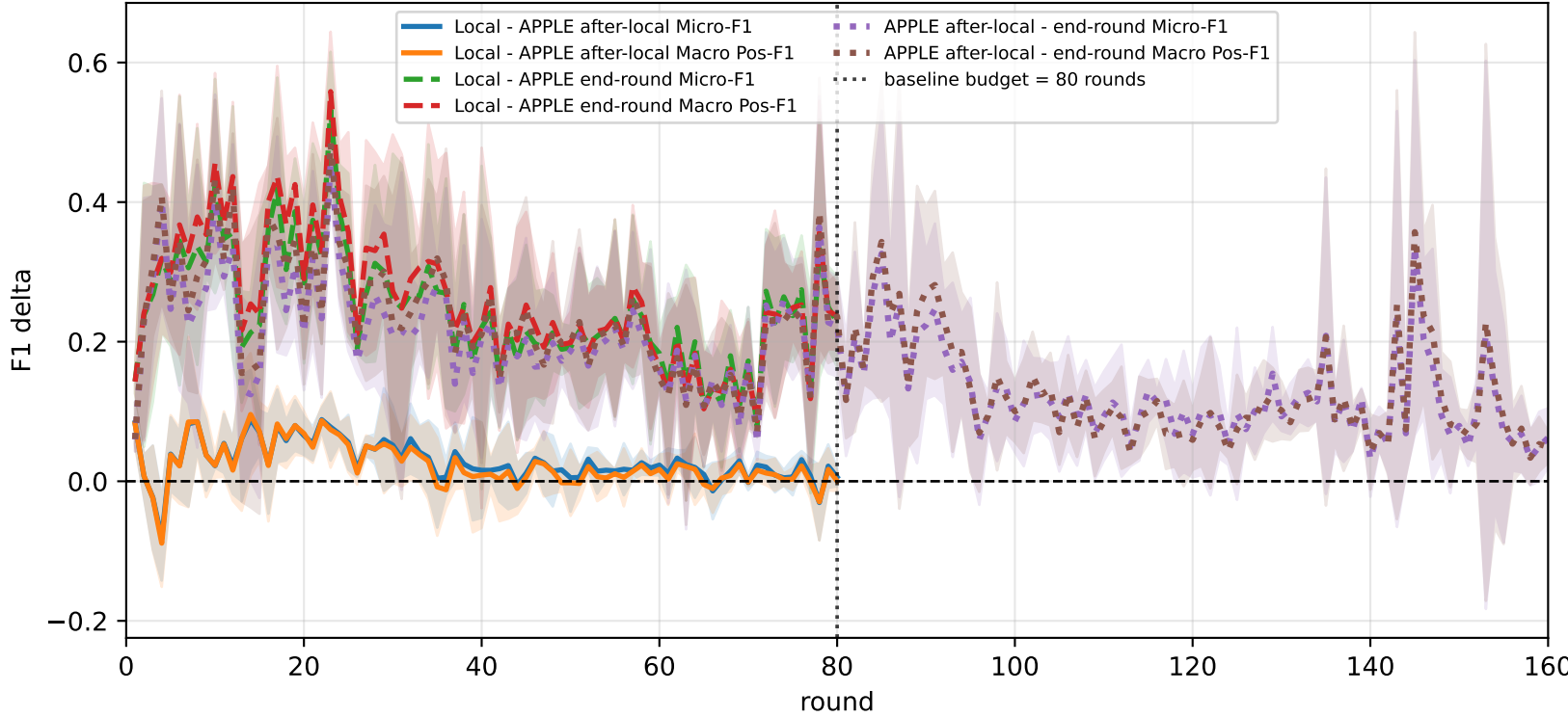
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



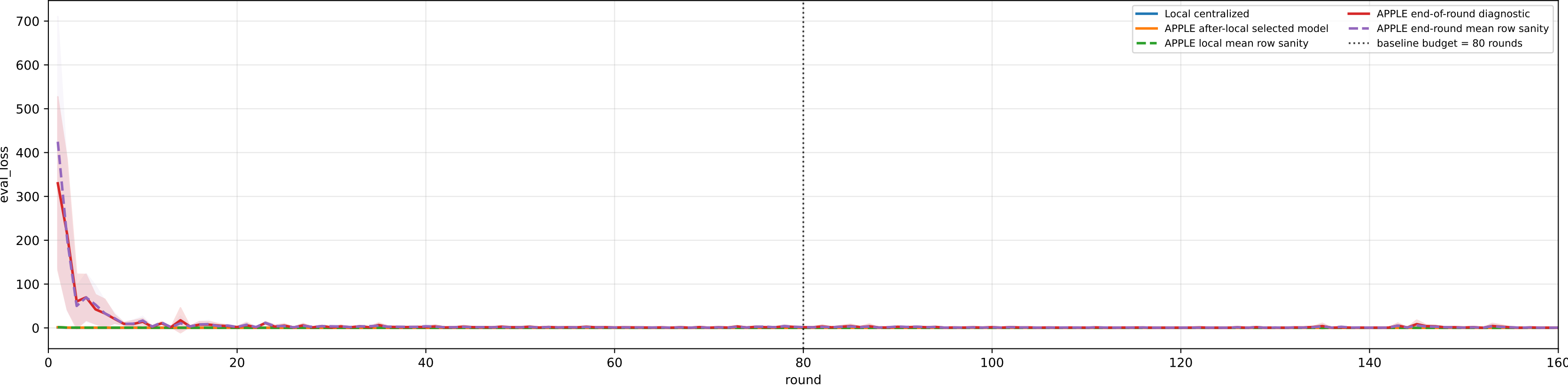
Controlled task-specialization level: High | APPLE diagnostics | family=specialized_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 | family_centroid_jsd_mean=0.162

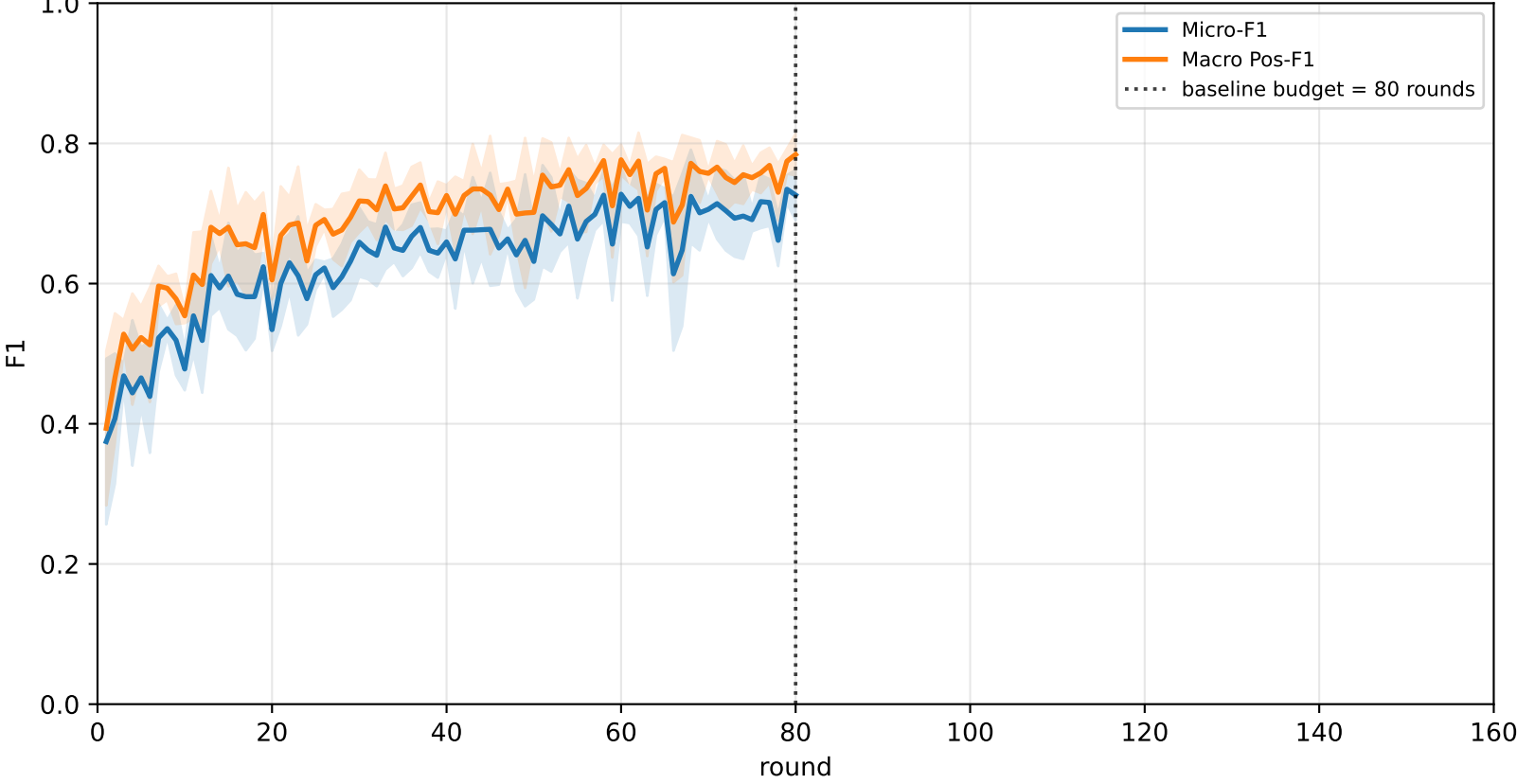
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	71.8% ± 2.2%	76.9% ± 0.8%	98.1% ± 0.8%	90.4% ± 2.4%	76.9% ± 4.5%	65.2% ± 5.3%	53.7% ± 4.1%
APPLE	77.6% ± 3.8%	81.1% ± 3.3%	96.9% ± 2.5%	90.9% ± 3.1%	82.0% ± 3.9%	75.6% ± 4.4%	60.0% ± 5.1%

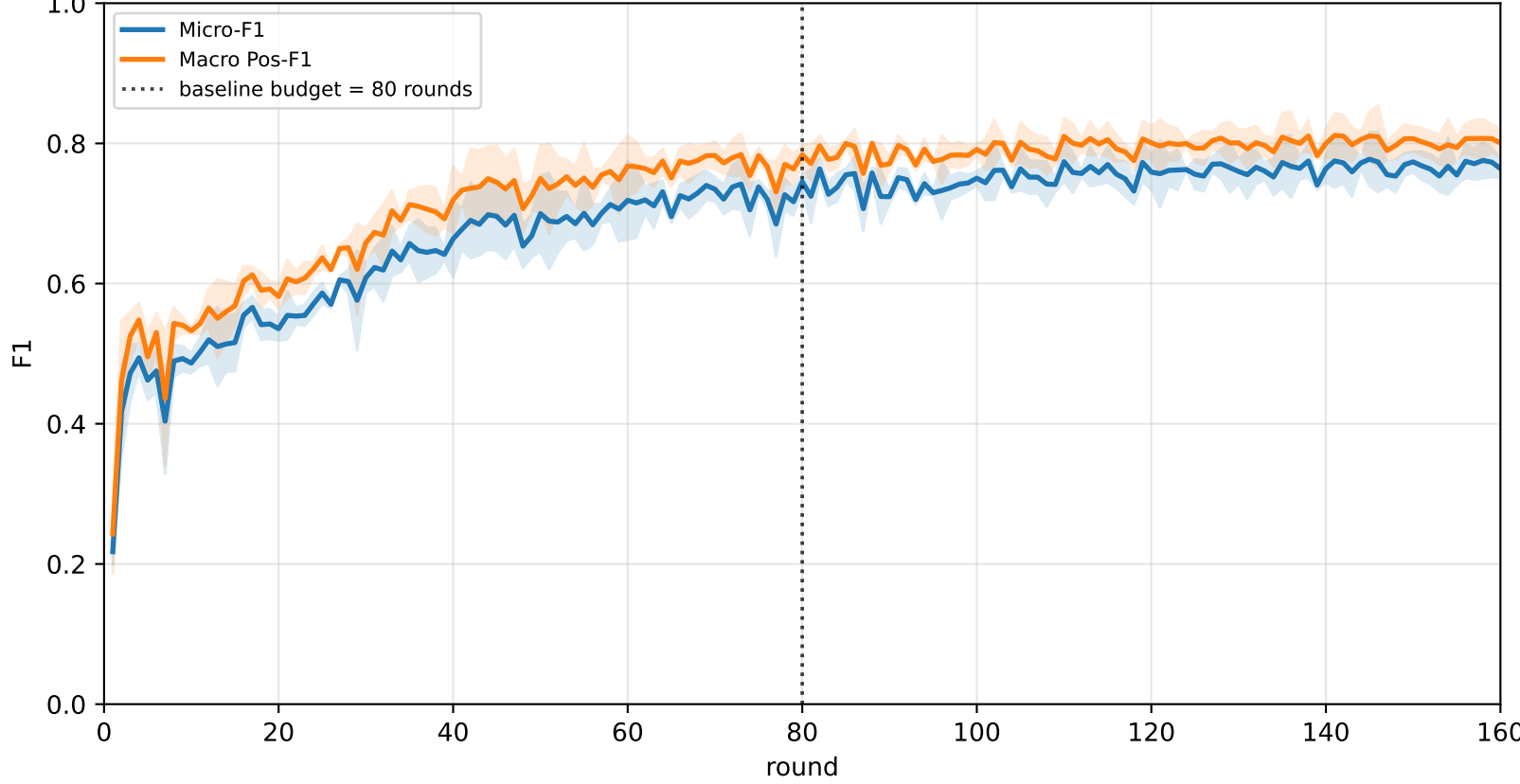
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



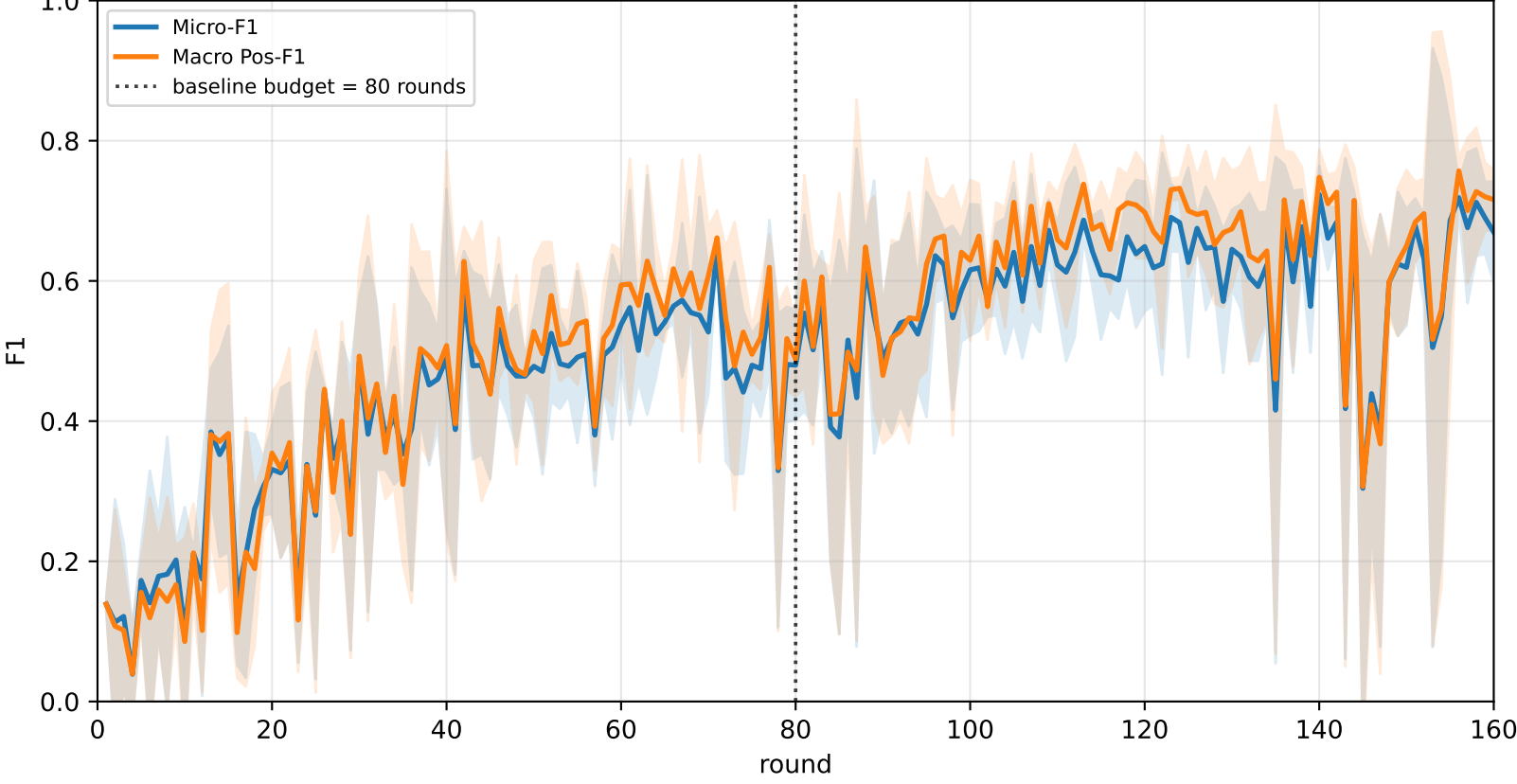
Pooled local centralized micro / macro F1



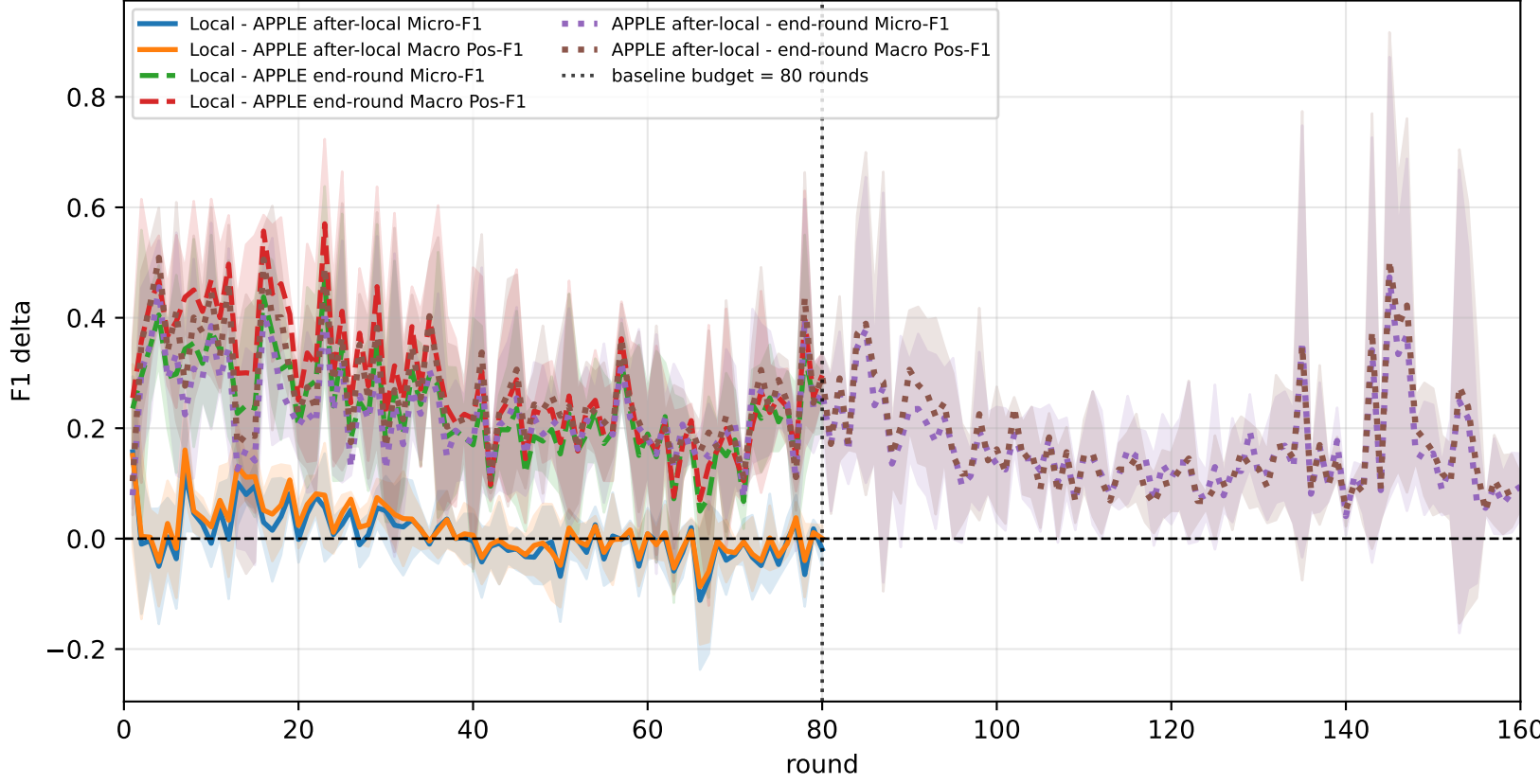
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



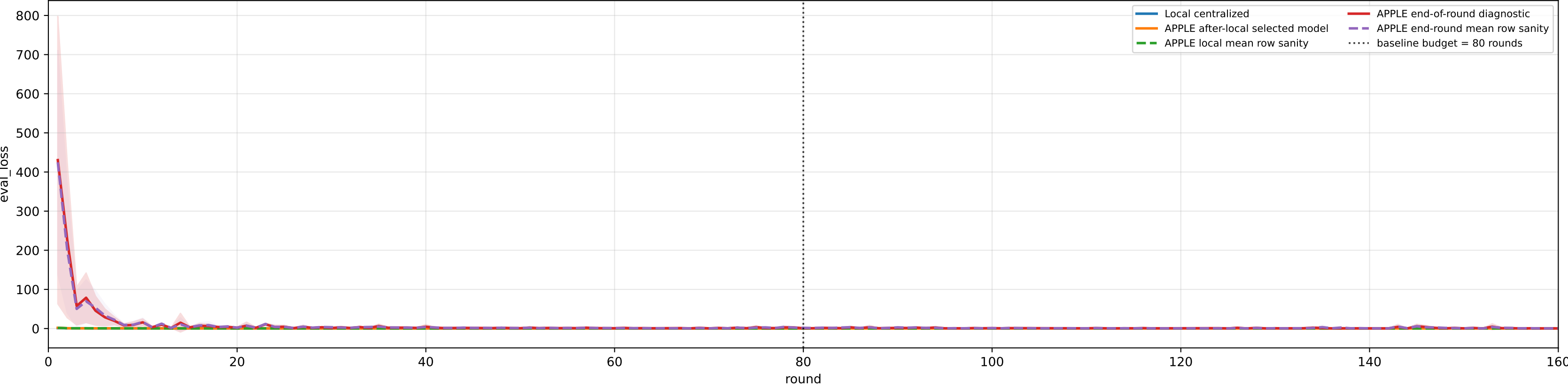
Controlled task-specialization level: High | APPLE diagnostics | family=specialized_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 | family_centroid_jsd_mean=0.162

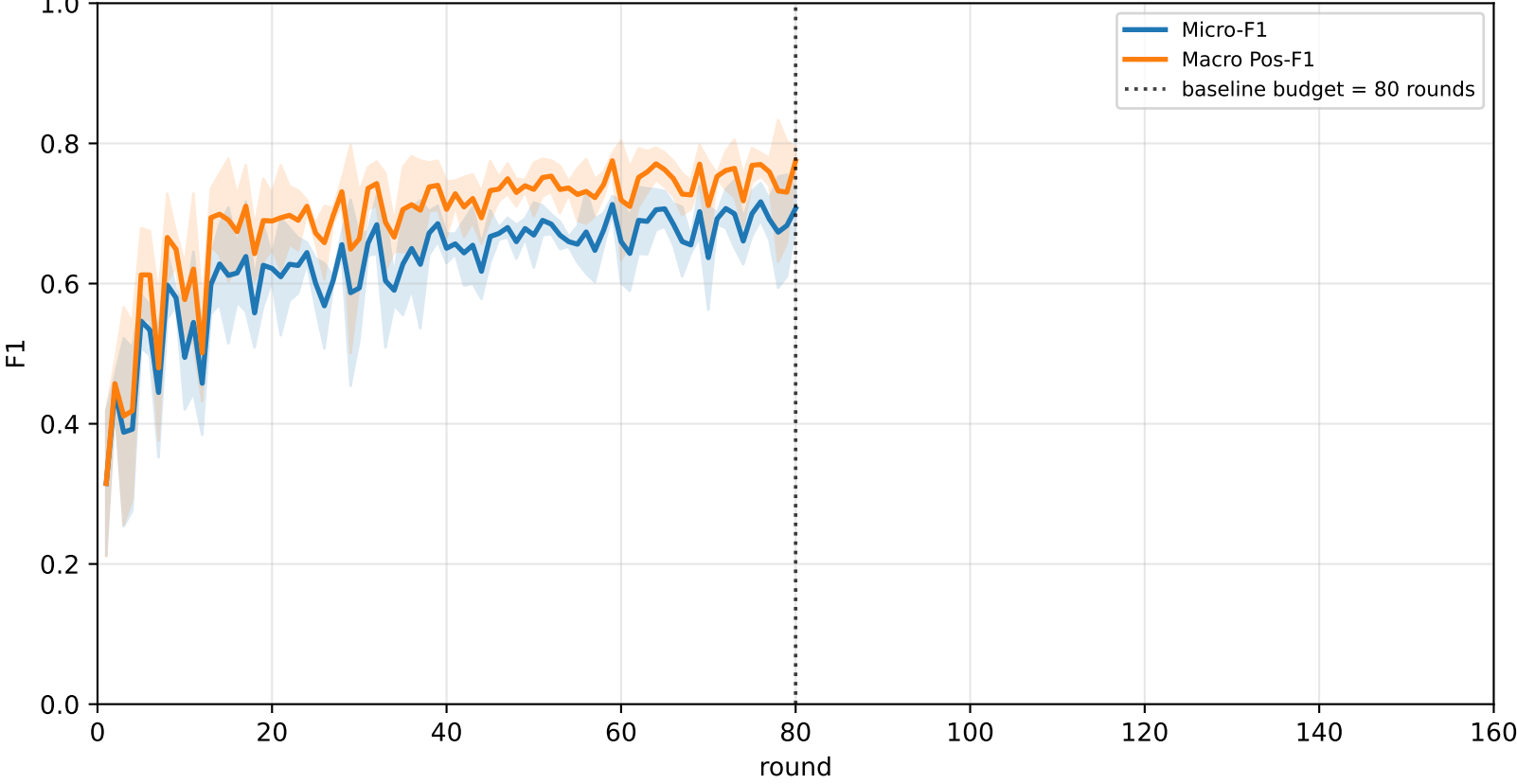
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	69.6% ± 3.7%	76.6% ± 3.0%	96.4% ± 1.1%	94.7% ± 1.9%	86.7% ± 5.4%	52.7% ± 5.2%	52.7% ± 4.1%
APPLE	75.6% ± 2.5%	78.9% ± 2.4%	91.1% ± 2.1%	93.4% ± 3.7%	78.7% ± 3.9%	72.2% ± 1.0%	59.2% ± 2.7%

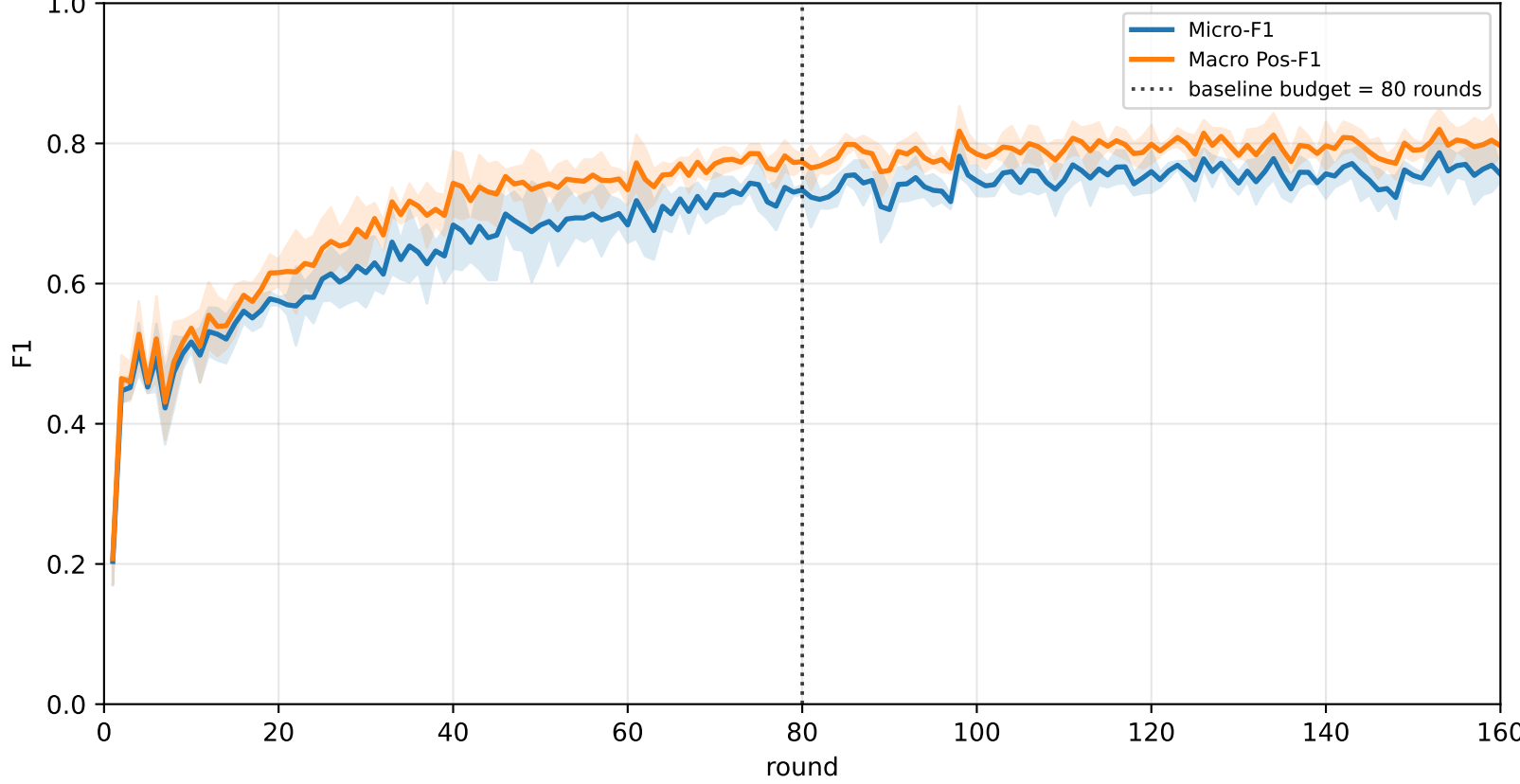
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



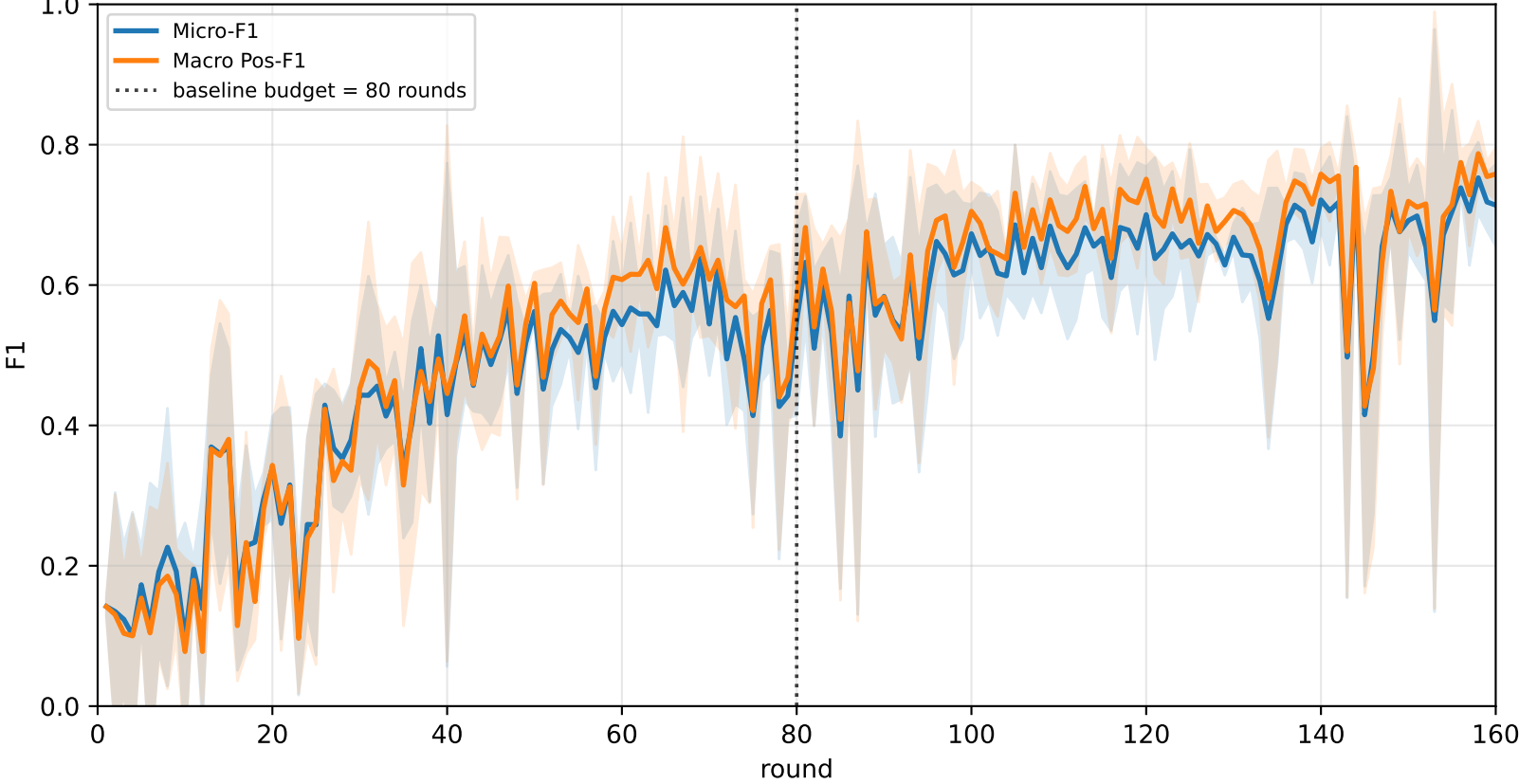
Pooled local centralized micro / macro F1



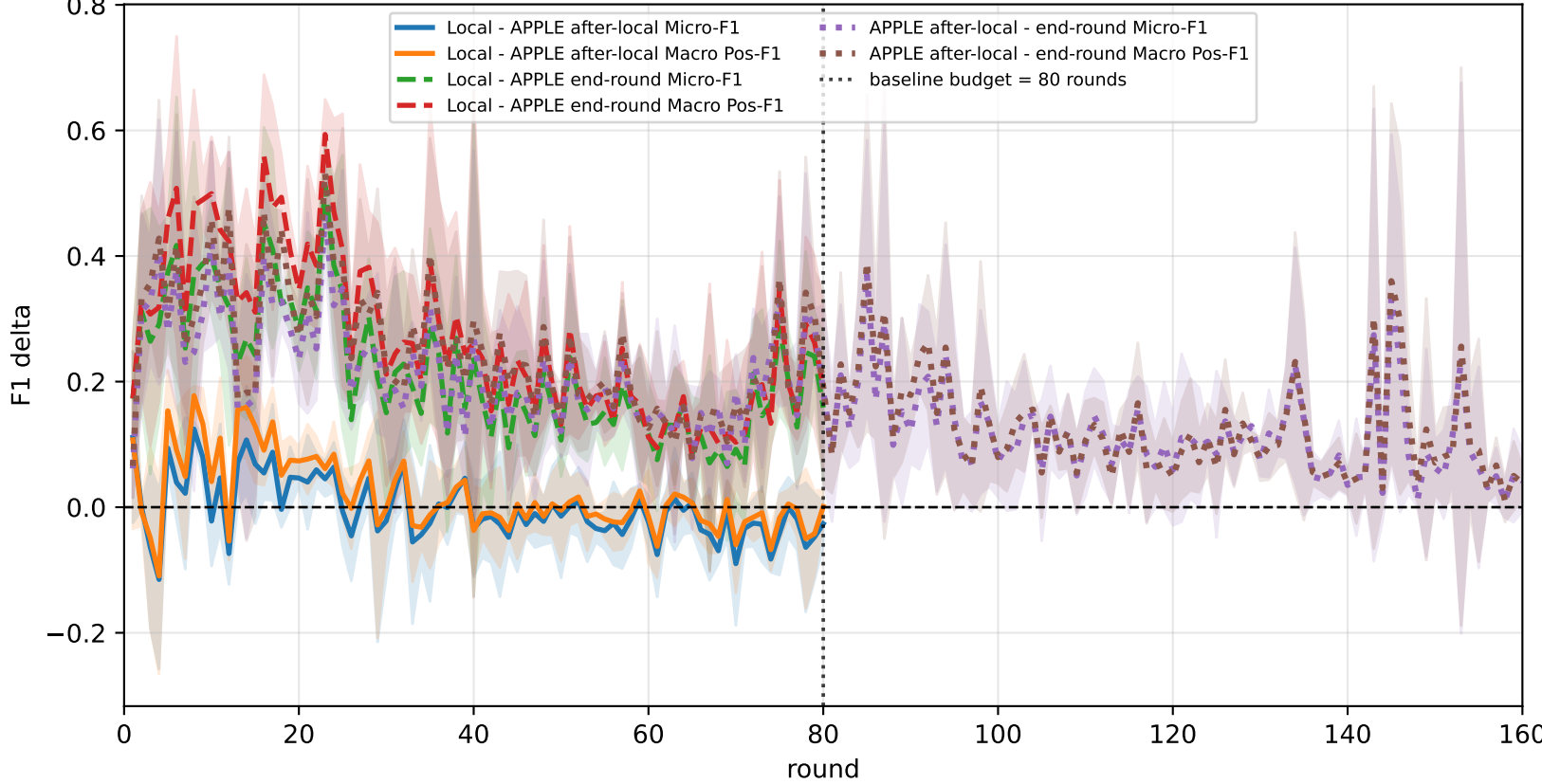
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



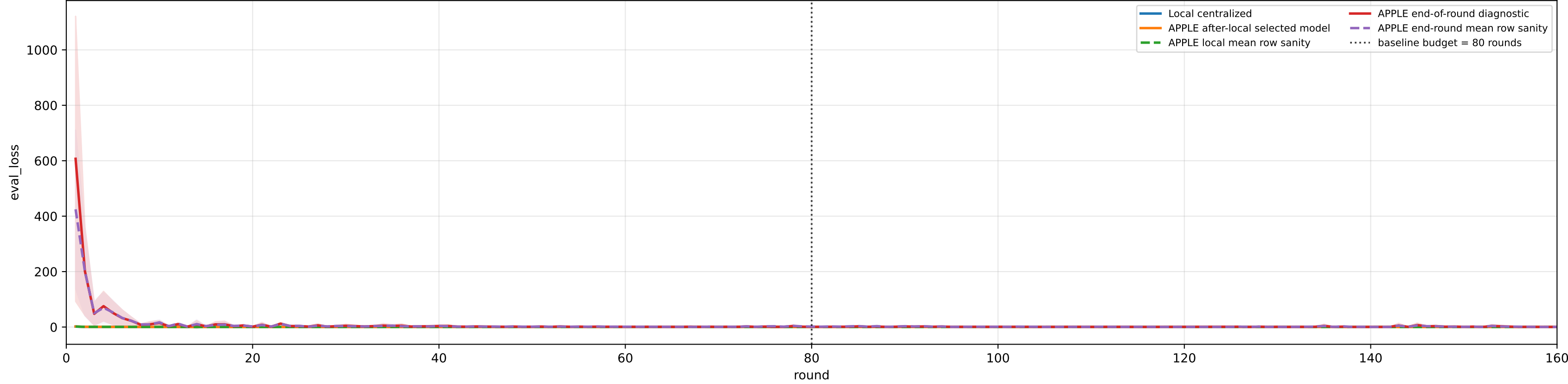
Controlled task-specialization level: High | APPLE diagnostics | family=specialized_cycle4 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 | family_centroid_jsd_mean=0.162

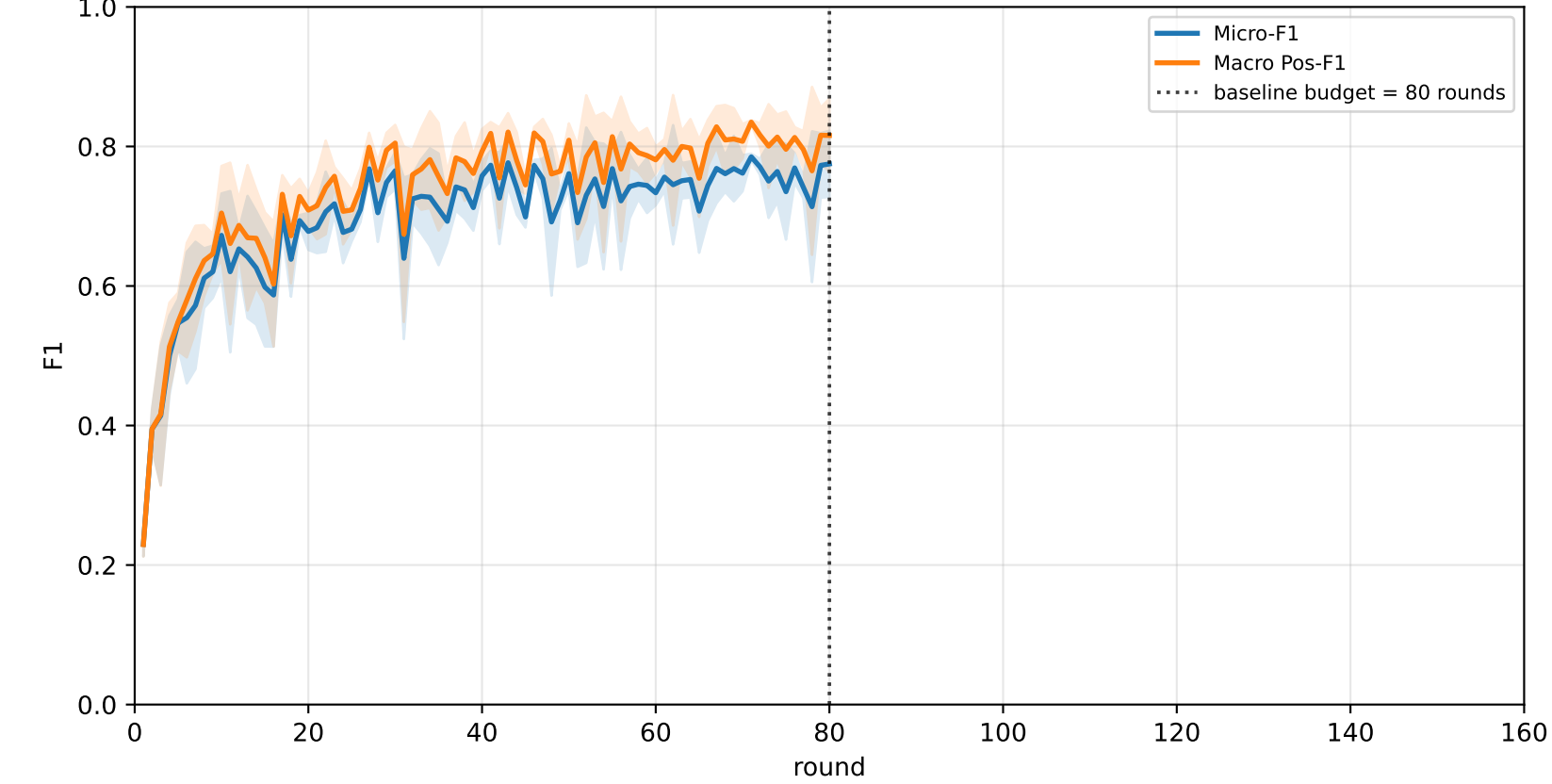
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	78.0% ± 1.1%	82.8% ± 1.1%	95.4% ± 2.6%	95.6% ± 1.0%	91.0% ± 2.8%	76.7% ± 2.4%	55.2% ± 0.7%
APPLE	76.8% ± 3.8%	80.4% ± 3.7%	94.4% ± 2.7%	89.1% ± 3.5%	85.2% ± 5.6%	76.2% ± 4.6%	57.1% ± 2.7%

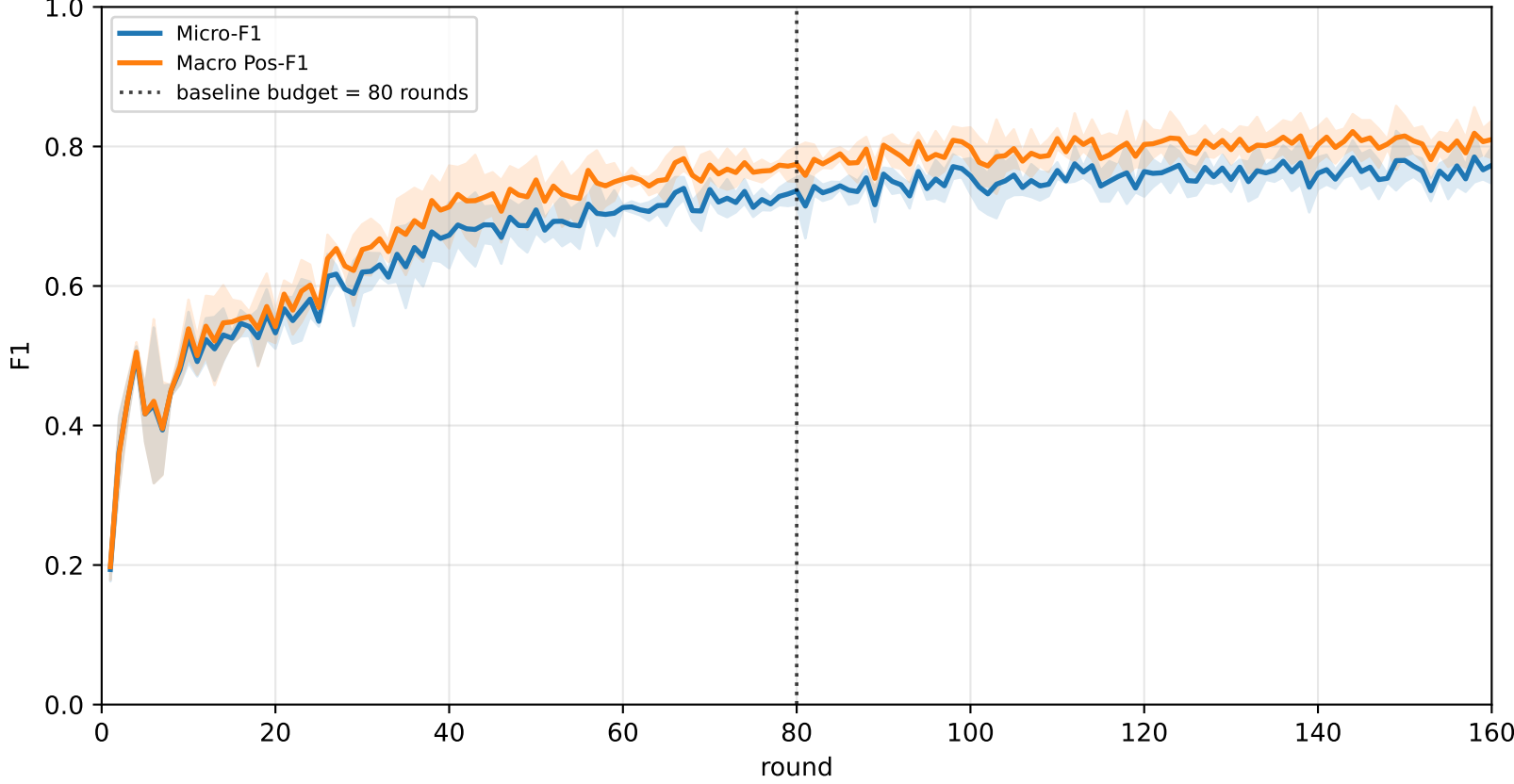
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



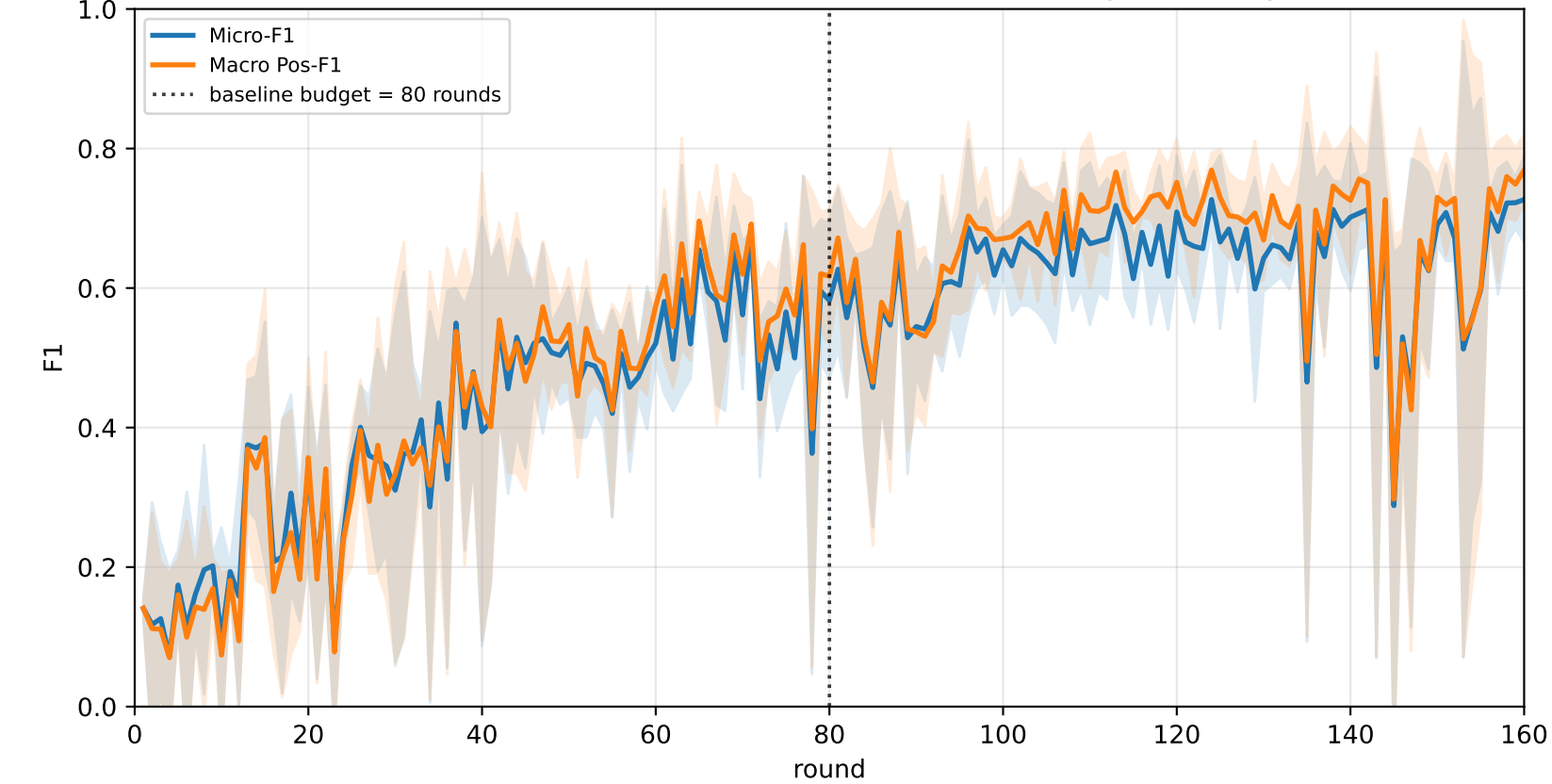
Pooled local centralized micro / macro F1



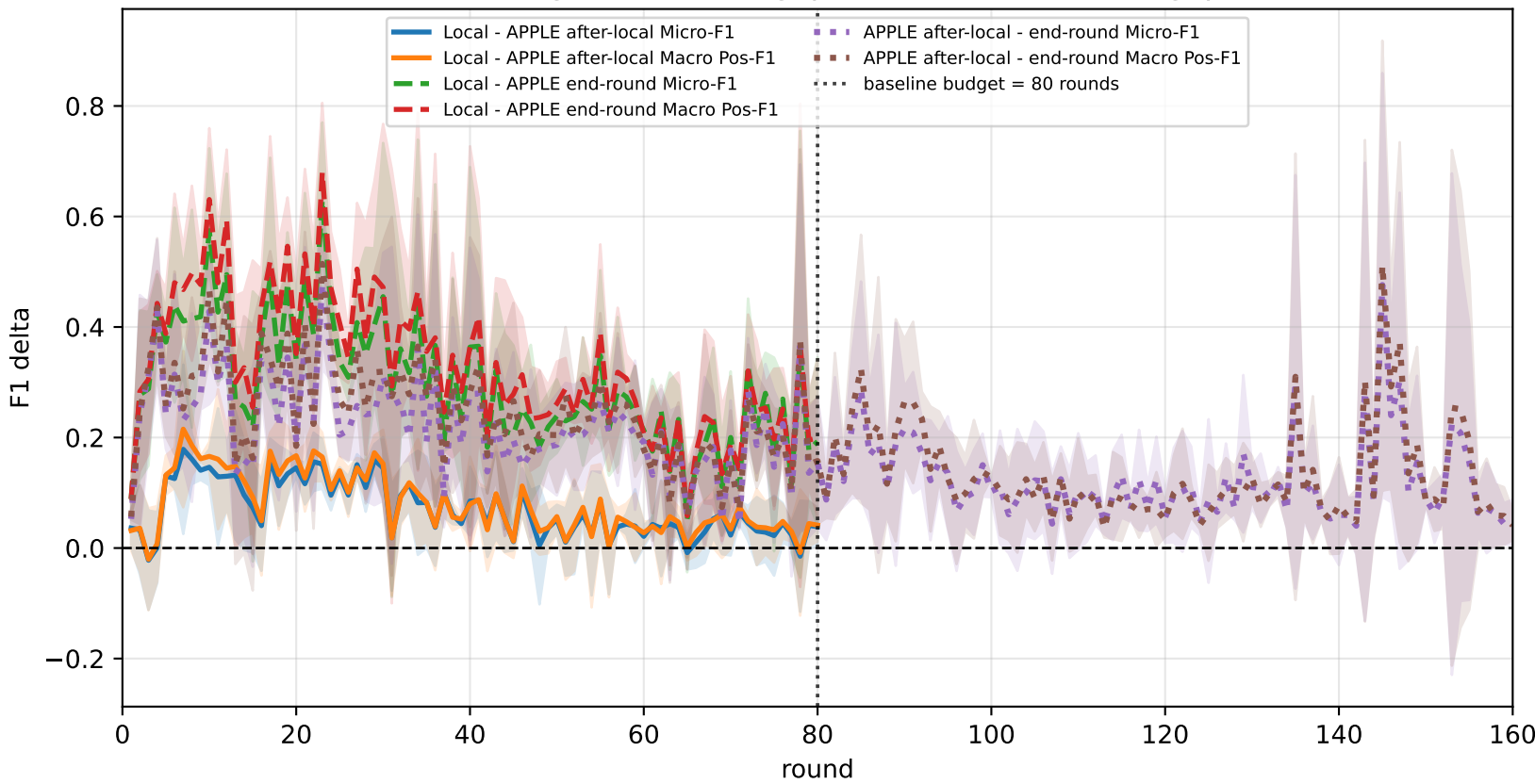
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



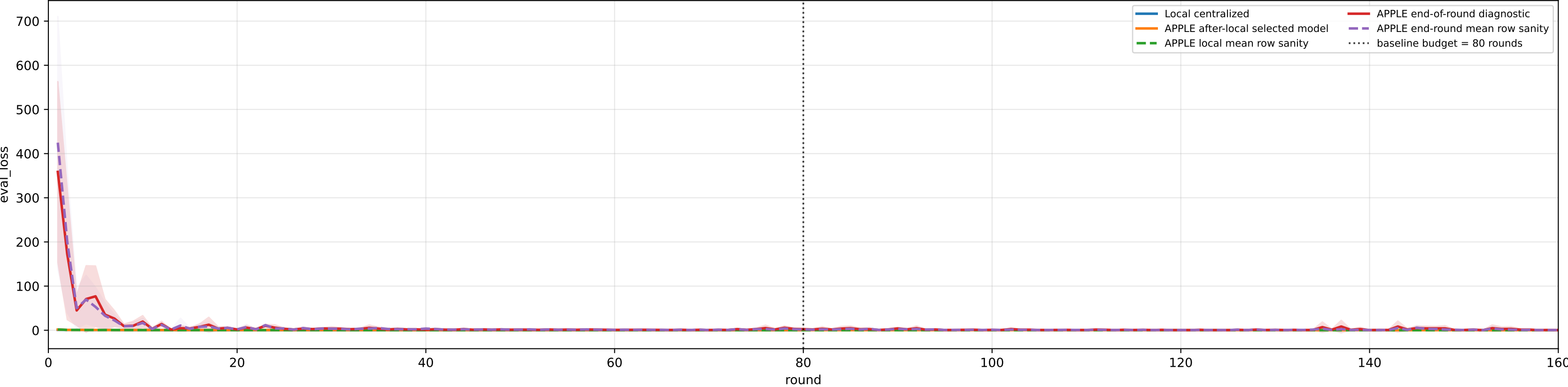
Controlled task-specialization level: High | APPLE diagnostics | family=specialized_cycle5 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 | family_centroid_jsd_mean=0.162

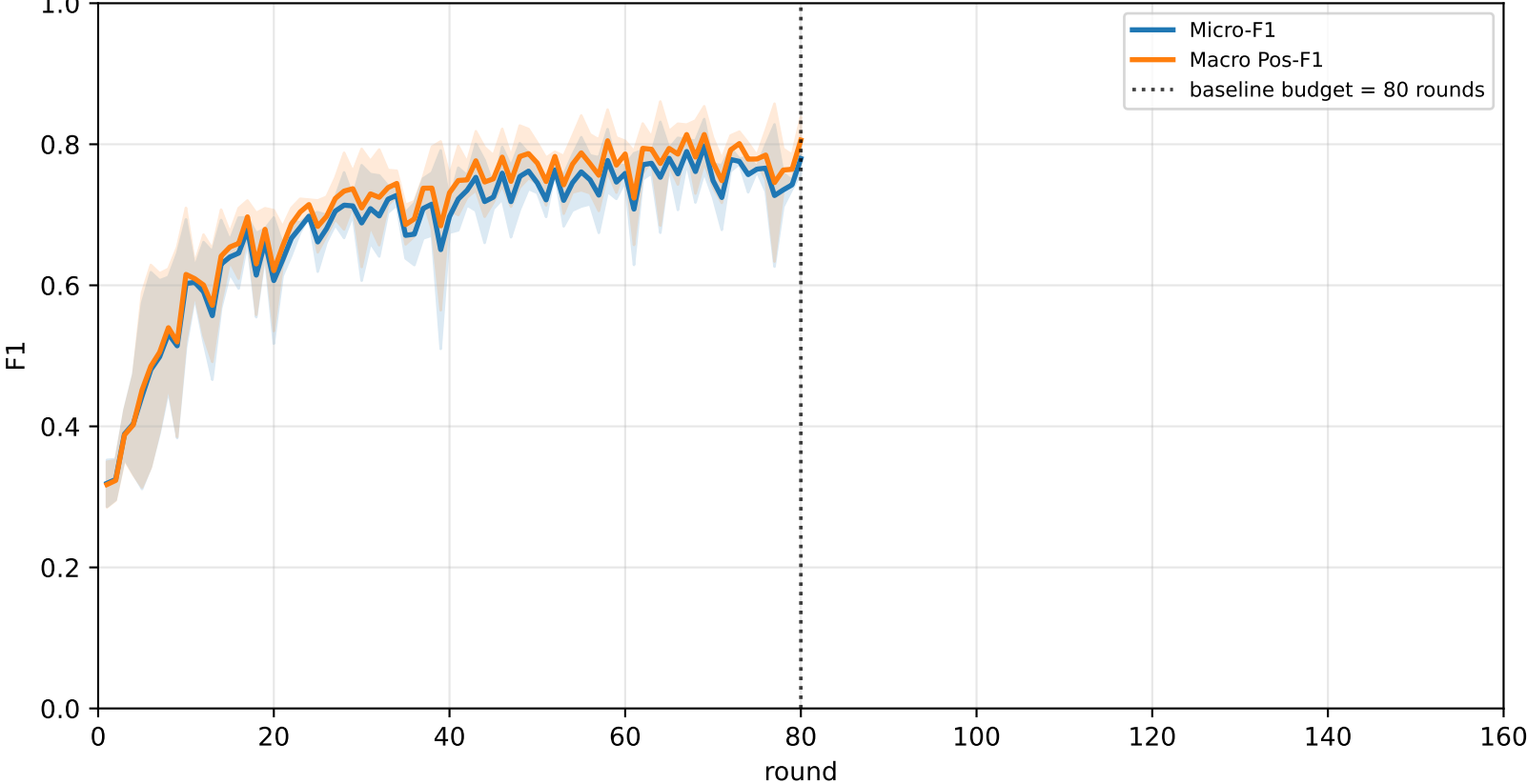
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	75.8% ± 0.9%	78.3% ± 1.5%	87.0% ± 1.3%	87.4% ± 3.9%	77.0% ± 3.6%	79.2% ± 5.6%	60.7% ± 3.5%
APPLE	76.3% ± 3.5%	79.5% ± 3.0%	93.0% ± 0.9%	87.9% ± 4.1%	80.0% ± 6.4%	77.6% ± 2.2%	58.8% ± 4.2%

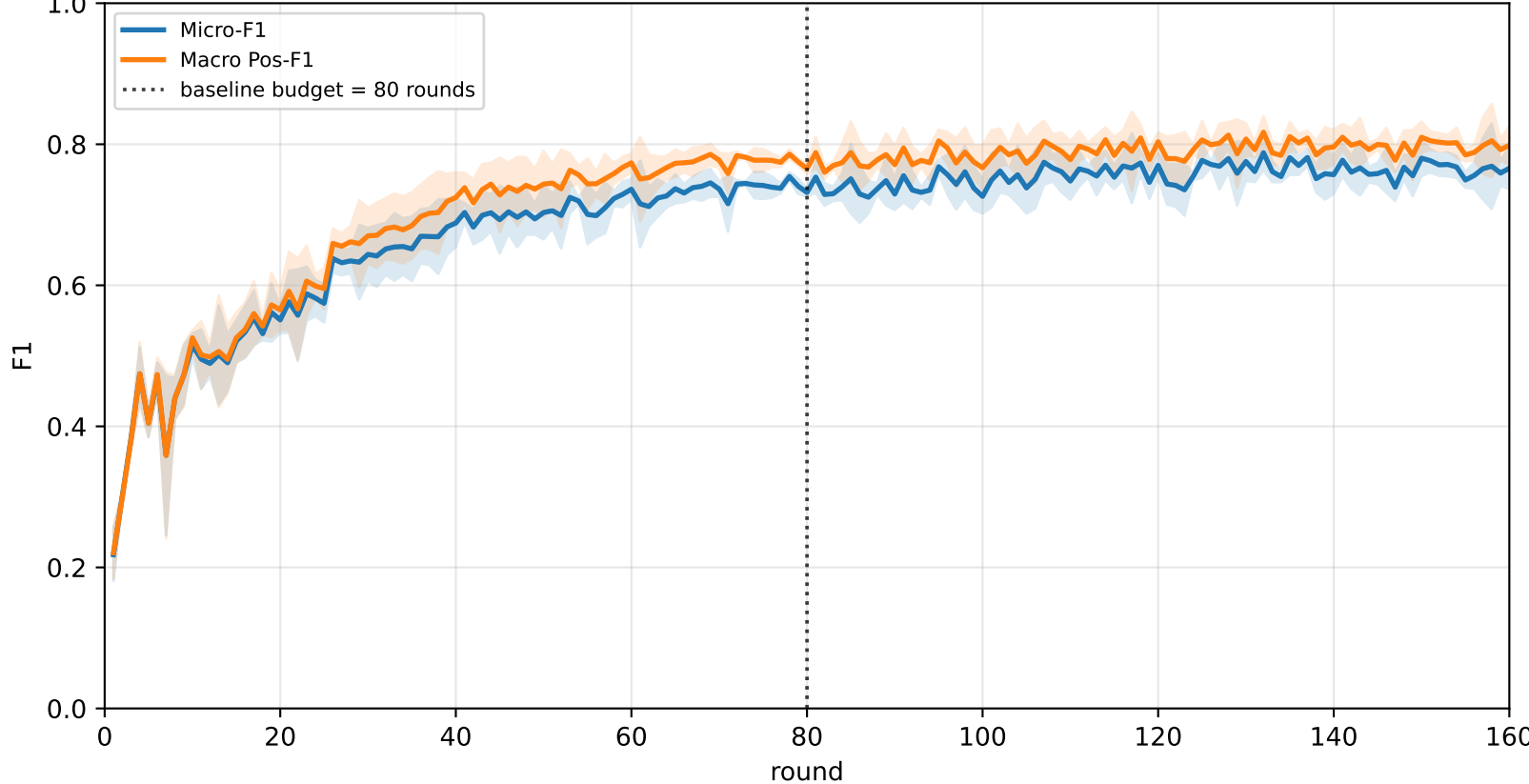
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



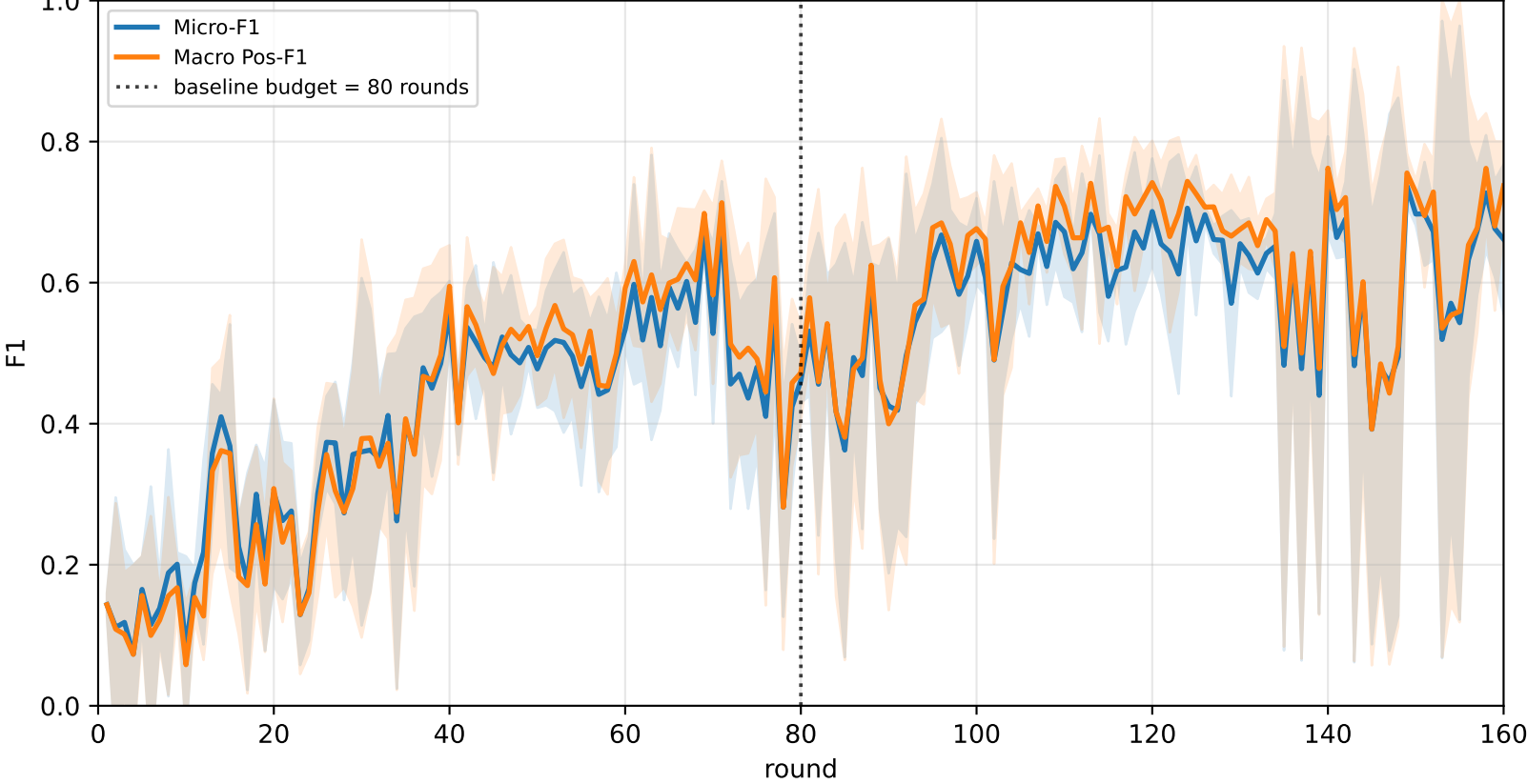
Pooled local centralized micro / macro F1



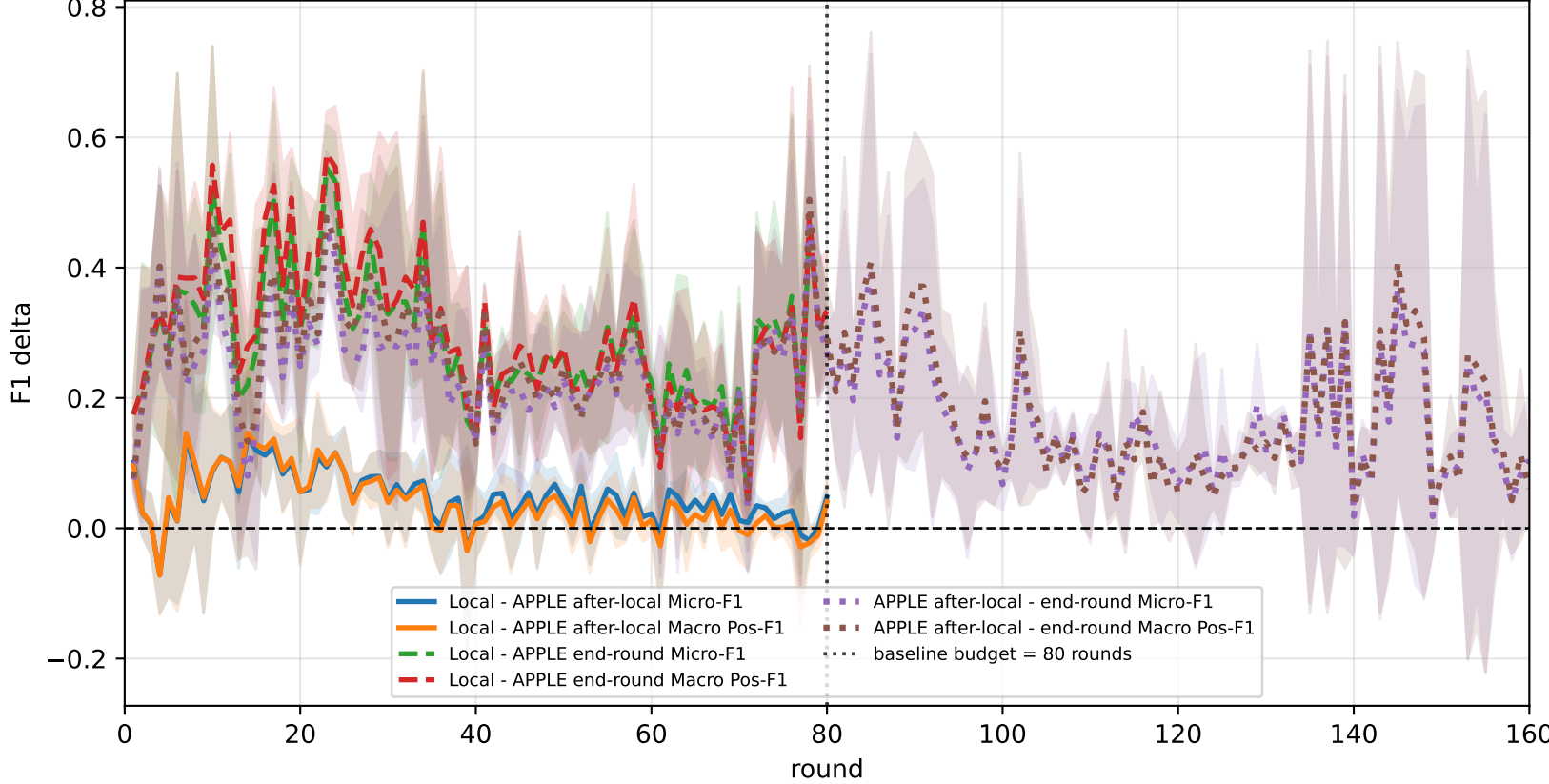
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: local gap and APPLE mismatch gap



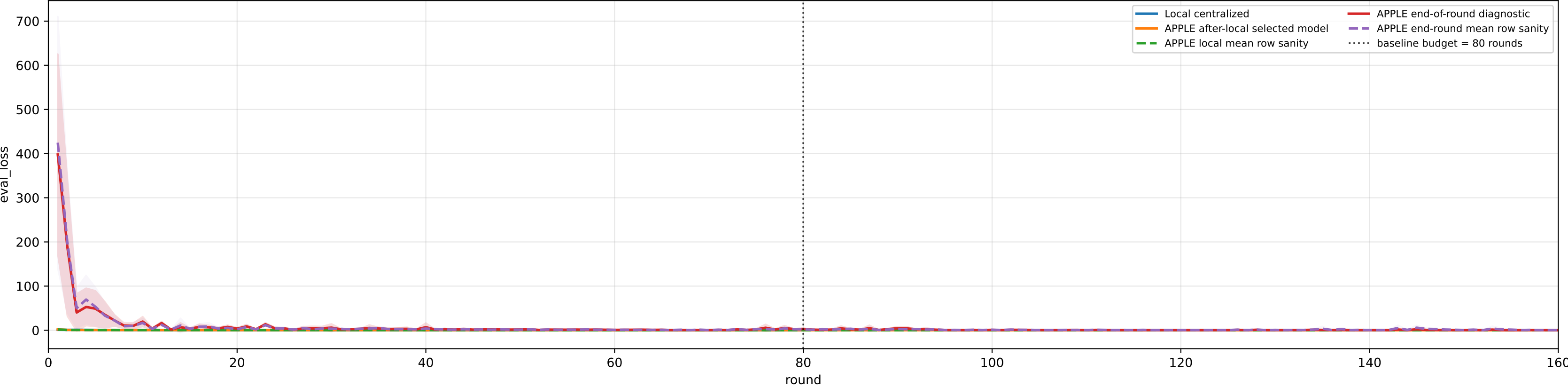
Controlled task-specialization level: High | APPLE diagnostics | family=specialized_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3% | specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 | family_centroid_jsd_mean=0.162

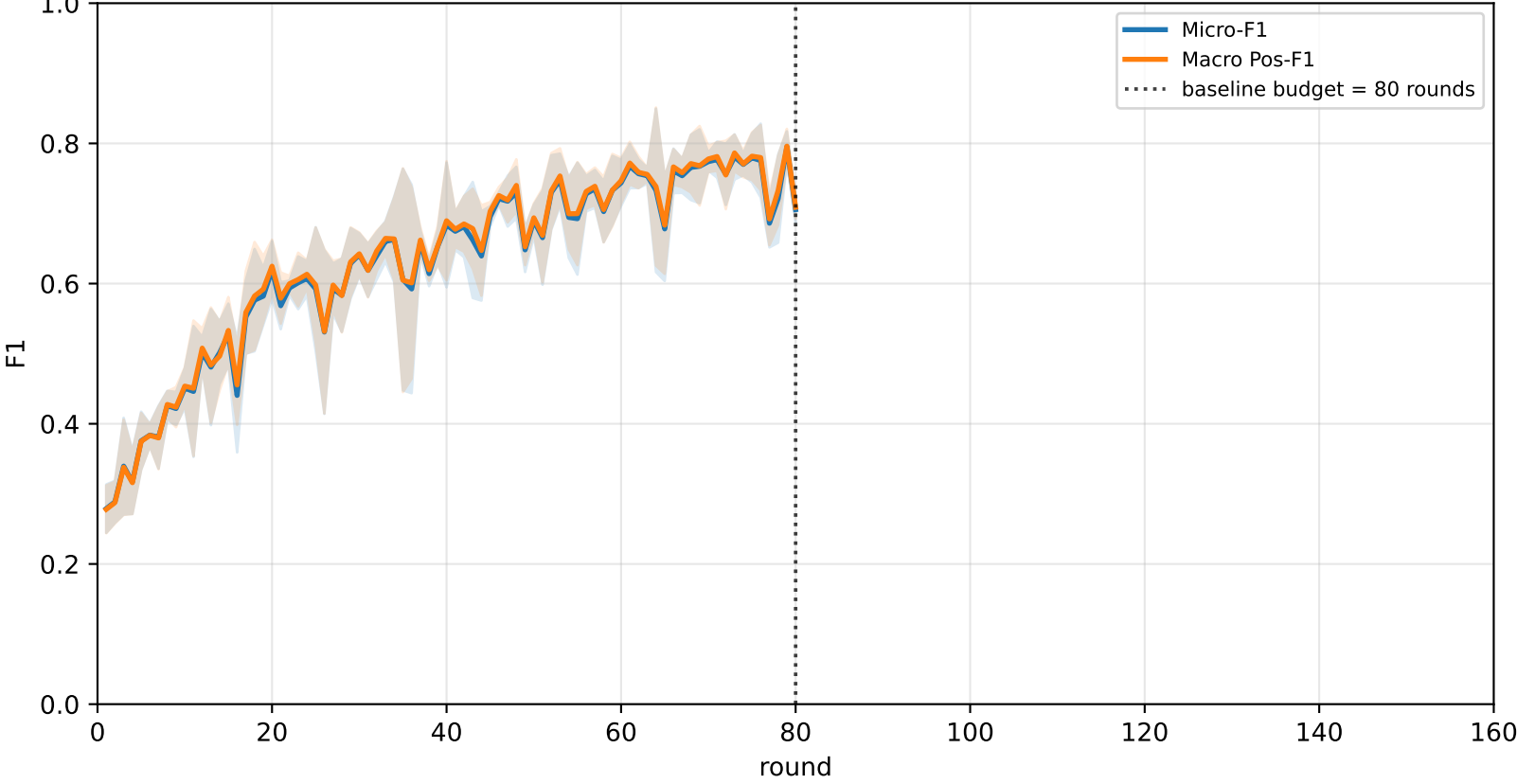
Best test performance (APPLE selected by after-local validation over 160 rounds; Local baseline 80 rounds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	78.5% ± 3.5%	79.1% ± 2.8%	83.3% ± 8.5%	79.7% ± 5.3%	76.3% ± 4.2%	77.7% ± 6.0%	78.4% ± 6.9%
APPLE	73.8% ± 2.9%	75.2% ± 2.3%	87.7% ± 3.2%	77.0% ± 3.4%	70.2% ± 0.8%	67.2% ± 4.3%	74.2% ± 10.0%

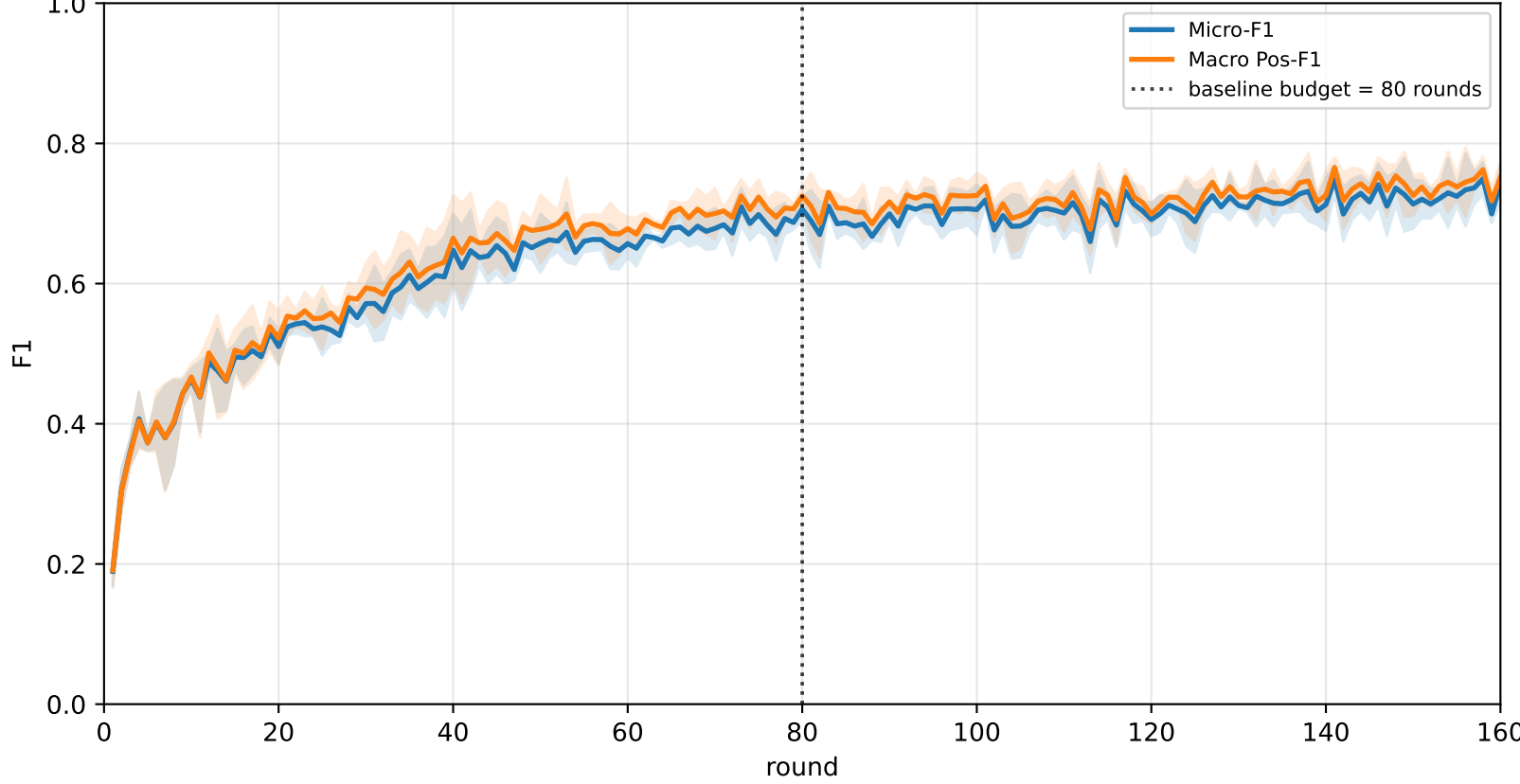
Validation loss: selected APPLE vs end-of-round mismatch (160-round APPLE run)



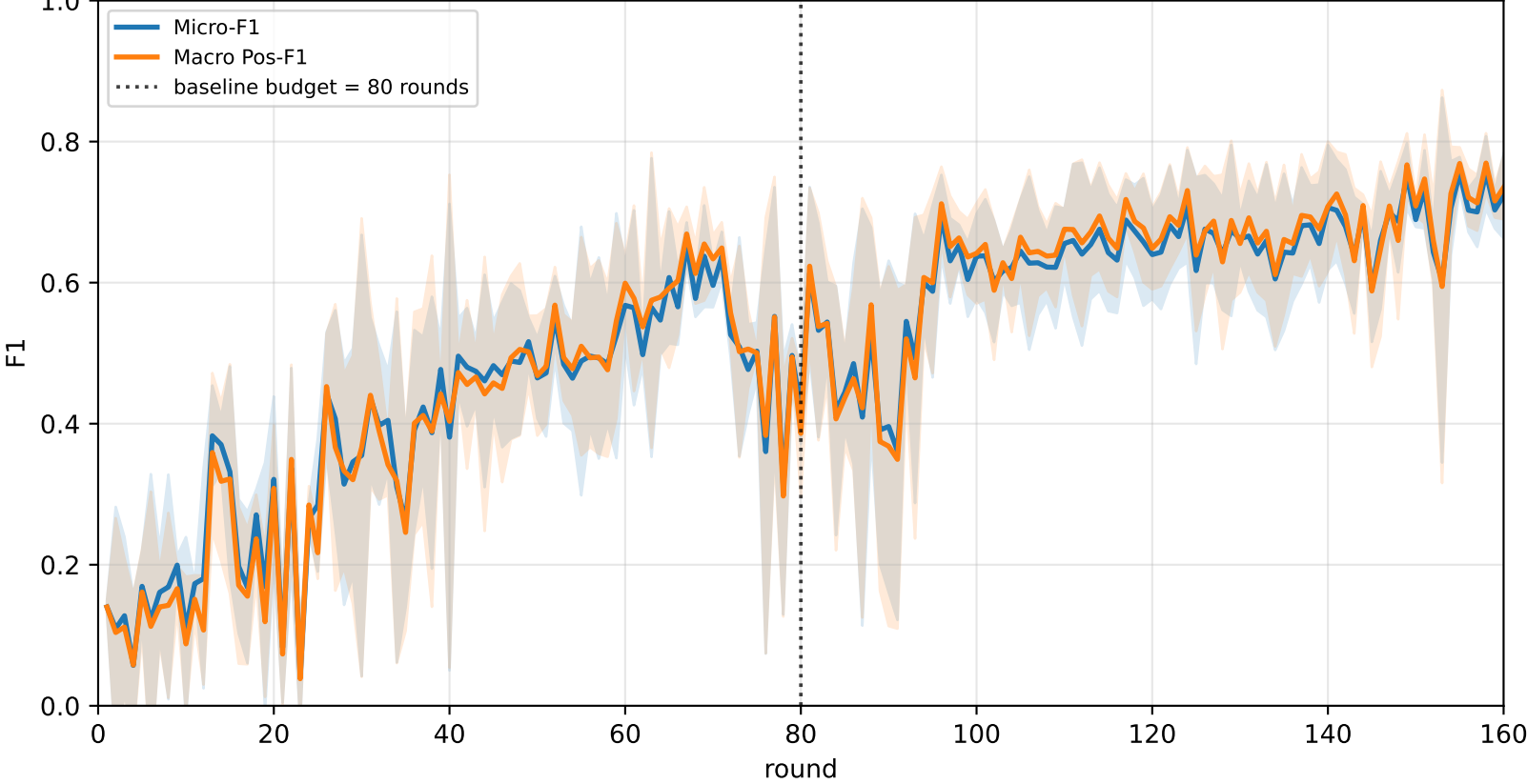
Pooled local centralized micro / macro F1



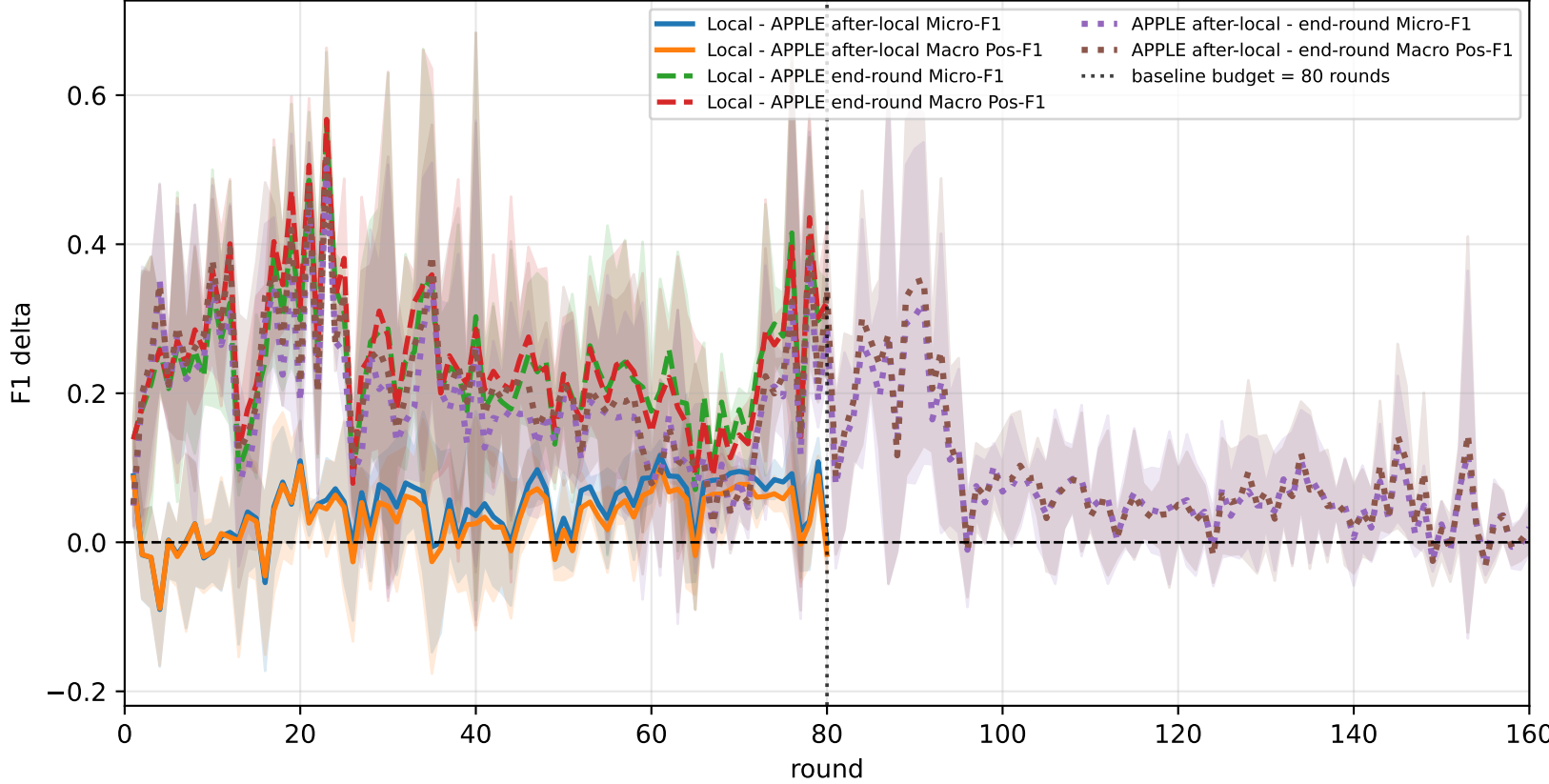
Pooled APPLE micro / macro F1 (after-local selected model)



Pooled APPLE micro / macro F1 (end-of-round diagnostic only)



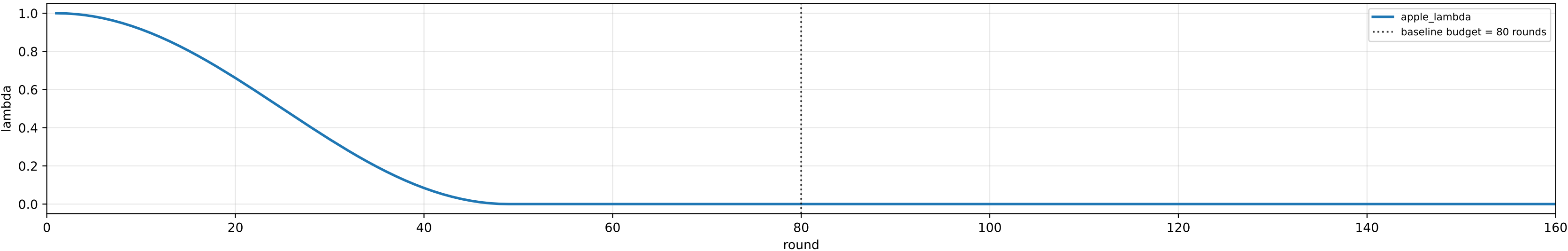
Delta diagnostics: local gap and APPLE mismatch gap



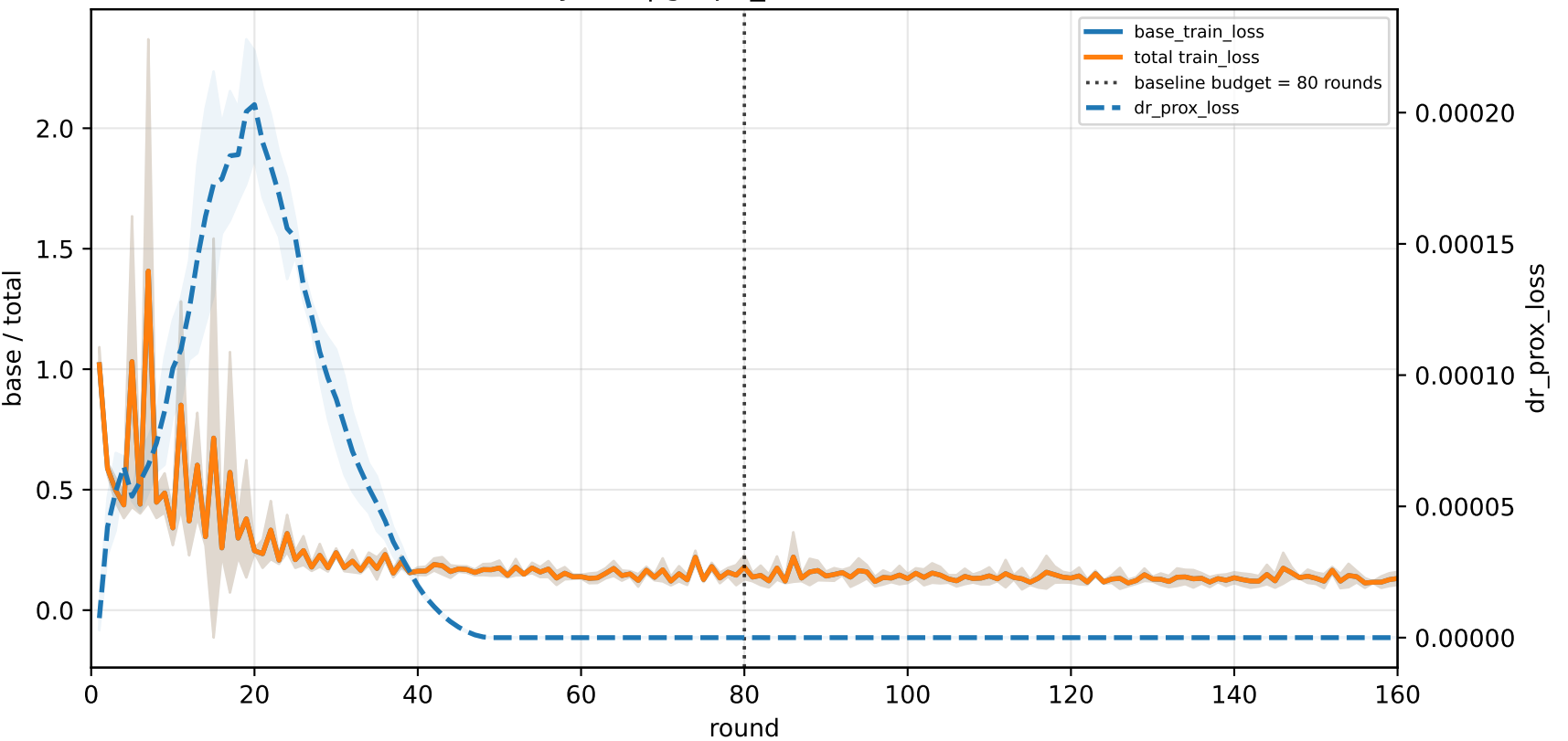
APPLE training-loss decomposition | High heterogeneity

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 |
family_centroid_jsd_mean=0.162

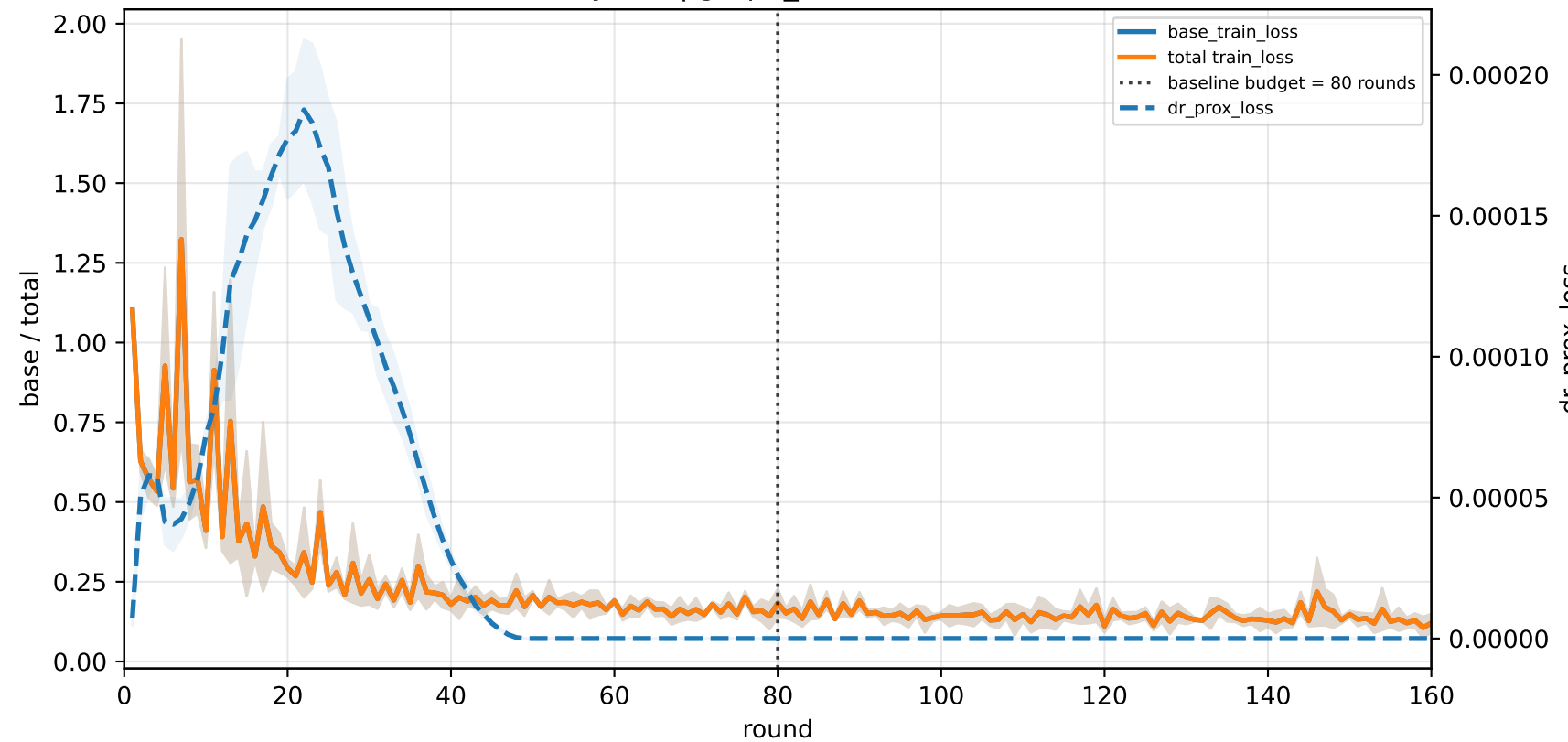
APPLE lambda scheduler | apple_mu=0.01



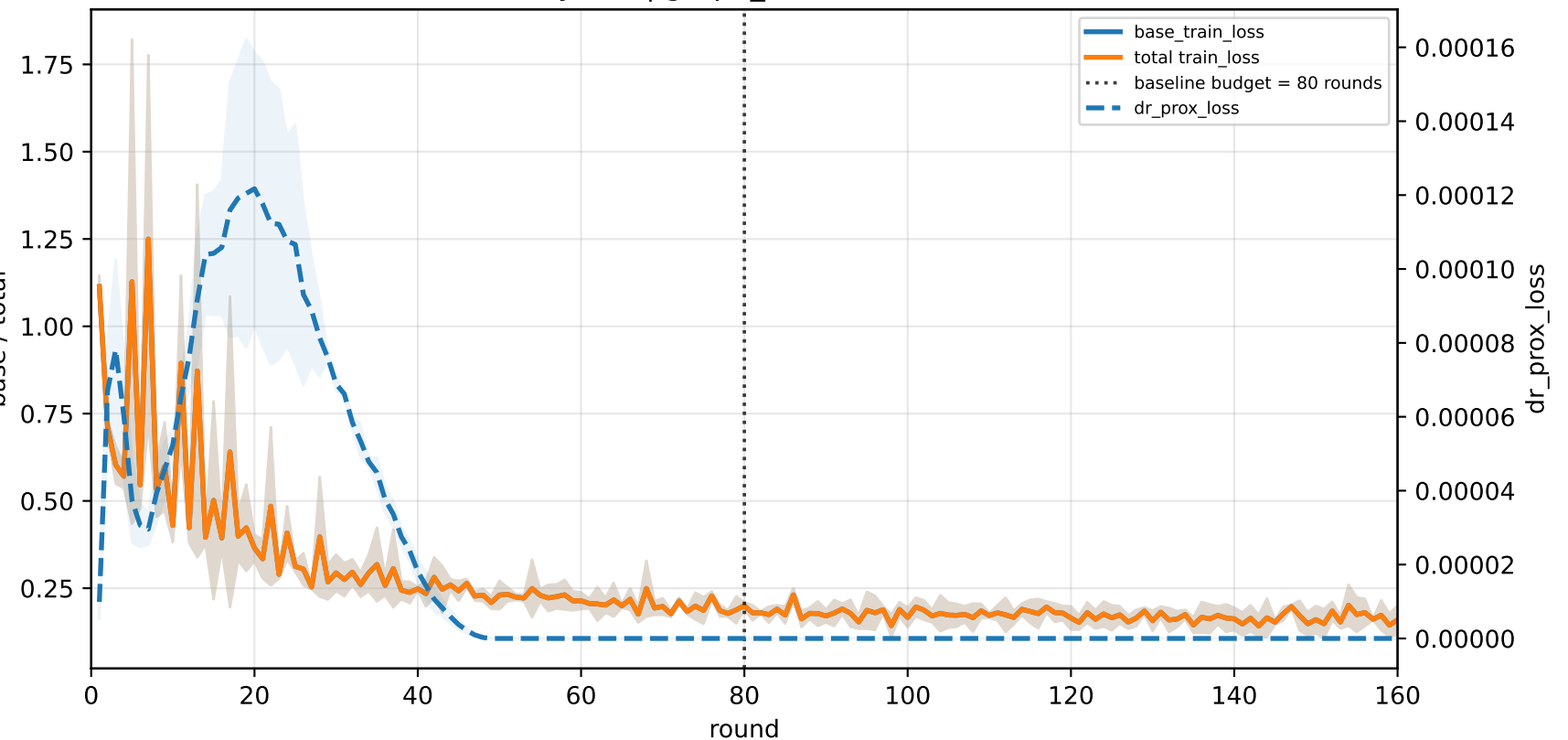
cycle2 | graph_id=2000010



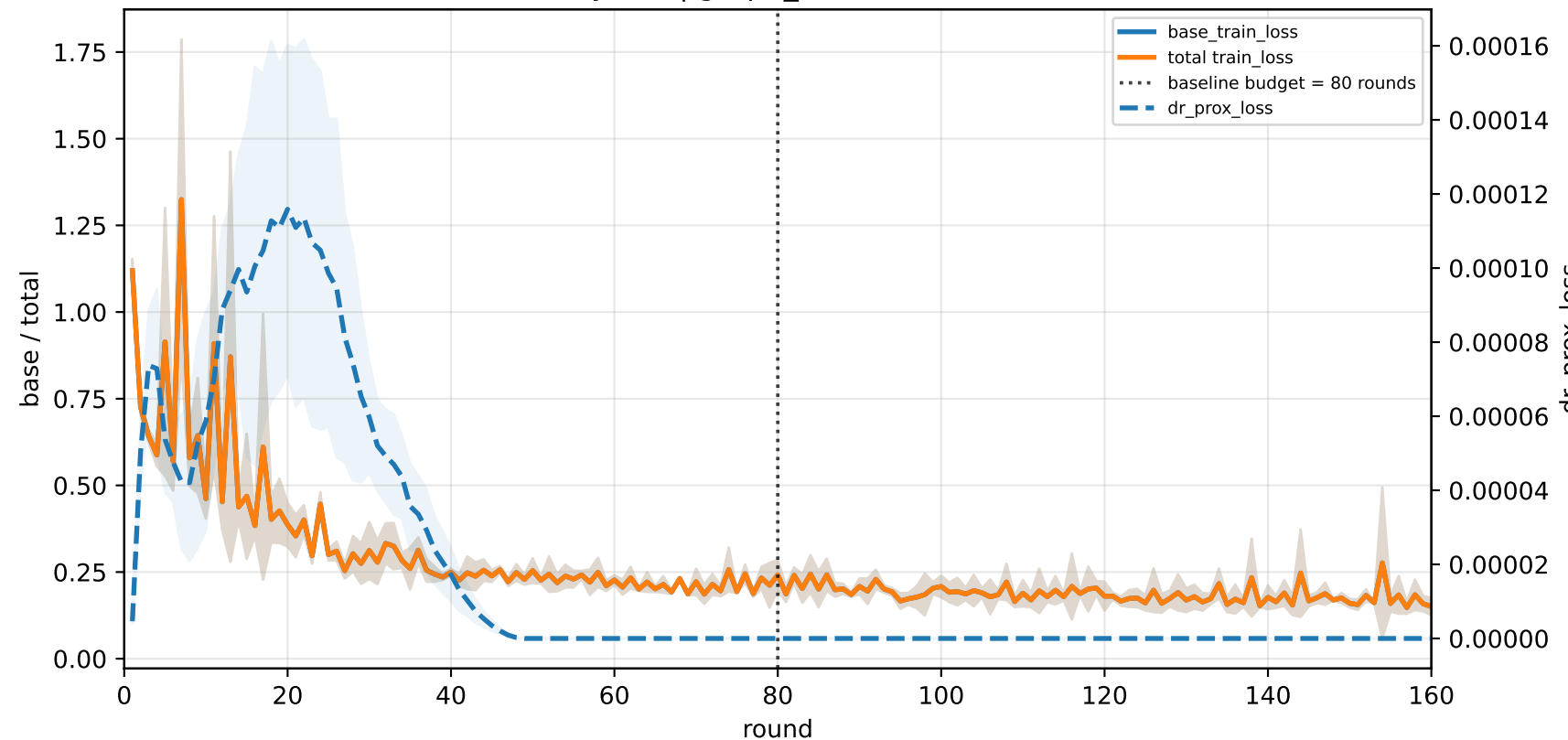
cycle3 | graph_id=2000011



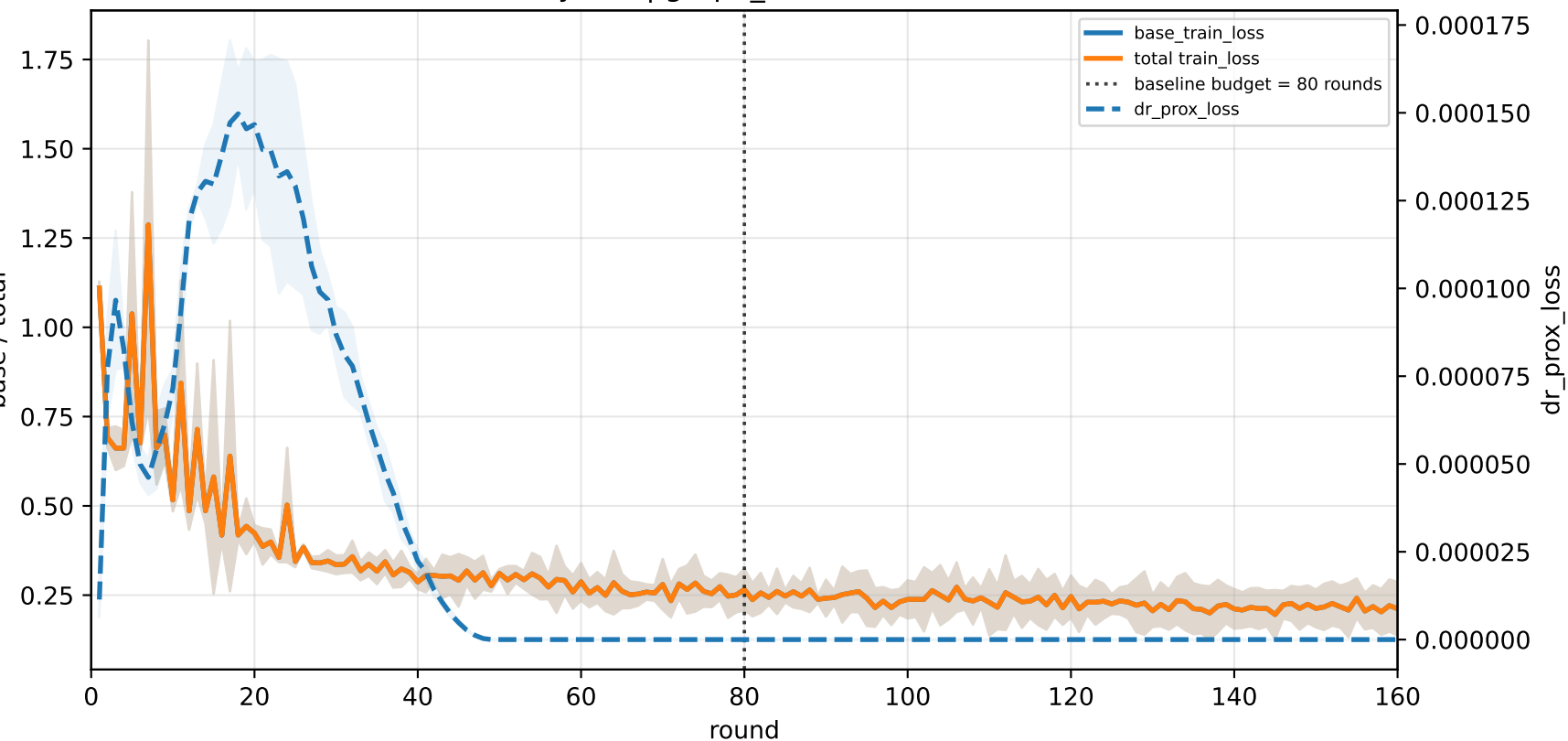
cycle4 | graph_id=2000012



cycle5 | graph_id=2000013



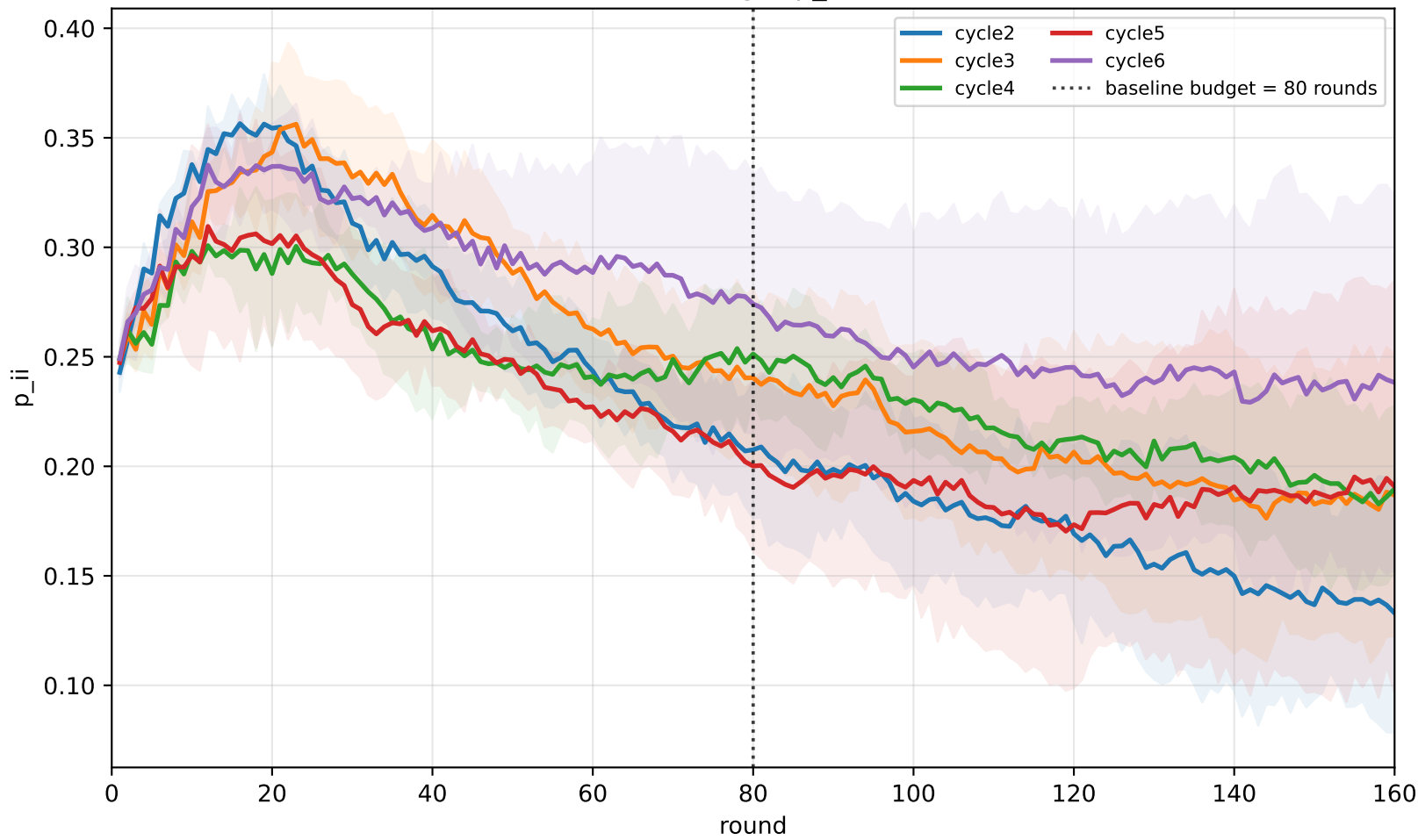
cycle6 | graph_id=2000014



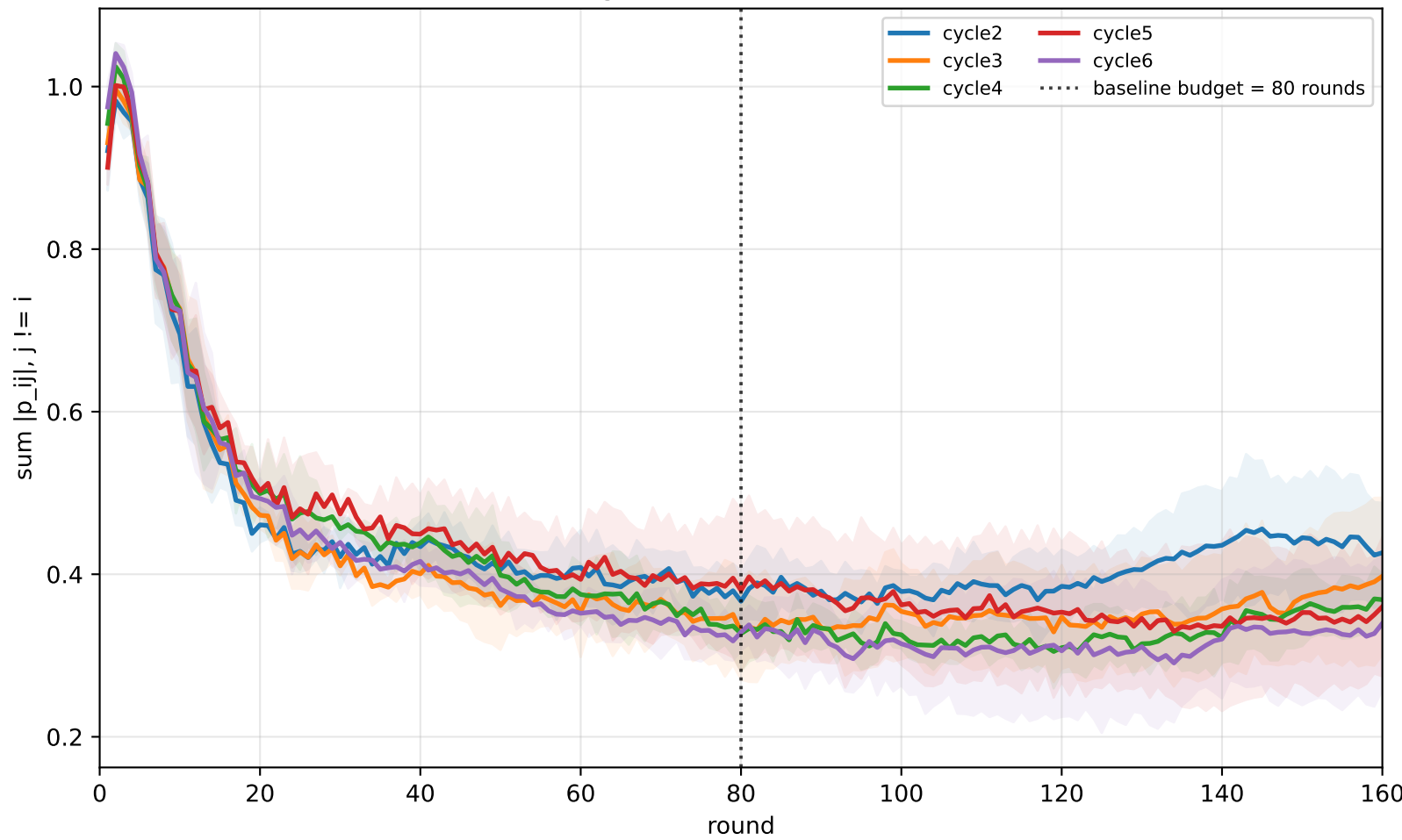
APPLE DR behavior | High heterogeneity

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 | family_centroid_jsd_mean=0.162

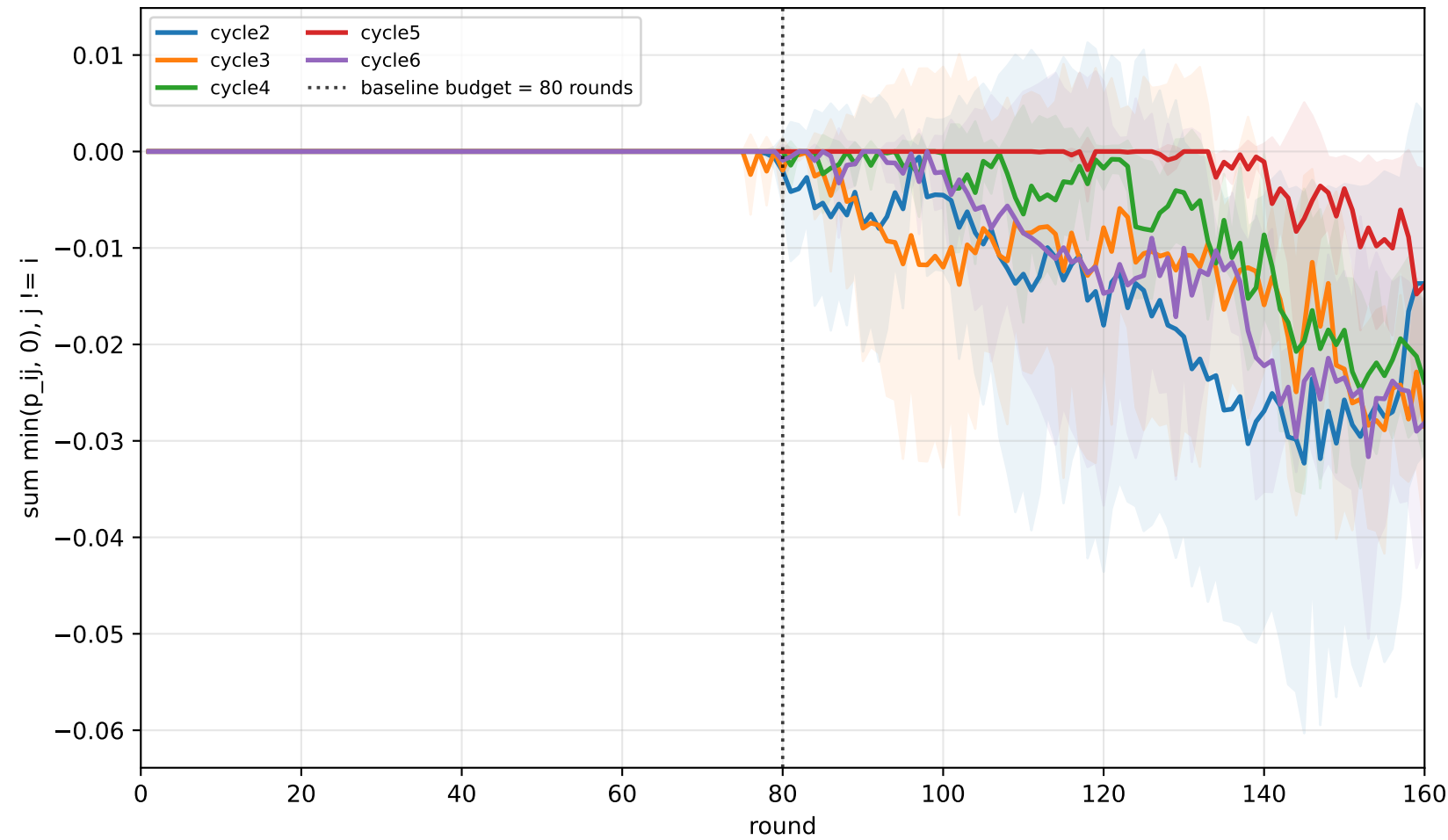
APPLE DR self-weight p_{ii} over rounds



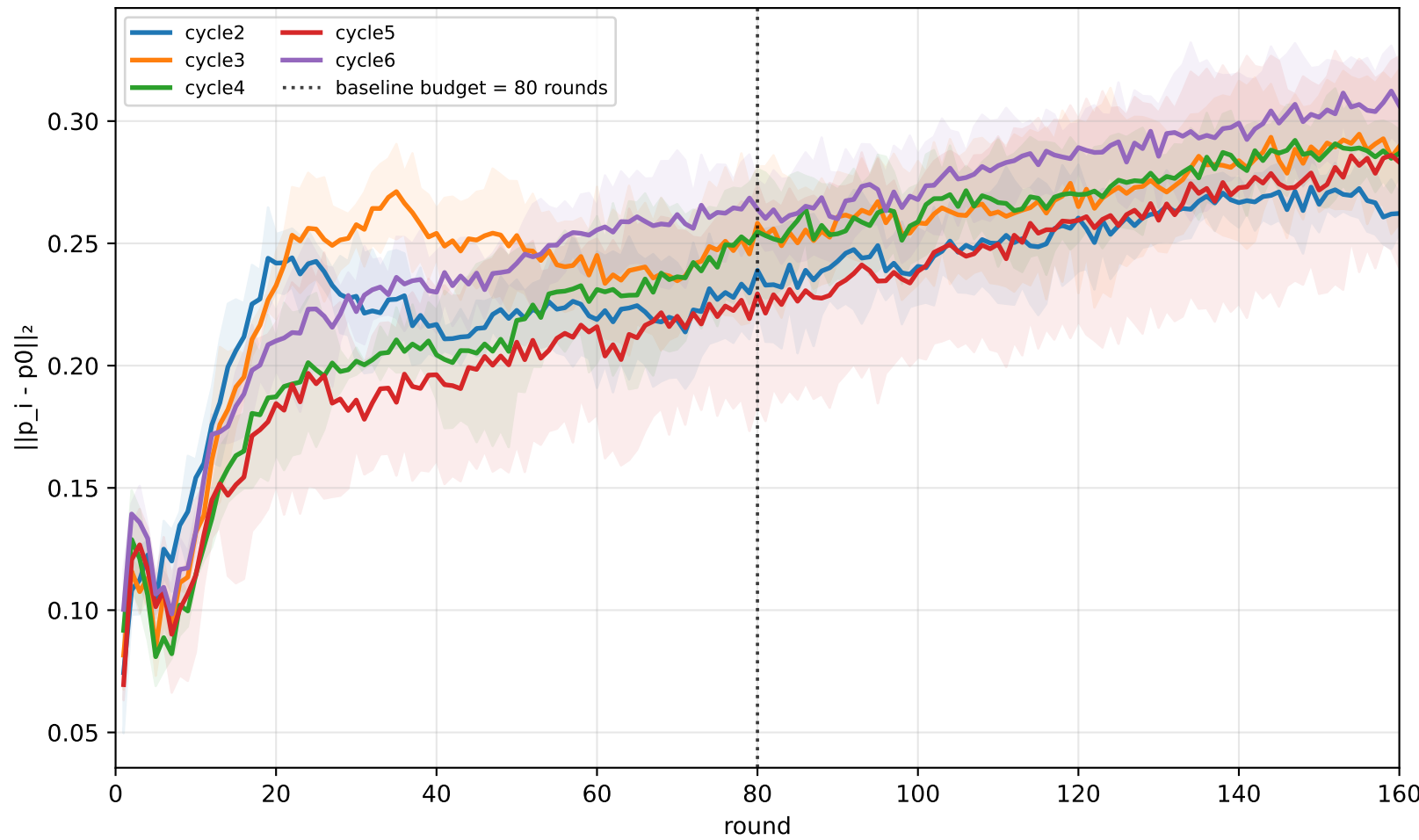
APPLE off-diagonal absolute mass over rounds



APPLE off-diagonal negative mass over rounds



APPLE DR movement away from p_0



Selected-best-round APPLE DR coefficients p_{ij} , mean over seeds

